

# Optimizing LLM Inference: Fluid-Guided Online Scheduling with Memory Constraints

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**Abstract.** Large language models now serve millions of users daily, with providers incurring costs exceeding \$700,000 per day. Each request requires token-by-token inference, making GPU scheduling central to latency, capacity, and cost. The difficulty is endogenous memory growth: generated tokens expand the Key-Value (KV) cache, and overflow can evict in-progress requests and waste prior computation. We formulate inference as a multi-stage online scheduling problem with endogenous memory growth, linear iteration times, and GPU-resident KV-cache constraints. We introduce a fluid model that characterizes equilibrium batch composition, memory requirement, and stability region. Guided by the fluid model, we design WAIT (Waiting for Accumulated Inference Threshold), a threshold-based admission rule for known output lengths, and Nested WAIT, which extends the rule to unknown output lengths by regulating how requests advance across decode-stage segments. Both algorithms achieve asymptotic performance comparable to the fluid benchmark. Nested WAIT uses an additional safety buffer of moderate scale to handle unknown output lengths and prevent memory overflow that would otherwise force evictions. In Vidur simulations configured for Llama-2-7B on an A100 GPU, with supplemental real-GPU validation reported in the appendix, the policies enlarge the empirically observed stable operating range relative to widely used baseline algorithms and reduce latency especially in near-overloaded and overloaded regimes.

**Key words:** Large Language Model, Key-value cache, Memory Constraint, Online scheduling

## 1. Introduction

Large Language Models (LLMs) now dominate natural language processing (NLP) (Devlin et al. 2019, Brown et al. 2020, Kaplan et al. 2020, Ouyang et al. 2022, Wei et al. 2022a,b, Touvron et al. 2023, OpenAI 2024), powering chatbots (Anthropic 2023, Bai et al. 2023, OpenAI 2024, DeepSeek-AI 2025), search engines (Google 2023, Microsoft 2023), and programming assistants

([GitHub 2022](#), [Anthropic 2025](#)). These models generate text through *inference*, which imposes significant computational demands on memory and processing resources ([García-Martín et al. 2019](#)). Systems like ChatGPT incur daily inference costs exceeding \$700,000 and contribute to substantial carbon emissions ([Patel 2023](#), [Patterson et al. 2021](#)). Efficient inference scheduling can reduce these operational costs and energy consumption ([Strubell et al. 2019](#), [Desislavov et al. 2021](#)) by balancing system throughput and response latency ([Wu et al. 2022](#)).

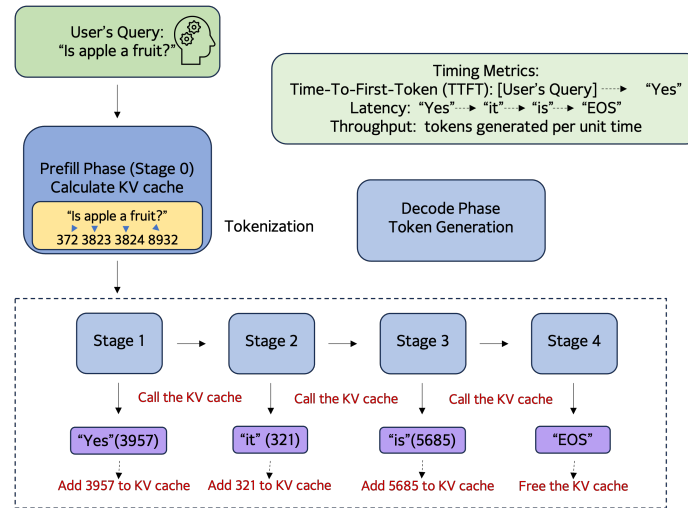
Figure 1 illustrates how a query is processed during LLM inference. The user submits a prompt (e.g., “Is apple a fruit?”), which then passes through two computational phases:

- *Prefill phase (stage 0)*. Upon receiving the prompt, the model first tokenizes the input into a sequence of discrete units (e.g., “Is”, “apple”, “a”, “fruit”, “?”). These tokens are then embedded and simultaneously processed in a single forward pass to compute the *Key-Value (KV) cache*, which stores intermediate representations (i.e., attention keys and values) for each token. These precomputed values enable efficient reuse during subsequent decoding steps. This phase corresponds to Stage 0 in Figure 1, where all prompt tokens are embedded and their KV representations are added to the cache.
- *Decode phase (stages 1 to  $l'$ )*. After the prefill phase, the model enters the decode phase, where it generates the output one token at a time. At each decode stage, the model queries the existing KV cache to compute the next token, appends the new token to the output sequence, and updates the KV cache with its key-value pair. For instance, the model might first generate “Yes” (Stage 1), then “it” (Stage 2), followed by “is” (Stage 3), and finally the “End of Sequence” (EOS) token (Stage 4). This progression is depicted in the lower part of Figure 1. Each token generation step involves both reading from and writing to the KV cache, so the KV cache grows linearly with the length of the generated sequence.

Figure 1 also highlights key performance metrics in LLM inference:

- *Time-to-first-token (TTFT)* measures the latency from user input to the first generated token.
- *Latency* refers to the total time required to complete the generation of all output tokens.
- *Throughput* captures the average number of tokens generated per unit time.

This inference structure, particularly the growing memory requirements and sequential decoding pattern, introduces fundamental constraints on scheduling and batching. Efficient scheduling must account for both prompt heterogeneity and KV cache dynamics to balance latency and resource use.



**Figure 1** An example of LLM inference.

Scheduling LLM inference tasks involves grouping prompts into *batches* processed concurrently on a GPU. Batching improves throughput: processing multiple prompts together amortizes the fixed overhead of each iteration and better uses GPU parallelism. However, this benefit comes with a fundamental tension: larger batches consume more memory through their combined KV caches. The KV cache improves efficiency, as it prevents the model from recalculating attention history for each new token; without it, computational cost would scale quadratically with sequence length. But the KV cache also grows dynamically during decode, making memory consumption unpredictable. This creates a core trade-off: *larger batches increase throughput but consume more memory, risking capacity overflow*.

Memory overflow can occur in several practical scenarios: (i) *bursty arrivals*, where sudden traffic spikes temporarily exceed capacity; (ii) *unknown output lengths*, where prompts generate longer responses than anticipated; and (iii) *multi-tenant environments*, where multiple users compete for GPU memory. The unknown output length problem is particularly challenging: since the scheduler cannot predict how much memory each prompt will eventually consume, it cannot reliably plan batch composition or admission decisions. Traditional scheduling methods such as Shortest Job First (SJF) assume fixed job sizes and known processing times, making them unsuitable when memory demands grow unpredictably (Tay et al. 2022, Kang et al. 2024, Hooper et al. 2025).

When the total KV cache exceeds GPU memory capacity, the system must evict some in-progress prompts. Two approaches exist: (i) *swapping* to CPU or SSD, which incurs I/O overhead (Sheng et al. 2023, Aminabadi et al. 2022); and (ii) *recomputation*, which discards the KV cache and

restarts the prompt from prefill (Kwon et al. 2023). We focus on recomputation because it is the default in modern production systems such as vLLM (vLLM Team 2025), and because our goal is to *minimize eviction* through scheduling rather than to optimize the eviction mechanism itself (Li et al. 2025a). Eviction creates a vicious cycle: it frees memory temporarily, but restarted prompts consume the memory again and can trigger further evictions. Operationally, this appears as failed or aborted generations under heavy load, including user-facing errors such as “Something went wrong.”

*Memory sufficiency alone does not guarantee stability.* The key to approaching the offered-load benchmark is *load balance*: arrivals, completions, and memory growth must remain aligned. Supporting such an operating point requires enough memory for the corresponding mix of active prompts, quantified later by the memory requirement  $M^*$ . Even when total capacity  $C$  exceeds this requirement, naive policies can push the stochastic system away from the corresponding composition of active prompts across prefill and decode stages. For instance, first-come-first-served (FCFS) can trigger cascading evictions that lead to a meaningful loss in effective throughput, measured as completed decode-token service net of evictions, even when capacity is theoretically sufficient for stability under the fluid equilibrium (Example 2). A slight imbalance causes one prompt type to overflow, triggering evictions; the freed memory then fills with another type, which itself overflows, creating a vicious cycle. Thus, a workload that lies inside the fluid stability region can still become unstable under poor scheduling.

Preventing eviction requires *approaching load balance through controlled admission*. The key scheduling decision is the composition of each inference iteration: which output-length classes and prefill/decode stages enter the next batch. This composition determines how much fixed overhead is shared in the current iteration, how much KV cache is carried forward, and whether future iterations approach the overflow boundary. Our algorithms use thresholds to regulate this composition. A prompt advances only when the corresponding threshold has enough room, keeping the in-service population close to the equilibrium composition across prefill and decode stages. In our experiments, operating near this composition *enlarges the empirically observed stable operating range*, because eviction cascades no longer destabilize workloads that are theoretically stable under the fluid equilibrium, and *reduces mean end-to-end latency most clearly near capacity and under overload*. In underloaded regimes, our algorithms and the baselines are often close; as load increases, baseline

latency grows sharply and completion rates saturate earlier, while our algorithms keep the system closer to the offered load. Section 6 documents these effects.

Recent system-level optimizations for LLM inference (Yu et al. 2022, Kwon et al. 2023, Agrawal et al. 2023, Pope et al. 2023, DeepSeek-AI 2024, Patel et al. 2024, Zhong et al. 2024) are primarily systems and implementation contributions. They motivate, but do not aim to characterize, the queueing structure created by stochastic arrivals, growing KV caches, and eviction-induced restarts. In practice, output length predictions are imprecise or costly (Fu et al. 2025), and algorithms designed for known output lengths can degenerate significantly when this assumption fails (see Proposition 5).

We develop a fluid approximation that formalizes this operating point as a fluid equilibrium, where prompt arrivals and completions balance, and characterizes the memory requirement  $M^*$  needed to support it. For a fixed capacity  $C$ , the condition  $M^* \leq C$  determines whether the arrival vector lies in the fluid stability region. This equilibrium guides the threshold design: when the stochastic system operates near the corresponding composition across prefill and decode stages, it converts the offered work into completed service while avoiding eviction. The analysis shows that memory is a resource to exploit rather than merely a constraint, because larger batches process more prompts per iteration and increase GPU utilization.

Achieving this equilibrium in the stochastic system is not automatic, because each admission decision changes both the current batch and the future KV-cache state. We therefore use the structure of the fluid equilibrium to construct threshold policies. With known output lengths, WAIT sets admission thresholds from the reference composition across prefill and decode stages: a prompt advances only when the corresponding prefill/decode-stage threshold is met, so admission restores the load-balanced allocation rather than creating downstream memory pressure. With unknown output lengths, Nested WAIT applies the same rule through nested decode segments: short prompts complete and release memory early, while prompts that remain active advance to later segments without output-length prediction. The threshold structure reduces the original memory-coupled control problem to lower-dimensional queueing dynamics while preserving the batching benefits needed to approach the offered-load benchmark.

### 1.1. Summary of Contributions

Our main contributions are as follows.

- *Memory-constrained scheduling model.* We develop a multi-stage online scheduling model with endogenous memory growth, where KV-cache memory grows during LLM inference and exceeding capacity triggers costly prompt evictions (Section 2). With fixed external arrivals, a stable policy completes work at the arrival rate, so the design goal is twofold: to keep the stochastic system inside the fluid stability region, and to operate near the equilibrium composition across prefill and decode stages supported by the memory requirement  $M^*$ . A fluid approximation from queueing theory lets us characterize both (Section 3).

- *Eviction-prevention scheduling algorithms.* We develop threshold-based algorithms that control memory growth and prevent eviction under the stated memory conditions. The WAIT algorithm (Section 4) handles known output lengths; the Nested WAIT algorithm (Section 5) extends to unknown output lengths through on-the-fly output-length classification. Endogenous memory growth makes the primitive state high-dimensional, because the scheduler must track resident prompts across output-length classes and prefill/decode stages. The threshold construction reduces this state to threshold and boundary queues that capture the memory-coupled dynamics. Under the linear multi-stage model with endogenous memory growth and the stated asymptotic scaling, the algorithms attain the fluid effective-throughput benchmark with the corresponding latency and TTFT guarantees.

- *Experimental validation.* We evaluate our algorithms on synthetic and real-world workloads using Vidur (Agrawal et al. 2024a), a widely used LLM-serving simulator, configured for Llama-2-7B on a single A100 GPU (Section 6). We compare against vLLM (Kwon et al. 2023) and Sarathi (Agrawal et al. 2023), two widely used open-source greedy FCFS-based scheduling baselines that represent, respectively, memory-reactive eviction and recomputation, and chunked-prefill scheduling. We also report real-GPU validations on A100 hardware, including direct timing measurements and end-to-end GPU experiments, providing implementation evidence as well as support for the simulator's accuracy (Section 2 and Appendix H.2). Consistent with this load-balancing mechanism, WAIT and Nested WAIT expand the empirically observed stable operating range compared with the tested baselines and reduce mean end-to-end latency most visibly near capacity and under overload.

Our batching theory also extends naturally to Prefill-Decode Disaggregated (PD) systems such as DistServe (Zhong et al. 2024) and Splitwise (Patel et al. 2024), in which the decode server handles decode-only workloads. In this setting, the linear iteration-time model in Section 2 is a

close approximation after calibration, because prefill attention effects are separated from decode service. The same threshold construction applies on the decode side, and Appendix H.4 reports numerical results on representative PD workloads.

## 1.2. Other Related Work

*Online Scheduling Problem and Queueing System.* Classical online scheduling problems focus on optimally assigning jobs that arrive sequentially, each with varying completion times. In settings with stochastic arrivals, foundational studies such as [Devanur and Hayes \(2009\)](#), [Vee et al. \(2010\)](#), [Cole and Roughgarden \(2014\)](#), [Balkanski et al. \(2016\)](#), [Lattanzi et al. \(2020\)](#) have developed algorithms that learn arrival patterns or distributions to refine scheduling strategies over time. Beyond individual job processing, works like [Im and Moseley \(2013\)](#), [Lucier et al. \(2013\)](#), [Liu and Lu \(2015\)](#), [Li et al. \(2020\)](#) have explored simultaneous batching and scheduling, grouping similar jobs to enhance efficiency. Within queueing theory, fluid models serve as first-order approximations of stochastic systems, enabling near-optimal control strategies [Mandelbaum et al. \(1998\)](#), [Maglaras \(2000\)](#), [B  uerle \(2002\)](#), [Liu and Whitt \(2011\)](#), while asymptotic analysis characterizes long-horizon behavior. LLM inference introduces a different control structure. As decode progresses, each prompt increases its KV-cache requirement, so the scheduler must track not only how many jobs wait, but also where admitted prompts sit in the decode pipeline and how their memory requirements will evolve. Capacity pressure can then trigger eviction and recomputation. These mechanisms enlarge both the state space and the action space relative to classical queueing-control formulations with exogenous service requirements, because the system may restart work due to endogenous memory growth. Stability therefore depends on how the scheduling rule controls memory growth before overflow occurs.

*LLM Inference.* A growing literature studies scheduling and systems design for Large Language Model (LLM) inference. Unlike classical scheduling, LLM inference combines stochastic arrivals, dynamic memory constraints, and multi-phase prefill/decode processing. System-level works such as Splitwise ([Patel et al. 2024](#)) and DistServe ([Zhong et al. 2024](#)) split inference into separate prefill and decode pipelines, while scheduling engines such as Orca ([Yu et al. 2022](#)) and Sarathi ([Agrawal et al. 2023, 2024b](#)) improve throughput through batching and admission control. Other system-level optimizations include multi-head latent attention with low-rank KV compression ([DeepSeek-AI 2024](#)) and Kendall's-Tau prioritization by predicted output length ([Fu et al. 2025](#)). These works advance system-level efficiency through implementation and engineering. We complement them



by analyzing the queueing-control problem created by endogenous memory growth: as admitted prompts decode, they increase the resident KV-cache load, and overflow can trigger eviction and restart. This mechanism links the fluid stability region and throughput loss to how the scheduler regulates the GPU-resident workload across prefill and decode stages. From an operations research perspective, several recent works analyze LLM inference scheduling under online-algorithm frameworks, focusing on competitive ratio and regret bounds. [Jaillet et al. \(2025\)](#) develop an online scheduling algorithm with near-optimal regret guarantees for settings with known output lengths. [Wang et al. \(2025\)](#) study optimization with variable prefill and decode lengths in pipeline-parallel settings. [Chen et al. \(2025\)](#) address robust optimization under prediction uncertainty, considering adversarial scenarios. Complementary theory-and-practice work by [Bari et al. \(2025\)](#) studies scheduling algorithms for LLM inference and discusses timing regimes induced by prefill, decode, and model-computation effects. A different theoretical angle is taken by [Li et al. \(2025b\)](#), whose stochastic processing model allows general batch processing times, including a piecewise-linear form that captures the compute-bound / memory-bound transition, and shows that work-conserving algorithms achieve the fluid-optimal throughput. On output length uncertainty, [Jaillet et al. \(2025\)](#) and [Wang et al. \(2025\)](#) assume lengths are known at arrival, [Chen et al. \(2025\)](#) considers adversarial unknown lengths, while our Nested WAIT algorithm handles stochastic unknown lengths through on-the-fly classification at segment boundaries. These works establish competitive-ratio, regret, or throughput-optimality guarantees under their respective timing and information models. Our work focuses on a complementary queueing mechanism: the scheduler must control the composition of the GPU-resident workload across prefill and decode stages when iteration time is memory dependent and overflow causes eviction and restart. The resulting fluid analysis links the stability region to the iteration-time model in Equation (1) and explains how threshold-based admission moves the realized stability region closer to that fluid prediction.

### 1.3. Notations

For integer  $n \geq 1$ , we denote  $[n] = \{1, 2, \dots, n\}$  as the set of integers from 1 to  $n$ . For  $x \in \mathbb{R}$ , denote  $\lceil x \rceil$  as the smallest integer not smaller than  $x$  and  $\lfloor x \rfloor$  as the largest integer not greater than  $x$ . Denote  $x_+ = \max\{x, 0\}$ . For set  $S$ , denote  $|S|$  as its cardinality. For two functions  $f(T)$  and  $g(T)$ , we use  $f(T) = O(g(T))$  if there exists constant  $c_1 > 0$  such that  $f(T) \leq c_1 g(T)$  as  $T \rightarrow +\infty$  and  $f(T) = \Omega(g(T))$  if there exists constant  $c_2 > 0$  such that  $f(T) \geq c_2 g(T)$  as  $T \rightarrow +\infty$ .



## 2. Model

We model Large Language Model (LLM) inference as an online scheduling system with stochastic arrivals, staged service, and a shared memory constraint. The key departure from classical scheduling is endogenous memory growth: a job's memory requirement is not fixed at admission, because as the model generates output tokens, the prompt's *Key-Value (KV) cache* grows over time. The notation is defined where it first appears, with a consolidated summary in Table 3 of Appendix B.

### 2.1. Prompts and Inference Process

Prompts arrive to a single Graphics Processing Unit (GPU) over time and require two phases of service. The *prefill* phase embeds the input context and initializes the prompt's KV cache. The *decode* phase then generates one output token per iteration, increasing the prompt's KV cache by one unit at each decode step. We focus on a single GPU as the basic scheduling unit; multi-GPU deployments can apply the same admission and batching logic within each GPU partition or pipeline stage.

We group prompts into  $m$  types indexed by  $j \in \{1, \dots, m\}$ . Type- $j$  prompts arrive according to a Poisson process with rate  $\lambda_j$ , have input length  $l_j$  after prefill, and require  $l'_j$  decode iterations before completion. Thus, a prompt of type  $j$  has the following length profile:

$$\begin{cases} \text{a length of } l_j \text{ tokens after prefill;} \\ \text{a length of } l_j + l'_j \text{ tokens after decode.} \end{cases}$$

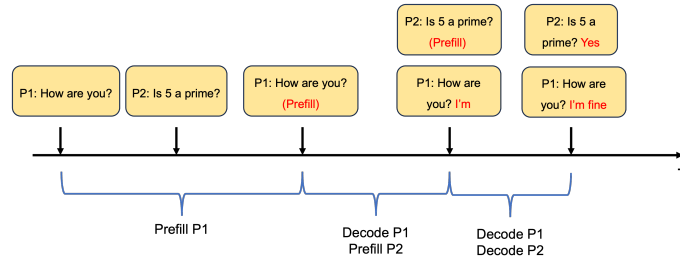
Therefore, a type- $j$  prompt undergoes one prefill iteration followed by  $l'_j$  decode iterations. To track service progress, we use  $s = 0$  for prompts awaiting prefill and  $s \in \{1, \dots, l'_j\}$  for prompts in decode. A prompt waiting outside the GPU at  $s = 0$  has no resident KV cache; processing its prefill allocates  $l_j$  units, and each generated decode token adds one unit. Thus a resident type- $j$  prompt with decode progress  $s$  occupies  $l_j + s$  units of KV cache memory. This memory requirement depends on whether the prompt is awaiting prefill or has advanced into decode, which is the source of the dynamic capacity constraint analyzed below.

Though the total number of different types  $m$  can be large (for example, from 1 to 10000), our algorithms and their theoretical guarantees exhibit only a weak (logarithmic) dependence on  $m$ . The analysis relies primarily on the total arrival rate across all types, ensuring that our approach scales efficiently regardless of the number of types.

While our main analysis focuses on fixed arrival rates  $\lambda_j$ , Appendix A discusses a finite-horizon extension in which thresholds are adjusted to bounded time-varying arrival rates.

## 2.2. Batching and Iteration Time

At each iteration, the scheduler forms a *batch* of prompts to process on the GPU. A batch may include newly admitted prompts in prefill and previously admitted prompts in decode, so the scheduling decision determines both how many prompts enter service and how many in-service prompts advance by one token. Because each iteration incurs a fixed launch and synchronization overhead, larger batches generally improve token throughput by spreading this overhead across more prompts. Figure 2 illustrates this timing with two sequential arrivals. Prompt  $P_1$  arrives first and is processed in prefill, which initializes its KV cache. Prompt  $P_2$  arrives later and also requires prefill. The displayed processing iterations are  $B^1 = \{(P_1, s = 0)\}$ ,  $B^2 = \{(P_1, s = 1), (P_2, s = 0)\}$ , and  $B^3 = \{(P_1, s = 2), (P_2, s = 1)\}$ , so the corresponding sums in Equation (1) are  $l_{j(P_1)}$ ,  $(l_{j(P_1)} + 1) + l_{j(P_2)}$ , and  $(l_{j(P_1)} + 2) + (l_{j(P_2)} + 1)$ , respectively. This example highlights the operational trade-off that drives the model: batching improves throughput by sharing fixed overhead across prompts, but it also increases the KV cache load carried by the system.



**Figure 2** Example of batching and scheduling with two prompts. The three GPU iterations shown are a prefill-only batch for  $P_1$ , a mixed batch that decodes  $P_1$  while prefilling  $P_2$ , and a decode-only batch for both prompts.

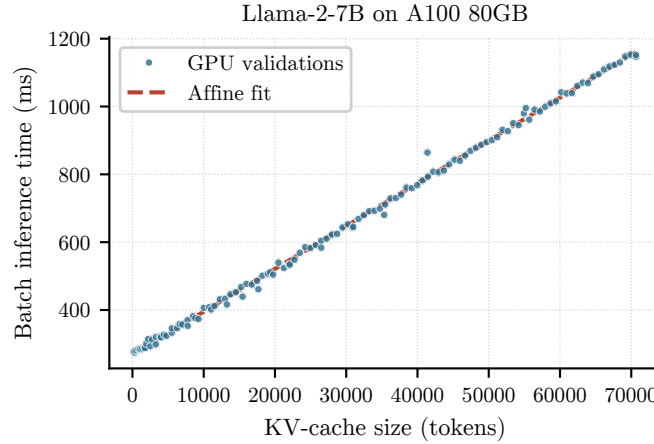
We next specify the service time of an iteration. In Transformer inference, prefill evaluates the input sequence and creates the initial KV cache; thereafter, each decode iteration produces one token per active prompt while attending to the prompt's cached history. For a decode-dominated batch, the batch KV-cache size is therefore a natural workload descriptor. Let  $B^t$  denote the batch processed at iteration  $t$ , let  $j(i)$  be the type of prompt  $i$ , and let  $s_i^t$  be its current prefill/decode stage. A prefill prompt has  $s_i^t = 0$  and initializes  $l_{j(i)}$  units of KV cache, while a decode prompt at stage  $s_i^t$  reads and extends a cache of size  $l_{j(i)} + s_i^t$ . Guided by profiling studies of LLM serving

workloads (Agrawal et al. 2023, Kwon et al. 2023, Zhong et al. 2024), we approximate iteration time by

$$\tau(B^t) = d_0 + d_1 \sum_{i \in B^t} (l_{j(i)} + s_i^t), \quad (1)$$

where  $d_0$  is the fixed per-iteration overhead and  $d_1$  is the marginal time cost per cached token. The linear form separates this fixed overhead from the marginal cost of serving larger KV caches. It captures the admission-control trade-off central to our model: admitting more prompts spreads  $d_0$  across more generated tokens and can therefore raise throughput, but it also lengthens subsequent iterations and consumes scarce GPU memory through larger KV caches. This regime is especially relevant because prompts typically spend most of their lifetime in decode, so admission decisions primarily determine the population and age distribution of active caches. Figure 3 reports A100 80GB validation results for Llama-2-7B with the prompt setting fixed at input length 256 and output length 20. We vary the number of requests in the batch from 1 to 256 and record batch inference time across batch states in which the active requests may be at different decode stages. The horizontal axis is the total number of tokens stored in the KV cache across active requests, corresponding to  $\sum_{i \in B^t} (l_{j(i)} + s_i^t)$  for a serving iteration  $t$ . Because requests may occupy different prefill/decode stages in the same iteration, batches with similar numbers of requests can have different total cached-token counts, producing the scattered validation points around the linear fit. Appendix H.2 reports additional simulator-validation and end-to-end GPU experiments for the same hardware setting.

The same formulation can accommodate other calibrated service-time curves. When prefill dominates, especially when prefill lengths are highly heterogeneous, input attention can add non-linear length effects; mixed prefill-decode batches and hardware saturation can create distinct compute-bound and memory-bound regimes. These effects can be represented by replacing (1) with a monotone function of batch composition, including piecewise-linear specifications such as  $\tau(b') = c + a \max\{0, b' - b_0\}$  (Li et al. 2025b). Prefill-decode disaggregated serving gives the complementary case: prefill work is routed to separate workers, and the decode server repeatedly executes one-token decode steps, making aggregate KV-cache usage an especially accurate state variable for iteration time (Patel et al. 2024, Zhong et al. 2024). Thus the linear model is both a close approximation in the KV-cache-driven regimes we study and a transparent first-order primitive for the fluid analysis. It keeps the central operational trade-off explicit: larger batches improve utilization, but they also increase iteration time and memory pressure. Even when a richer calibrated



**Figure 3** Batch inference time of Llama-2-7B on a single NVIDIA A100 80GB GPU. We fix the prompt length at 256 tokens and the output length at 20 tokens, and vary the number of requests in the batch from 1 to 256. Each validation point corresponds to a batch state; requests in the same batch may be at different decode stages, so the horizontal axis reports the resulting total number of tokens stored in the KV cache across active requests. The vertical axis reports measured batch inference time. The linear fit indicates an approximately linear relationship between batch inference time and total cached tokens.

service-time model is needed, the same perspective remains useful: the scheduler must control the composition of resident prompts across prefill and decode stages because this composition determines both memory pressure and future service times.

### 2.3. Memory Constraint and Challenges

The service-time model above describes how a batch uses the GPU in one iteration. The second constraint is intertemporal: a prompt that remains in service carries its KV cache across iterations, and this cache grows as decode progresses. Hence the scheduling state must track not only the current batch  $B^t$ , but all prompts whose KV caches are currently resident on the GPU.

**Memory Constraint:** At all times  $t$ , the total KV cache stored on the GPU must not exceed capacity  $C$ :

$$\sum_{i \in \mathcal{G}^t} (l_{j(i)} + s_i^t) \leq C, \quad (2)$$

where  $\mathcal{G}^t$  is the set of prompts whose KV caches are resident on the GPU at time  $t$ . This set includes the current batch  $B^t \subseteq \mathcal{G}^t$  as well as prompts that are waiting between iterations while retaining their KV caches on the GPU. Here  $j(i)$  is the type of prompt  $i$ , and  $s_i^t \in \{0, 1, \dots, l'_{j(i)}\}$  is its current decode progress. Prompts that have not yet been prefilled are not in  $\mathcal{G}^t$ ; once their

KV cache is allocated, their memory contribution grows from  $l_{j(i)}$  with each generated token. If the constraint (2) cannot be satisfied, the server must *evict* resident prompts by discarding their KV caches. We model eviction as last-in-first-out: the most recently admitted prompts are evicted first until enough memory is freed. An evicted prompt loses its accumulated computation and restarts from prefill, re-entering the scheduling queue as a new job. This restart rule can create an eviction cascade: each eviction relieves memory pressure temporarily, but the restarted prompts return to the queue, raise future admission pressure, and can force additional evictions. The following example illustrates this mechanism.

**EXAMPLE 1 (DYNAMIC MEMORY GROWTH AND EVICTION).** Consider the running toy prompt “Hello,” which receives the one-token response “Hi.” This prompt type has prefill length  $l = 1$  and decode length  $l' = 1$ . A prompt waiting for prefill uses no KV cache. Once its prefill is processed, the prompt becomes resident and uses one unit of KV cache; after it generates its single decode token, it uses  $l + 1 = 2$  units just before completion. Suppose the GPU has capacity  $C = 12$ . One full-memory iteration can process four prefills and four decodes: the four newly prefilled prompts require  $4 \times 1 = 4$  units, and the four prompts already in decode require  $4 \times 2 = 8$  units, for a total of 12 units and four completions.

Now suppose three prompts are already decoding, using  $3 \times 2 = 6$  units, and the scheduler admits six new prompts for prefill, using another  $6 \times 1 = 6$  units during the iteration. After the iteration, the three old prompts complete and release their caches, while the six newly admitted prompts move into decode and now require  $6 \times 2 = 12$  units, filling memory. If four more prompts arrive next, admitting their prefills requires four additional units. The only way to create this space is to evict two resident decode prompts, each freeing two units. These evicted prompts lose their KV caches and return to the queue as new jobs, where they compete with subsequent arrivals and can force further evictions.

This example highlights the **eviction cascade** and the admission trade-off behind it. A larger batch can raise throughput because the fixed per-iteration overhead is shared across more generated tokens, but every admitted prompt also creates a KV cache that stays on the GPU and grows one token at a time until completion. If the scheduler admits too many prompts now, later decode iterations may leave too little memory for new arrivals, forcing eviction and restart. When output lengths are unknown, this decision must be made before the scheduler knows how long each prompt will remain in memory. Section 3 formalizes the balanced admission level at which arrivals, completions, and memory growth can be kept in balance.

## 2.4. Objective and Policy Space

We evaluate a scheduling policy over a finite horizon  $[0, T]$ . The primary efficiency metric, denoted  $\text{Throughput}^{(T, \pi)}$ , is *effective throughput*: the average number of decode tokens from prompts that complete under policy  $\pi$ . Tokens generated before a prompt is evicted do not count, because eviction discards the prompt's KV cache and returns it to prefill. This convention makes memory overflow part of the performance criterion rather than a separate implementation detail. For a fixed arrival process, effective throughput is capped by the offered decode-token load. A policy falls below this cap when it leaves capacity idle or admits work that later restarts instead of completing. Admission control must therefore balance utilization against memory feasibility: admitting enough prompts to exploit the GPU, but not so many that endogenous KV-cache growth forces eviction. Maintaining this balance over a wider range of arrival rates expands the realized stability region, and under overload it keeps completed work close to the server's usable capacity. We also evaluate  $\text{Latency}^{(T, \pi)}$ , the average elapsed time from prompt arrival to completion, and  $\text{TTFT}^{(T, \pi)}$ , the average elapsed time from arrival to the first generated output token. These delay metrics capture service quality beyond aggregate output. In particular, TTFT rules out policies that keep completion counts high by repeatedly prioritizing short prompts while delaying the first service of longer prompts. Later sections introduce the fluid model and use it to establish throughput, latency, and TTFT guarantees for the proposed policies.

Let  $\Pi$  denote the class of admissible online policies. At each decision epoch  $t$ , a policy  $\pi \in \Pi$  observes the prompts that have arrived, their input lengths, their current prefill/decode stages  $s_t^i$ , and whether their KV caches are resident on the GPU. The policy is non-anticipating: it cannot use future arrivals or a prompt's output length before that length is revealed by completion.

Given this information, the policy chooses which waiting prompts to admit, which resident prompts to batch in the current iteration, and whether to preempt or evict prompts. Preemption here means scheduler-side pausing: the prompt remains GPU-resident and keeps its KV cache, so it can resume without recomputation. We do not model CPU or SSD swap-out with a separate swap-in latency. Eviction removes the KV cache to free memory; under the LIFO rule used here, the most recently admitted resident prompts are evicted first, and each evicted prompt returns to stage  $s = 0$  and must restart from prefill. All decisions must satisfy the memory constraint (2) after the resulting state transition.

### 3. Fluid Model and Equilibrium

Section 2 identifies the central admission-control trade-off created by endogenous memory growth: the scheduler should admit enough prompts to exploit batching, but not so many that their KV caches later overflow memory. This section derives the balanced operating point for that trade-off. The key state variable is the composition of GPU-resident prompts across prefill and decode stages. As prompts move deeper into decode, each prompt carries a larger KV cache; if too many prompts accumulate at late stages, the system can run out of memory and restart work through eviction.

We develop a fluid model that replaces stochastic sample paths with average flows and characterizes a fluid equilibrium in which arrivals, prefill/decode transitions, and completions occur at matching rates. For a given arrival vector, this equilibrium gives two reference quantities: the memory requirement  $M^*$  needed to support the equilibrium and the effective throughput **Throughput\*** achieved when the offered load can be completed without eviction. Equivalently, for a fixed memory capacity  $C$ , the fluid stability region consists of arrival vectors whose equilibrium memory requirement satisfies  $M^* \leq C$ . We derive these quantities first for a single prompt type, which isolates the accumulation of KV cache over prefill and decode, and then for multiple prompt types sharing one memory constraint. The same equilibrium also supplies the design target for WAIT in Section 4: keep the batch composition close to the implied composition across prefill and decode stages, thereby avoiding eviction-induced restarts while approaching the largest throughput attainable under the offered load and memory constraint.

#### 3.1. Single-Type Fluid Model

Consider a single prompt type  $j$ . A resident prompt uses  $l_j$  units of KV cache immediately after prefill and  $l_j + s$  units at decode stage  $s = 1, \dots, l'_j$ ; after decode stage  $l'_j$ , the prompt completes and releases its cache. Let  $n_j^*$  denote the equilibrium number of type- $j$  prompts at each prefill/decode stage. The balanced state therefore has  $n_j^*$  prompts in the prefill stage and in each decode stage.

Before deriving the equilibrium, we identify the load condition under which batching can be effective. Adding prompts to a batch is useful only if the additional completions outweigh the longer iteration time. Across its prefill and decode stages, a type- $j$  prompt contributes memory-dependent work proportional to  $(l'_j + 1)(l_j + l'_j/2)$ . If arrivals are high enough that this memory-dependent work alone saturates the server, batching cannot restore stability.



**PROPOSITION 1.** *When the condition  $\lambda_j d_1 (l'_j + 1) \left(l_j + \frac{l'_j}{2}\right) \geq 1$  is satisfied, the system is unstable in the sense that the expected average latency under any scheduling policy with arrival rate  $\lambda_j$  grows linearly with time. Formally,*

$$\mathbb{E}[\text{Latency}^{(T, \pi)}] = \Omega(T) \quad \text{as } T \rightarrow \infty.$$

The proof is a service-capacity cut: each completed prompt must consume  $d_1 \sum_{s=0}^{l'_j} (l_j + s)$  units of memory-dependent processing time over its lifetime, and the fixed positive overhead  $d_0$  leaves no slack at the boundary under finite memory capacity (Appendix C.1). We therefore focus on the nontrivial regime  $\lambda_j d_1 (l'_j + 1) \left(l_j + \frac{l'_j}{2}\right) < 1$ , where batching can support a fluid equilibrium.

At equilibrium, the  $l'_j + 1$  stages each contain  $n_j^*$  prompts, so the memory usage is

$$M_j^* = n_j^* \sum_{s=0}^{l'_j} (l_j + s) = n_j^* (l'_j + 1) \left(l_j + \frac{l'_j}{2}\right), \quad (3)$$

where  $l_j + l'_j/2$  is the average KV-cache usage of a type- $j$  prompt across its prefill and decode stages.

Flow balance requires arrivals during one iteration to match completions. Since one iteration takes time  $d_0 + d_1 M_j^*$ ,  $\lambda_j (d_0 + d_1 M_j^*)$  prompts arrive and  $n_j^*$  prompts complete, giving

$$\underbrace{(d_0 + d_1 M_j^*)}_{\text{Time per iteration}} \lambda_j = \left( d_0 + d_1 n_j^* (l'_j + 1) \left(l_j + \frac{l'_j}{2}\right) \right) \lambda_j = n_j^*. \quad (4)$$

Solving gives  $n_j^* = d_0 \lambda_j / [1 - d_1 \lambda_j (l'_j + 1) (l_j + l'_j/2)]$ . Substituting this value into the memory expression yields

$$M_j^* = n_j^* (l'_j + 1) \left(l_j + \frac{l'_j}{2}\right) = \frac{d_0 \lambda_j (l'_j + 1) \left(l_j + \frac{l'_j}{2}\right)}{1 - d_1 \lambda_j (l'_j + 1) (l_j + \frac{l'_j}{2})}. \quad (5)$$

The denominator is the slack in the batching feasibility condition; as it approaches zero, the inventory required for balance diverges. Equation (5) also shows why long responses are especially demanding. For fixed input length and arrival rate, away from the critical denominator, the memory needed to support the fluid equilibrium grows on the order of  $(l'_j)^2$ , because the number of resident

prefill/decode stages and the average KV-cache size across those stages both increase with decode length. As the denominator approaches zero, this requirement diverges. Thus the memory condition defining the fluid stability region is highly sensitive to long-decode traffic, making control of the composition across prefill and decode stages central to the scheduling problem. The balanced throughput is completed decode tokens per iteration divided by iteration time:

$$\text{Throughput}_j^* = \frac{n_j^* \cdot l'_j}{d_0 + d_1 n_j^* (l'_j + 1) \left( l_j + \frac{l'_j}{2} \right)} = \lambda_j l'_j, \quad (6)$$

where the equality follows from the flow-balance condition (4).

### 3.2. Multiple-Type Fluid Model

We now extend the balance calculation to  $m$  prompt types. Type  $j$  has input length  $l_j$ , decode length  $l'_j$ , and arrival rate  $\lambda_j$ . Because all types draw from the same memory pool, the equilibrium is determined by the joint allocation of active prompts across prefill and decode stages and the aggregate KV-cache usage it induces.

The balance equations are written at the scheduler's iteration scale: an iteration selects a batch, advances active prompts by one prefill/decode stage, and updates their KV-cache usage. Let  $n_j^*$  denote the equilibrium number of type- $j$  prompts at each prefill/decode stage. The resulting memory usage is

$$M^* = \sum_{j=1}^m n_j^* (l'_j + 1) \left( l_j + \frac{l'_j}{2} \right) \quad (7)$$

Flow balance requires  $(d_0 + d_1 M^*) \lambda_j = n_j^*$  for each type  $j$ , because  $n_j^*$  type- $j$  prompts complete in one iteration. Substituting these identities into (7) gives

$$M^* = \frac{d_0 \sum_{j=1}^m \lambda_j (l'_j + 1) \left( l_j + \frac{l'_j}{2} \right)}{1 - d_1 \sum_{j=1}^m \lambda_j (l'_j + 1) \left( l_j + \frac{l'_j}{2} \right)} \quad (8)$$

The aggregate term in (8) inherits the same quadratic dependence on decode length. A small amount of long-decode traffic can therefore account for a disproportionate share of the memory requirement needed for stability, which is precisely the pressure that the threshold policies regulate. The corresponding iteration time is

$$\begin{aligned} \Delta T^* &= d_0 + d_1 M^* \\ &= \frac{d_0}{1 - d_1 \sum_{j=1}^m \lambda_j (l'_j + 1) \left( l_j + \frac{l'_j}{2} \right)}, \end{aligned} \quad (9)$$

and the fluid throughput benchmark is

$$\mathbf{Throughput}^* = \sum_{j=1}^m \frac{n_j^* \cdot l'_j}{d_0 + d_1 M^*} = \sum_{j=1}^m \lambda_j l'_j \quad (10)$$

When the system reaches the fluid equilibrium, the equality  $\mathbf{Throughput}^* = \sum_j \lambda_j l'_j$  is the throughput delivered by its balanced batch composition. The contribution of the fluid calculation is not this accounting identity alone, but the accompanying description of the composition needed to realize it: how many prompts of each type should be resident at each prefill and decode stage, and how much memory is needed to sustain that composition. Useful completed decode work cannot exceed the decode work brought by exogenous arrivals. We state the benchmark result in the regime  $M^* \leq C$ , where the corresponding fluid equilibrium fits in memory and the offered-load value is the relevant stabilizable target. If  $M^* > C$ , the arrival vector is outside this fluid stability region, and the system is overloaded relative to the full offered-load benchmark.

**PROPOSITION 2.** *Suppose that the memory capacity satisfies  $C \geq M^*$ , where  $M^*$  denotes the memory requirement needed to support the fluid equilibrium in (8). Then, for any online policy  $\pi \in \Pi$ , the expected throughput satisfies*

$$\mathbb{E}[\mathbf{Throughput}^{(T,\pi)}] \leq \mathbf{Throughput}^*,$$

where  $\mathbf{Throughput}^*$  is given in (10).

The scheduling question is how closely an online policy can approach this benchmark while respecting the memory dynamics when  $M^* \leq C$ . This focus on a stabilizable fluid operating point, and on behavior near the boundary of that region, follows the standard fluid-control viewpoint for queueing systems (Maglaras 2000, Bäuerle 2002, Liu and Whitt 2011). In standard queueing models with exogenous service requirements, the workload created by an admitted job is fixed. However, in this LLM inference system, current admission and batching decisions determine future KV-cache memory usage, because prompts that advance in decode remain resident and carry larger caches. Over-admission shifts future batches toward memory-heavy late-decode prompts and can force eviction before arrivals become completed service. Overly conservative admission leaves the server short of completions. The fluid equilibrium therefore identifies the composition a policy should maintain: roughly  $n_j^*$  type- $j$  prompts at each prefill and decode stage, with total memory requirement  $M^*$ . Section 4 uses  $M^*$  and  $n_j^*$  to define thresholds that keep the batch composition near this equilibrium.

## 4. Scheduling with Known Output Lengths: The WAIT Algorithm

This section develops the *Waiting for Accumulated Inference Threshold* (WAIT) algorithm for the case in which output lengths and arrival rates are known when thresholds are set. WAIT uses the fluid equilibrium from Section 3 as a batching target: for each output-length class  $j$ , it waits until enough prompts have accumulated before admitting a batch, so that the batch composition remains close to the equilibrium levels  $n_j^*$  across prefill and decode stages. This threshold structure is the key dimensionality reduction in the known-output-length case: instead of tracking every resident prompt's future KV-cache growth, the scheduler controls threshold-sized flows across prefill and decode stages. Under the corresponding memory feasibility condition, this rule prevents eviction cascades while allowing the server to approach the fluid throughput benchmark. The throughput guarantees below assume  $C \geq M^*$ , where  $M^*$  is the memory requirement needed to support that fluid equilibrium in (8); this is the stable or near-boundary regime in which the load-balanced equilibrium allocation across prefill and decode stages fits in memory. Section 6 separately examines overloaded regimes, where the full offered-load benchmark is no longer feasible but the same load-balancing principle helps the scheduler use available capacity more effectively and reduce eviction-induced waste.

### 4.1. Motivation: Why Naive Scheduling Fails

The condition  $C \geq M^*$  is a capacity condition, not a scheduling rule. It identifies when the fluid equilibrium can fit in memory, but it does not ensure that an online policy will keep work balanced across prefill and decode stages.

Consider a First-Come-First-Serve (FCFS) policy that admits prompts as soon as they arrive. Even when  $C = M^*$ , so the fluid equilibrium fits exactly in memory, FCFS can move too much work into downstream decode stages and create a persistent throughput loss. The following proposition gives a worst-case lower bound for this failure mode.

**PROPOSITION 3.** *When  $C = M^*$ , there exists an instance in which First-Come-First-Serve (FCFS) incurs a throughput gap  $\text{Throughput}^* - \mathbb{E}[\text{Throughput}^{(T, \pi)}] = \Omega(1)$  as  $T \rightarrow \infty$ .*

The proof is provided in Appendix C.4. The example below is a simpler single-type illustration of the same eviction cascade.

**EXAMPLE 2 (EVICTION CASCADE UNDER FCFS).** Consider a single prompt type: a prompt such as “Hello” with a one-token response “Hi,” so  $l = 1$  and  $l' = 1$ . Each prompt uses 1 unit of memory after prefill and 2 units after decode. Let  $C = 12$  and  $\lambda = 4$ . The fluid equilibrium keeps 4 prompts at the prefill stage and 4 prompts at the decode stage, uses  $4 \times 1 + 4 \times 2 = 12$  units of memory, and completes 4 prompts per iteration.

Now suppose the system starts slightly imbalanced with 6 prompts at stage 0 (awaiting prefill) and 3 prompts at stage 1 (in decode). Under FCFS:

1. **Iteration 1:** FCFS processes all 9 prompts. The 3 decode prompts complete, while the 6 prefill prompts advance to decode and require  $6 \times 2 = 12$  units of memory. The system is full after only 3 completions.
2. **Iteration 2:** About 4 new prompts arrive during the previous iteration. Admitting them for prefill requires 4 additional memory units, so the server evicts 2 decode prompts to free  $2 \times 2 = 4$  units. The system returns to 4 prefill prompts and 4 decode prompts, again using  $4 + 8 = 12$  units.
3. **Iteration 3:** The 4 decode prompts complete. The 2 evicted prompts have lost their KV caches and must restart from prefill, re-entering the queue alongside genuinely new arrivals. The restart mechanism perpetuates the imbalance.

The eviction-restart cycle yields approximately 3–3.5 completions per iteration instead of the 4 completions sustained by the fluid equilibrium. Thus  $C = M^* = 12$  is sufficient to support the fluid balance, but FCFS does not maintain that balance and loses effective throughput.

The operational implication is that having enough memory to support the fluid equilibrium does not by itself guarantee that the stochastic system will operate near that equilibrium. The scheduler must also regulate the composition of prompts across prefill and decode stages. FCFS admits prompts greedily, lets this composition drift away from the fluid balance, and thereby creates the eviction-restart cycle.

## 4.2. The WAIT Algorithm

The FCFS example points to the missing control variable: the composition of the in-service population across prefill and decode stages. WAIT regulates this composition by assigning each type  $j$  a threshold  $n_j$ , interpreted as the target number of type- $j$  prompts to serve from each prefill/decode stage in an iteration. If fewer than  $n_j$  new type- $j$  prompts are waiting at entry, WAIT

does not admit that type. Once the threshold is reached, the server processes up to  $n_j$  prompts from each prefill/decode stage of that type. In this way, new work enters the system only when it can replenish a balanced allocation across prefill and decode stages, rather than pushing the system toward the downstream buildup seen under FCFS.

The threshold rule also gives the analysis its tractable structure. Once the thresholds are fixed, each type evolves as a threshold queue observed at batch-completion epochs, while the shared memory interaction is captured by the feasibility of the threshold vector. The waiting requirement is therefore not merely a safety device against eviction; it also preserves the batching benefit created by the fixed overhead  $d_0$ . Batches form at a scale large enough to use the server effectively, while the thresholds keep memory usage within capacity.

*Algorithm Description.* For each prompt type  $j \in [m]$ , WAIT sets a threshold  $n_j$  derived from the fluid model. At decision epoch  $t$ , let  $n_{js}^t$  denote the number of type- $j$  prompts at stage  $s$ . Type  $j$  is eligible for the next batch only when

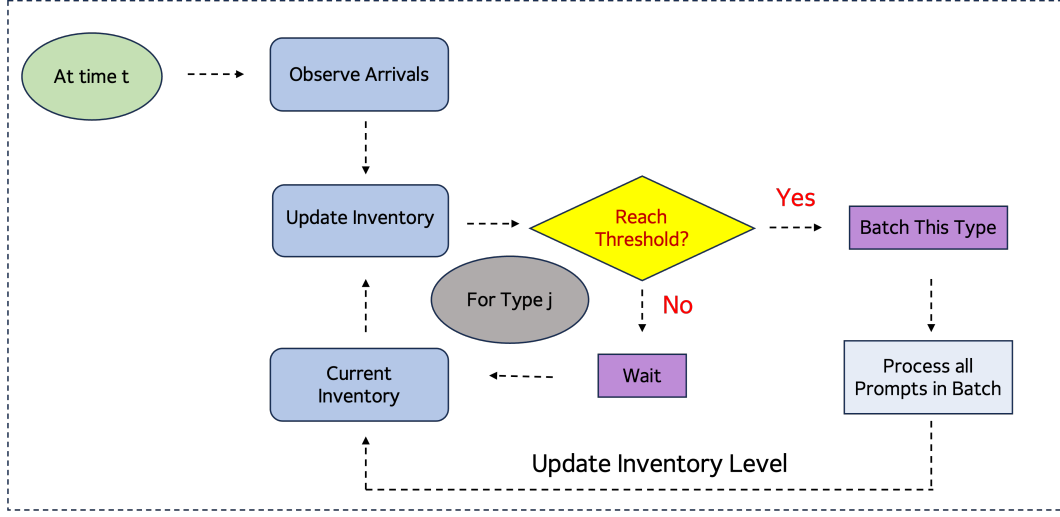
$$n_{j0}^t \geq n_j.$$

When this condition holds, WAIT selects  $\min\{n_j, n_{js}^t\}$  type- $j$  prompts from each prefill/decode stage  $s$ . If no type is eligible, the server waits for more arrivals. Prompts that are not selected remain in their current queues. A prompt waiting outside the GPU before prefill has no resident KV cache, but any prompt that has already been admitted and is waiting between decode iterations keeps its KV cache on the GPU and continues to count against memory. Algorithm 1 and Figure 4 describe the policy.

### 4.3. Asymptotic Guarantees

We analyze WAIT in an asymptotic scaling that averages out stochastic fluctuations while preserving the fixed memory constraint. For  $\zeta \geq 1$ , arrival rates scale as  $\lambda_j^{(\zeta)} = \zeta \lambda_j$  and processing times scale as  $(d_0^{(\zeta)}, d_1^{(\zeta)}) = (\zeta^{-1} d_0, \zeta^{-1} d_1)$ , with memory capacity  $C$  fixed. For WAIT, the required physical memory is  $M_{\text{req}}^{(\zeta, \pi)} = M^\pi$ , which is independent of  $\zeta$  because no downstream boundary buffer is needed. For a policy  $\pi$ , let  $N_j^{(\zeta, \pi)}$  be the number of completed decode tokens from type- $j$  prompts over this horizon, and define

$$\text{Throughput}^{(\zeta, \pi)} = \frac{1}{\zeta T} \sum_{j=1}^m N_j^{(\zeta, \pi)}.$$



**Figure 4** Batch formation under WAIT for a fixed type  $j$ . The scheduler updates the type- $j$  inventory and batches this type only when the prefill-entry queue reaches  $n_j$ ; otherwise, it waits and carries the inventory forward. When batched, up to  $n_j$  prompts from each prefill/decode stage are advanced.

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#### Algorithm 1 WAIT: Waiting for Accumulated Inference Threshold

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**Input:** Memory  $C$ , arrival rates  $\lambda_j$ , thresholds  $n_j$  satisfying (11),  $\forall j \in [m]$

- 1: Initialize prompt inventory  $n_{js} \leftarrow 0$  for all  $j \in [m], s \in \{0, 1, \dots, l'_j\}$
  - 2: Initialize event queue with arrival events for each type  $j$
  - 3: Set current time  $t \leftarrow 0$
  - 4: Set server status to idle
  - 5: **while** True **do**
  - 6:   Wait for the next event (arrival or batch completion)
  - 7:   **if** event is an arrival of type  $j$  **then**
  - 8:     Update inventory:  $n_{j0} \leftarrow n_{j0} + 1$  ▷ Add new prompt to waiting queue of prefill phase
  - 9:   **else if** event is a batch completion **then**
  - 10:     Retrieve the selected counts  $a_{js}$  stored with the completed batch
  - 11:     Simultaneously advance selected prompts one stage:  $n_{js} \leftarrow n_{js} - a_{js}$  and  $n_{j,s+1} \leftarrow n_{j,s+1} + a_{js}$  for  $s < l'_j$
  - 12:     Clear KV caches for the  $a_{j,l'_j}$  prompts completing after stage  $l'_j$  ▷ Free memory for completed prompts
  - 13:     Set server status to idle
  - 14:   **end if**
  - 15:   Set  $J(t) \leftarrow \{j \in [m] : n_{j0} \geq n_j\}$  ▷ Eligible types
  - 16:   **if** server is idle and  $J(t) \neq \emptyset$  **then**
  - 17:     For every  $j \in J(t)$  and stage  $s$ , set  $a_{js} \leftarrow \min\{n_j, n_{js}\}$
  - 18:     Form batch  $B$  from all selected prompts and store  $a_{js}$  with  $B$
  - 19:     Process batch  $B$ ; keep already resident prompts outside the batch waiting with their KV caches retained
  - 20:     Set server status to busy until the completion event for  $B$
  - 21:   **end if**
  - 22: **end while**
- 

For delay metrics, let  $\text{Latency}_{\text{cal}}^{(\zeta, \pi)}$  and  $\text{TTFT}_{\text{cal}}^{(\zeta, \pi)}$  denote the elapsed calendar-time averages over this scaled system. The theorem reports the service-normalized delays  $\text{Latency}^{(\zeta, \pi)} := \zeta \text{Latency}_{\text{cal}}^{(\zeta, \pi)}$  and  $\text{TTFT}^{(\zeta, \pi)} := \zeta \text{TTFT}_{\text{cal}}^{(\zeta, \pi)}$ , so the effective horizon in the bounds is  $\zeta T$ . The benchmark is the fluid throughput  $\text{Throughput}^* = \sum_{j=1}^m \lambda_j l'_j$  in (10).



Assume  $C \geq M^*$ , with  $M^*$  given by (8). Consider a WAIT policy  $\pi = [n_1, \dots, n_m]$ , where  $n_j$  is the type- $j$  threshold applied at each prefill/decode stage. The threshold vector must satisfy a load-balance condition and a memory condition:

$$\begin{aligned} \Delta T(n_{1:m}) &:= d_0 + d_1 M^\pi \leq \frac{n_j}{\lambda_j}, \forall j \in [m], \\ M^\pi &= \sum_{j=1}^m n_j (l'_j + 1) (l_j + \frac{l'_j}{2}), \\ M^* &\leq M_{\text{req}}^{(\zeta, \pi)} := M^\pi \leq C \end{aligned} \tag{11}$$

The first line is the arrival-completion balance for each type: during an iteration of length  $\Delta T(n_{1:m})$ , the expected number of type- $j$  arrivals is  $\lambda_j \Delta T(n_{1:m})$ , and WAIT serves  $n_j$  type- $j$  prompts from each active stage. The memory condition then asks the threshold vector to be large enough to support the fluid equilibrium, but not so large that the required memory  $M_{\text{req}}^{(\zeta, \pi)}$  exceeds the available physical capacity  $C$ . The condition  $M^* \leq C$  says that the workload is stabilizable in the fluid model; the theorem below shows that WAIT realizes this stability in the stochastic system by keeping memory growth under control.

**THEOREM 1.** *If the thresholds  $\pi = [n_1, \dots, n_m]$  satisfy (11), then*

$$\begin{aligned} \text{Throughput}^* - \mathbb{E} [\text{Throughput}^{(\zeta, \pi)}] &= O((\zeta T)^{-\frac{1}{2}}), \\ \mathbb{E} [\text{Latency}^{(\zeta, \pi)}], \mathbb{E} [\text{TTFT}^{(\zeta, \pi)}] &= O((\zeta T)^{\frac{1}{2}}). \end{aligned}$$

*Moreover, if  $\Delta T(n_{1:m}) < n_j / \lambda_j$  for all  $j \in [m]$ , then*

$$\begin{aligned} \text{Throughput}^* - \mathbb{E} [\text{Throughput}^{(\zeta, \pi)}] &= O(1/(\zeta T)), \\ \mathbb{E} [\text{Latency}^{(\zeta, \pi)}], \mathbb{E} [\text{TTFT}^{(\zeta, \pi)}] &= O(1). \end{aligned}$$

The theorem rests on the control structure created by WAIT. The difficult step is not to prove recurrence after a stable queue has been isolated. Instead, it is to design a batching rule that keeps endogenous KV-cache growth under control before memory overflows. Each admitted prompt creates a KV cache that grows throughout decode. If an unstructured batching rule admits too much work, these caches can exceed memory in later iterations, forcing eviction and restart. The restarted prompts then return as future work and can reduce effective throughput even when the fluid memory requirement fits within capacity. WAIT prevents this failure mode by fixing a target allocation across prefill and decode stages through  $n_{1:m}$ . Given this threshold vector, every full batch has memory

usage  $M^\pi$  and iteration length  $\Delta T(n_{1:m})$ , and the main stochastic component for type  $j$  is the time until its entry queue reaches  $n_j$ . The proof uses the threshold structure to define an auxiliary embedded full-threshold process observed at review epochs and then couples it to the event-driven WAIT sample path type by type: although realized WAIT batches may contain different subsets of eligible types, each type's threshold batch receives a corresponding service opportunity within one full-threshold review interval. This coupling turns the original memory-coupled controlled process into threshold queues tied together by the deterministic feasibility condition  $M^\pi \leq C$ . The complete proof is in Appendix D.

The square-root delay bound at the boundary is also unavoidable among policies that keep up with the fluid throughput benchmark. When the threshold queues have no slack, their fluctuations cannot be averaged away by any non-predictive online rule, even if the total memory equals the fluid memory requirement. The next proposition gives the matching lower bound for latency and TTFT.

**PROPOSITION 4.** *There exists an instance where  $C = M^*$ , and for any non-predictive online policy  $\pi$  whose normalized effective-throughput gap is  $o(1)$ , the expected average latency and TTFT satisfy*

$$\mathbb{E}[\text{Latency}^{(\zeta, \pi)}], \quad \mathbb{E}[\text{TTFT}^{(\zeta, \pi)}] = \Omega((\zeta T)^{1/2}).$$

The strict inequalities in Theorem 1 correspond to slack in these threshold queues. If  $\Delta T(n_{1:m}) < n_j/\lambda_j$ , the type- $j$  entry queue has negative drift, and the average latency and TTFT of type- $j$  prompts remain  $O(1)$ .

The policy in this section assumes that output lengths and arrival rates are known when the thresholds are set. In practice, these quantities can be estimated from predictors (Fu et al. 2025), but an online serving system may not know a request's decode length at admission. The next section relaxes this information requirement by developing a nested version of WAIT for unknown output lengths.

## 5. Scheduling with Unknown Output Lengths: The Nested WAIT Algorithm

The WAIT algorithm assumes that output lengths are known when thresholds are set. Nested WAIT relaxes this requirement by using output-length information only as it is revealed by the prompt's own decode trajectory. The *Nested Waiting for Accumulated Inference Threshold* (Nested

WAIT) algorithm organizes decoding into nested segments and controls how prompts advance across those segments. In this setting, endogenous memory growth is harder to control because the scheduler does not know, at admission, which resident prompts will continue into later decode stages. Nested WAIT reduces this unknown-output-length state to segment-boundary queues that track the surviving prompts most likely to create downstream memory pressure.

### 5.1. Algorithm Overview

We first return to the running example to show how unknown output lengths complicate policy design and memory control. The same admitted prompts may either finish at an early decode boundary or continue to later decode stages with larger KV caches, so neither an optimistic nor a conservative admission rule is satisfactory.

**EXAMPLE 3 (UNKNOWN OUTPUT LENGTHS).** Continuing Example 2, suppose output lengths are now *unknown* at arrival: each “Hello” prompt receives either “Hi” (1 token) or “Hi there!” (2 tokens). The scheduler faces a dilemma. If it optimistically treats all outputs as short, then capacity  $C = 12$  appears to allow 6 prompts, because each would use 2 units after one decode token. If some prompts instead generate two tokens and require 3 units each, the same admission decision can overflow memory and trigger eviction. A conservative rule that treats every prompt as long avoids this overflow, but admits only 4 prompts and wastes capacity whenever most prompts are short.

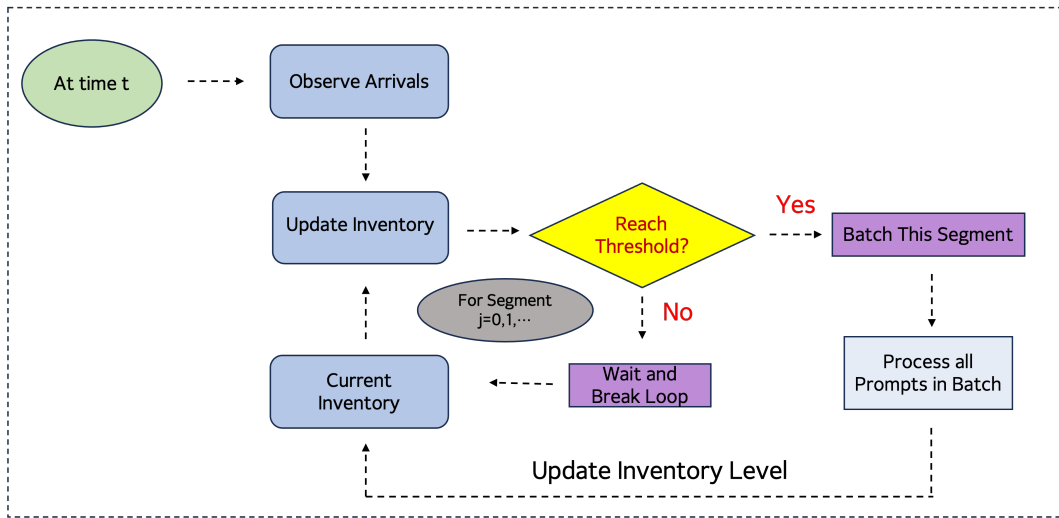
Nested WAIT avoids both extremes by pooling prompts in early decode stages and separating them only at completion boundaries. Segment 1 serves all prompts until the first possible output length  $l'_1$ ; prompts that complete there leave the system and release memory, while the remaining prompts enter Segment 2. The same logic repeats across later boundaries. The policy therefore does not reserve memory for long outputs before they are observed, but still lets short prompts complete without waiting behind longer ones.

Formally, let output lengths satisfy  $l'_1 < l'_2 < \dots < l'_m$ , where type  $j$  has decode length  $l'_j$ , and set  $l'_0 = 0$ . Nested WAIT defines  $m$  segments, with segment  $k$  covering the decode work between boundaries  $l'_{k-1}$  and  $l'_k$ . The segment receives types  $k, k+1, \dots, m$ , whose combined arrival rate is  $\lambda_k + \dots + \lambda_m$ . The threshold vector  $\pi = [n_1, \dots, n_m]$  has two roles:  $n_k$  caps the number of prompts processed per prefill/decode stage in segment  $k$ , and it is also the number of prompts that must be available at the segment’s entry boundary before that segment begins service.

The hidden output-length labels are i.i.d. across prompts according to the arrival mix, and Nested WAIT is nonanticipating: before a prompt reaches a segment boundary, the scheduler does not use whether that prompt will complete there or continue downstream.

Thus, if segment  $k^*$  has fewer than  $n_{k^*}$  prompts at its entry boundary, Nested WAIT pauses segment  $k^*$  and all later segments while retaining their KV caches on the GPU. This is a scheduling preemption, not a memory swap: resident prompts waiting at a downstream boundary or inside a segment continue to count against GPU memory. Only prompts still outside the GPU before prefill, namely the external entry queue of Segment 1, have no resident KV cache. Earlier segments  $k < k^*$  can continue batching up to  $n_k$  prompts per prefill/decode stage, so new arrivals can keep moving through the shared early segment. Once enough prompts reach the boundary of segment  $k^*$ , that segment resumes. Algorithm 2 formalizes this procedure, and Figure 5 illustrates the resulting pipeline.

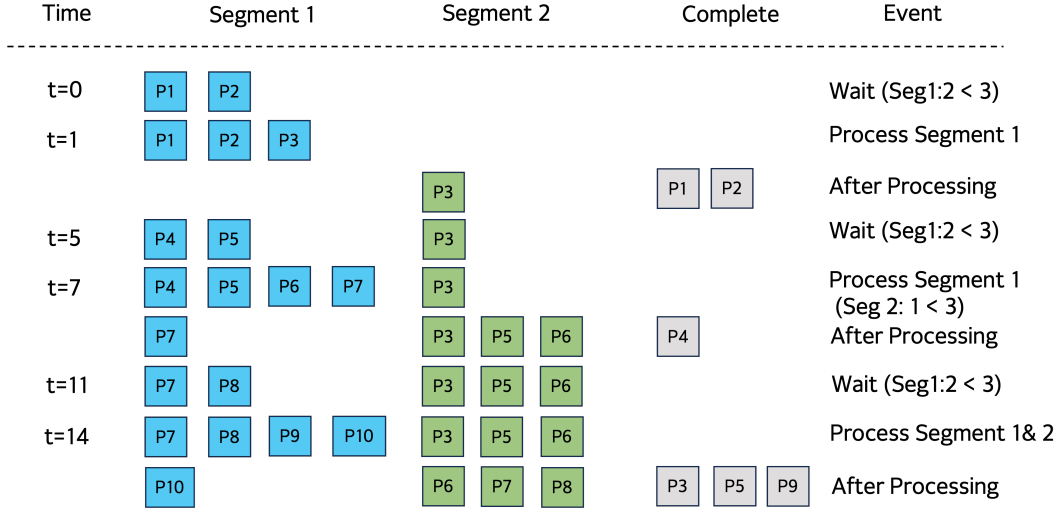
Nested WAIT's segmentation rule is based on decode progress. For the main analysis, we take a common prefill length  $l$  to keep the memory expressions transparent; Appendix A discusses heterogeneous prefill lengths and coarser segmentations.



**Figure 5** Nested WAIT pipeline. Prompts enter the first segment without output-length labels; short prompts complete and exit at early boundaries, while longer prompts advance to later segments where separate thresholds regulate batching.

In the two-type instance from Example 3, Nested WAIT uses two segments. Segment 1 processes all newly arrived prompts through prefill/decode stages 0 and 1 at combined rate  $\lambda_1 + \lambda_2$ ; type-1 prompts then complete, while type-2 prompts enter Segment 2 and are processed at rate  $\lambda_2$ .

Figure 6 shows this operation over time. Segment 1 waits until enough newly arrived prompts have accumulated, and Segment 2 waits until enough prompts have survived the first segment. When both thresholds are met, the active segments are scheduled together. Thus early segments provide shared batching, while later segments regulate service for prompts that have been identified as longer jobs.



**Figure 6** Two-segment Nested WAIT example. Segment 1 batches newly arrived prompts once its threshold is met; Segment 2 starts only after enough prompts have survived the first segment and revealed themselves as longer jobs.

Algorithm 2 gives the formal description of Nested WAIT. Completion events determine which prompts exit and which continue downstream; thresholds determine when each active segment has enough work to process without sacrificing batching efficiency.

## 5.2. Main Theorem

The analysis uses the same asymptotic scaling and scaled delay metrics as Section 4: arrivals and service speed scale with  $\zeta$ , and the effective horizon in the bounds is  $\zeta T$ . As in WAIT,  $M^\pi$  denotes the base threshold memory and is independent of  $\zeta$ . The required physical memory  $M_{\text{req}}^{(\zeta, \pi)}$ , however, also includes the logarithmic safety buffer in (14); this buffer is the finite-horizon cost of controlling downstream boundary queues. Equivalently, for a fixed physical memory capacity, the theorem applies to horizons and failure probabilities for which  $C \geq M_{\text{req}}^{(\zeta, \pi)}$ . The new difficulty is that downstream segments receive endogenous input. The prompts entering segment  $k$  are exactly those that survived earlier segments, so their arrivals depend on upstream batching, completion events, and

**Algorithm 2** Nested WAIT: Nested Waiting for Accumulated Inference Threshold**Input:** Memory capacity  $C$ , arrival rates  $\lambda_j$  for  $j \in [m]$ , thresholds  $n_k$  for segments  $k \in [m]$ **Input:** Output lengths  $l'_1 < l'_2 < \dots < l'_m$  for output-length classes  $j \in [m]$ ▷  $Q_{1,0}$  is the external prefill queue; for  $k \geq 2$ ,  $Q_{k,l'_{k-1}}$  is the resident boundary queue▷ The threshold-controlled interior of segment  $k \geq 2$  consists of stages  $l'_{k-1} + 1, \dots, l'_k$ 

```

1: Initialize all segment-entry, boundary, and interior queues to zero
2: Initialize event queue; set current time  $t \leftarrow 0$ 
3: Set server status to idle
4: while True do
5:   Wait for the next event (arrival or batch completion)
6:   if event is an arrival then
7:      $Q_{1,0} \leftarrow Q_{1,0} + 1$                                      ▷ New prompts enter segment 1 at stage 0
8:   else if event is a batch completion for prefix  $1, \dots, k$  then
9:     Retrieve the selected counts  $a_{k',s}$  stored with the completed prefix batch
10:    Update all affected queues simultaneously using the stored selected counts
11:    for each processed segment  $k' \leq k$  do
12:      for each selected stage  $s$  with stored count  $a_{k',s}$  do
13:        if  $k' \geq 2$  and  $s = l'_{k'-1}$  then
14:          Move selected boundary prompts to the first interior stage  $l'_{k'-1} + 1$ 
15:        else if  $s < l'_{k'}$  then
16:          Move selected interior prompts to stage  $s + 1$ 
17:        else
18:          if  $k' < m$  then
19:            Complete prompts whose output ends at  $l'_{k'}$ ; move survivors to  $Q_{k'+1,l'_{k'}}$ 
20:          else
21:            Complete and clear KV caches
22:          end if
23:        end if
24:      end for
25:    end for
26:    Set server status to idle
27:  end if
28:  Find largest  $k$  such that  $Q_{k',l'_{k'-1}} \geq n_{k'}$  for all  $k' \leq k$                                      ▷ All segments up to  $k$  meet threshold
29:  if server is idle and such  $k$  exists then
30:    Set  $S_1 = \{0, \dots, l'_1\}$ 
31:    For  $k' = 2, \dots, k$ , set  $S_{k'} = \{l'_{k'-1}, \dots, l'_{k'}\}$ 
32:    For every  $k' \leq k$  and  $s \in S_{k'}$ , set  $a_{k',s} \leftarrow \min\{n_{k'}, Q_{k',s}\}$ 
33:    Form the prefix batch from all selected prompts and store  $a_{k',s}$ 
34:    Process batch; already resident prompts outside the batch wait with their KV caches retained
35:    Set server status to busy until the completion event for this prefix batch
36:  end if
37: end while

```

the resident KV caches carried across segment boundaries. We study throughput  $\text{Throughput}^{(\zeta, \pi)}$ , latency  $\text{Latency}^{(\zeta, \pi)}$ , and time to first token  $\text{TTFT}^{(\zeta, \pi)}$  under a threshold vector  $\pi = [n_1, \dots, n_m]$ .

The need for additional memory is already visible at  $C = M^*$ . At this capacity, the fluid equilibrium fits in memory if output lengths are known. A non-predictive policy, however, must admit prompts before knowing whether they will finish at an early boundary or continue downstream. Optimistic admission can overflow memory once prompts reveal longer outputs; conservative admission

avoids that risk but leaves capacity unused. The resulting uncertainty creates a non-vanishing throughput loss.

**PROPOSITION 5.** *When memory capacity is exactly at the fluid stability boundary,  $C = M^*$ , there exists an instance where any online non-predictive policy  $\pi$ , unaware of output lengths upon arrival, incurs a throughput gap  $\text{Throughput}^* - \mathbb{E}[\text{Throughput}^{(\zeta, \pi)}] = \Omega(1)$ .*

This lower bound shows that uncertainty has a memory cost:  $M^*$  supports the fluid equilibrium with known output lengths, but does not provide enough room for the boundary queues created by unknown output lengths. The proof constructs a two-output-length instance in which the number of prompts continuing past the first boundary has binomial fluctuations (Appendix C.5). Theorem 2 shows that Nested WAIT closes this gap with a logarithmic safety buffer. The construction has two parts: choose thresholds that give these boundary queues negative drift, and reserve enough memory for their residual stochastic fluctuations.

The threshold vector must make each segment queue stable while keeping batches large enough to exploit the GPU. Let  $\Delta T_{[1, \dots, m]}(n_1, \dots, n_m)$  denote the iteration time when segment  $k$  contains exactly  $n_k$  prompts at each prefill/decode stage in that segment. The required drift conditions are

$$\begin{aligned} \Delta T_{[1, \dots, m]}(n_1, \dots, n_m) &< \frac{n_1}{\sum_{j=1}^m \lambda_j}, \\ p_k &< \frac{n_k}{n_{k-1}} < 1, \quad \forall k \in \{2, \dots, m\}, \end{aligned} \tag{12}$$

where  $p_k = (\sum_{j=k}^m \lambda_j) / (\sum_{j=k-1}^m \lambda_j) < 1$  for  $k \geq 2$ . The first inequality gives negative drift for the entry queue of Segment 1. The second gives negative drift for each downstream boundary queue, since only a  $p_k$  fraction of prompts leaving segment  $k - 1$  continue to segment  $k$ ; the upper bound  $n_k < n_{k-1}$  keeps the boundary queue in the nondegenerate binomial-tail regime used in the martingale-root bound. If  $n_k \geq n_{k-1}$ , the downstream boundary queue is deterministically bounded and can be handled separately without this logarithmic tail term. Segments with  $p_k = 0$  have no stochastic downstream input and can be omitted or merged. For memory accounting, define

$$\delta_1 = l'_1 + 1, \quad \bar{s}_1 = \frac{l'_1}{2}, \quad \delta_k = l'_k - l'_{k-1}, \quad \bar{s}_k = \frac{l'_{k-1} + 1 + l'_k}{2} \quad (k \geq 2).$$

The base memory used by threshold-sized segment batches is

$$M^\pi = \sum_{k=1}^m n_k (l + \bar{s}_k) \delta_k. \tag{13}$$

Here  $\delta_k$  is the number of prefill/decode stages covered by segment  $k$ , and  $\bar{s}_k$  is the average decode progress across those stages. The term  $M^\pi$  is the threshold analogue of the fluid memory requirement  $M^*$ : if  $n_k$  is chosen on the scale of the cumulative fluid population that survives to segment  $k$ , then  $M^\pi$  is on the same scale as  $M^*$ , up to rounding and segment aggregation.

For  $k \geq 2$ , the resident queue at boundary  $l'_{k-1}$  is measured after the threshold review for segment  $k$ . When segment  $k$  is selected, the  $n_k$  boundary prompts removed from this queue are charged to the first interior stage of segment  $k$  in  $M^\pi$ , which is a safe upper bound because that stage has one additional decode token. The additional memory buffer in the theorem counts only boundary prompts not selected in the current prefix batch. Appendix A.2, especially Figures 13–15, gives a numerical decomposition showing this base term relative to the safety buffers. With these definitions, the theorem below gives the performance guarantee.

Thus  $M^* \leq C$  is the fluid-model stabilizability condition when output lengths are known. With unknown output lengths, Nested WAIT also needs enough memory for the downstream boundary queues created as prompts reveal whether they continue to later segments. The theorem below shows that, with this additional safety buffer, Nested WAIT realizes the corresponding stability guarantee in the stochastic system.

**THEOREM 2.** *Assume thresholds  $\pi = [n_1, \dots, n_m]$  satisfy (12), and the physical memory capacity  $C$  satisfies*

$$\begin{aligned} C &\geq M_{\text{req}}^{(\zeta, \pi)} := M^\pi + \sum_{k=2}^m (l + l'_{k-1}) \left( n_k + \theta_k^{-1} \ln \left( \frac{m(1 + \zeta T/d_0)}{\delta} \right) \right) \\ &= O \left( M^\pi + \sum_{k=2}^m (l + l'_{k-1}) \theta_k^{-1} \ln \left( \frac{m(1 + \zeta T/d_0)}{\delta} \right) \right), \end{aligned} \quad (14)$$

where  $\theta_k > 0$  is the unique solution of  $e^{-\theta_k n_k} (1 - p_k + p_k e^{\theta_k})^{n_{k-1}} = 1$ ; in particular,  $\theta_k \geq 8(n_k - n_{k-1} p_k)/n_{k-1}$  for  $k \geq 2$ . For the corresponding no-overflow Nested WAIT dynamics:

$$\begin{aligned} \text{Throughput}^* - \mathbb{E}[\text{Throughput}^{(\zeta, \pi)}] &= O((\zeta T)^{-1}), \\ \mathbb{E}[\text{Latency}^{(\zeta, \pi)}], \mathbb{E}[\text{TTFT}^{(\zeta, \pi)}] &= O(1). \end{aligned}$$

Moreover, the no-overflow dynamics are feasible on the physical GPU, i.e., memory is not exceeded, with probability at least  $1 - \delta$  over horizon  $[0, T]$ .

Unknown output lengths make this memory-control problem harder. At admission, the scheduler does not know which prompts will finish early and which will continue into later, more memory-intensive decode stages. Yet endogenous memory growth starts as soon as a prompt's KV cache



becomes resident on the GPU. As Example 3 illustrates, prompts reveal their output lengths only by either completing at a boundary or continuing beyond it. Completed prompts leave and release memory, while surviving prompts carry larger KV caches downstream. Without segment-level control, these surviving prompts can create late-stage buildup, overflow memory, and trigger eviction-induced restarts. Nested WAIT therefore regulates both admission and advancement through prefill and decode while the output-length mix is still being revealed.

The proof follows the prompts at these segment boundaries, where output-length information is revealed and downstream memory growth must be controlled. Conditional on a threshold-sized batch leaving segment  $k - 1$ , the number that continue to segment  $k$  follows binomial thinning with probability  $p_k$ . This coupling turns the unknown-output-length process into boundary queues. These queues track how much surviving work is allowed to continue into later segments, which is the part of the memory-growth trace that can cause overflow. In the memory condition (14),  $M^\pi$  covers threshold-sized segment batches, while the remaining terms keep retained KV caches at downstream boundaries from causing overflow and eviction. The theorem states the clean case with one boundary per output length. In practice, the number of possible output lengths can be large, and the same construction can be implemented with coarser decode segments that group nearby lengths and maintain thresholds only at the segment boundaries. Appendix A discusses this general segment design, and Section 6 evaluates it in simulated serving workloads.

## 6. Numerical Experiments

We evaluate WAIT and Nested WAIT on synthetic Poisson workloads (Section 6.1) and a real workload sampled from lmsys-chat-1m (Section 6.2). We benchmark against vLLM (Kwon et al. 2023) and Sarathi (Agrawal et al. 2023), both greedy FCFS-based serving policies in our experiments; Sarathi differs by splitting long prefills into chunks that can be interleaved with decode work. We use Vidur (Agrawal et al. 2024a) as the discrete-event simulator, configured for a single NVIDIA A100 80 GB serving Llama-2-7B. Appendix H validates Vidur's iteration-time predictions against physical A100 measurements over the profiled range and at an extrapolation point  $B = 256$ , and also reports end-to-end real-GPU runs of WAIT. The workloads below stress decode-side KV-cache growth, the regime targeted by the linear iteration-time approximation in Section 2.

With exogenous arrivals at rate  $\lambda$ , policies in the stable region complete requests at the offered load, so completion-rate differences appear only when a policy approaches or exceeds the arrival

rates it can sustain. The main figures plot mean end-to-end latency against  $\lambda$ , averaging each point over 10 independent simulation replications. Within each replication, latency is averaged over a central measurement window after the initial transient, using the same window across policies. We use  $\lambda^*$  to mark the observed transition from the near-overloaded to the overloaded regime: it is the largest reported rate at which latency remains controlled and effective completion rate still tracks the offered load. Above that range, completion rate saturates or declines because the policy suffers overload or eviction-induced restarts, and the paired completion-rate panel records this loss of sustainable service.

**Hardware.** All experiments use Vidur to simulate a single NVIDIA A100 80 GB serving Llama-2-7B (NVIDIA 2026, Hugging Face 2025). A FP16 KV-cache token for this model occupies 2 (key/value)  $\times$  32 layers  $\times$  32 KV heads  $\times$  128 head dimension  $\times$  2 bytes, or 524,288 bytes (0.5 MiB) (Atharva 2026). After model weights and the baseline 1% memory margin, the baseline-memory runs use a KV-cache cap of approximately  $1.37 \times 10^5$  tokens. If resident KV cache exceeds the cap, the server evicts and restarts in-progress requests. The iteration-time model is Equation (1); its A100 validation is reported in Appendix H.2.

**Baseline configurations (vLLM, Sarathi).** Both baselines use their default Vidur configurations and greedily admit arrivals in FCFS order. They also use the same local capacity-reservation rule: before admitting a new request, the scheduler requires a small memory buffer (1% of KV-cache capacity) to remain unused. This rule protects against admitting a request when the server is already essentially full, but it is not a dynamic limit on the total number of jobs in service. Once admitted requests continue decoding, their KV caches can still grow beyond the available capacity, in which case the scheduler evicts and restarts an in-progress request. The difference between the baselines is in how prefill is batched. vLLM processes a newly admitted request's full prefill in one iteration. Sarathi bounds the prefill work per iteration to chunks of at most 512 tokens, allowing partial prefills to be interleaved with decode iterations. This chunked-prefill rule reduces head-of-line blocking from long prefills, but it does not control the cumulative decode-side population. Thus neither baseline enforces the load-balancing threshold used by WAIT.

**WAIT and Nested WAIT settings.** In the experiments below, the main tuning variable for WAIT and Nested WAIT is a system-wide batch-size cap  $tl$ , which controls how many requests can be served concurrently across prefill and decode stages. This cap is not itself one of the per-stage

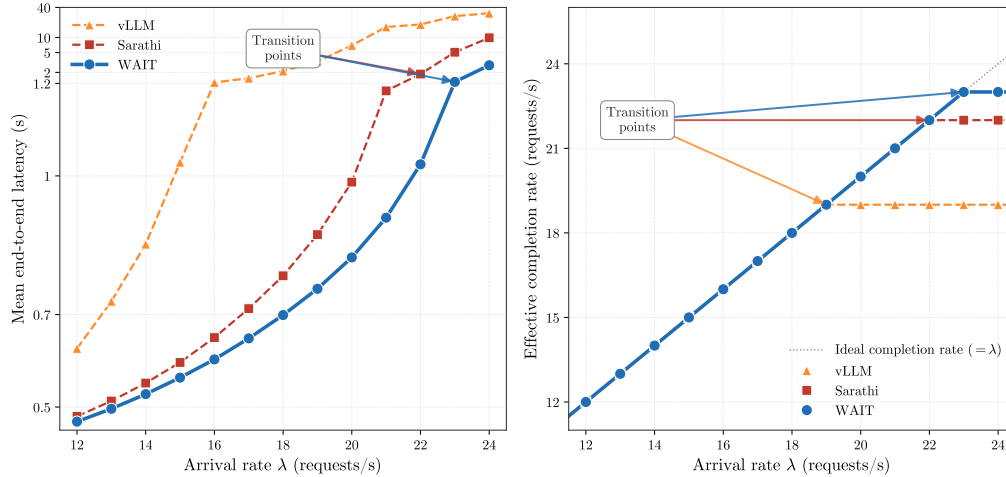
scheduling thresholds. In a single-type WAIT run,  $tl$  is distributed across the prefill/decode stages of that workload, yielding the integer per-stage thresholds used by the scheduler. In a Nested WAIT run,  $tl$  is first allocated across decode-stage segments according to the segment lengths and continuation probabilities, and the resulting segment-level caps are then rounded and spread across the stages inside each segment. For each workload, the corresponding subsection specifies the parameter grid and the distribution-aware threshold and segment design used in the experiment. The calibration uses only the workload distribution and the offered arrival rate; the online scheduler does not use realized future arrivals or unrevealed final output lengths. Under the baseline settings used for the arrival-rate figures, WAIT and Nested WAIT do not trigger eviction.

### 6.1. Synthetic Workloads

We consider three synthetic Poisson workloads. The single-type workload  $p_{512d20}$  has prefill length 512 and decode length 20, isolating the effect of WAIT's threshold mechanism (Algorithm 1). The two-type workload  $(0.7p_{512d20}, 0.3p_{512d50})$ , meaning that each arrival is  $p_{512d20}$  with probability 0.7 and  $p_{512d50}$  with probability 0.3, tests the segment design of Nested WAIT (Algorithm 2) on prompts that share the prefill profile but differ in decode length. The long-decode workload  $p_{512d1000}$  keeps the same prefill length but increases the decode length to 1000, allowing us to examine the same threshold mechanism when KV cache growth is much more pronounced.

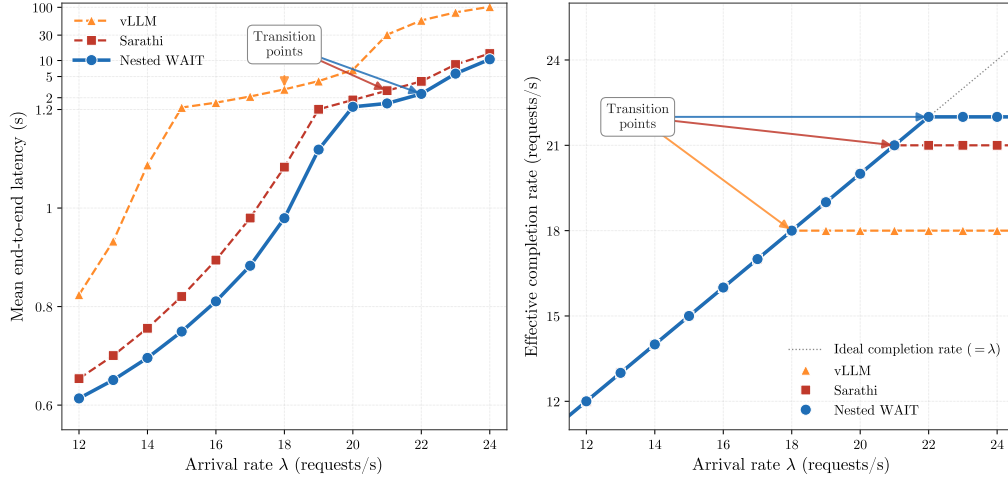
*Single-type workload (WAIT).* For this workload, WAIT calibrates  $tl$  on the finite grid  $\{20, 30, \dots, 200\}$  and converts the selected cap into the corresponding per-stage limits. Figure 7 shows mean end-to-end latency (left, on a stretched-symlog scale that expands the stable region) and effective completion rate (right), i.e., completed requests per second net of eviction-induced restarts, versus arrival rate  $\lambda$ . Because all requests are homogeneous, this workload highlights the effect of threshold-based admission without the additional segment structure used in Nested WAIT. WAIT keeps latency controlled up to  $\lambda \approx 23$ , whereas Sarathi and vLLM lose stability near  $\lambda \approx 22$  and  $\lambda \approx 19$ , respectively. At low load the three policies are close, but the gap widens rapidly as  $\lambda$  approaches the boundary: WAIT attains the smallest mean latency at every tested rate in  $\lambda \in \{12, \dots, 24\}$ , with the gap to Sarathi growing from  $\sim 2\%$  at  $\lambda = 12$  to over 70% at  $\lambda = 23$ . Because there is only one request type, the gain does not come from type separation; it comes from threshold-based admission itself, which keeps the batch composition closer to the balanced prefill/decode composition and delays the onset of instability. The right panel shows the same insight

in completion-rate terms: WAIT follows the ideal  $\lambda$ -line up to a higher arrival rate than Sarathi and vLLM.



**Figure 7** Single-type workload (p512d20, Poisson arrivals): mean end-to-end latency (left) and effective completion rate (right) versus arrival rate  $\lambda$ . The arrows mark each policy's transition from the near-overloaded to the overloaded regime; the dotted line in the right panel indicates the ideal completion rate equal to  $\lambda$ .

*Two-type workload (Nested WAIT).* For the two-type workload, Nested WAIT uses the same system-wide batch-size cap  $tl$  and two decode-stage segments, 1–20 and 21–50. The second segment receives the prompts that survive the first 20 decode stages, so its continuation ratio is 0.3. Thus the implementation sets real-valued per-stage thresholds in the ratio  $n_2/n_1 = 0.3$  and solves  $20n_1 + 30n_2 = tl$  before rounding the corresponding segment-level caps to integers; for example,  $tl = 30$  gives segment-level caps approximately 21 and 9. Figure 8 shows the same comparison for the two-type workload (0.7p512d20, 0.3p512d50). The stability ordering is again vLLM first ( $\lambda^* \approx 18$ ), Sarathi second ( $\lambda^* \approx 21$ ), and Nested WAIT last ( $\lambda^* \approx 22$ ). Nested WAIT attains the lowest mean latency over these tested rates, with the gap to Sarathi growing from  $\sim 6\%$  at low load to over 40% near the boundary. The mechanism also differs from the single-type case. Here the two request classes have the same prefill length and differ only in decode length, so the main source of congestion is the accumulation of the long-decode minority in later decode stages. Nested WAIT's segment thresholds keep that downstream inventory from expanding too far, which prevents those delays from propagating to the short-decode majority. The resulting gain is therefore not only at entry: by controlling downstream buildup, Nested WAIT preserves low latency for the short jobs and enlarges the stable operating region beyond what a single global cap can achieve.

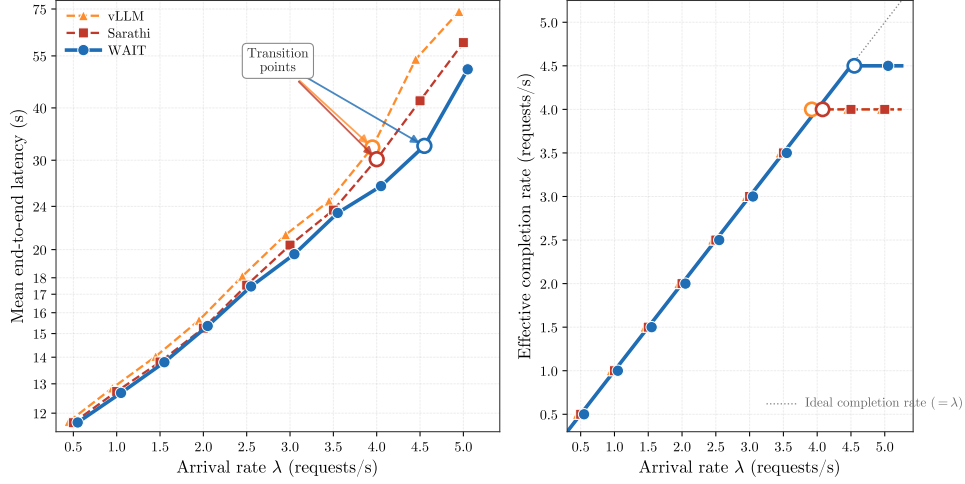


**Figure 8** Two-type workload (0.7p512d20, 0.3p512d50) (Poisson arrivals): mean end-to-end latency (left) and effective completion rate (right) versus  $\lambda$ . The transitions from near-overloaded to overloaded are visible in both panels: vLLM ( $\lambda^* = 18$ ) < Sarathi ( $\lambda^* = 21$ ) < Nested WAIT ( $\lambda^* = 22$ ).

*Long-decode regime.* We also study the long-decode workload p512d1000 with Poisson arrivals at  $\lambda = 0.5, 1.0, \dots, 5.0$ . Relative to the single-type workload p512d20, each request now occupies about 30× more KV cache, so all three policies lose stability at much smaller arrival rates. Figure 9 shows that at low rates  $\lambda \leq 2.5$  the three policies are nearly indistinguishable, but near the boundary WAIT again has the lowest latency: at  $\lambda = 4.5$  it attains 32.3 s versus 41.7 s for Sarathi and 53.8 s for vLLM. The intuition is that long-decode requests stay in the system for many more iterations, so excess admission persists longer and makes it harder to keep arrivals and completions in balance. WAIT performs better in this regime because it keeps the in-system inventory closer to that balance point. The same pattern is already visible at  $\lambda = 4.0$ , where the baseline-memory setting gives 26.4 s for WAIT versus 30.2 s for Sarathi. Table 1 reports the accompanying eviction-induced restart evidence at the same arrival rate: Sarathi incurs a 2.88% restart rate, whereas WAIT still avoids eviction altogether. Under the server’s last-in-first-out eviction rule, each evicted request loses its partial decode progress and begins again from prefill, so the restart gap is another manifestation of the same imbalance.

| Metric                                            | Sarathi | WAIT (ours) |
|---------------------------------------------------|---------|-------------|
| Mean end-to-end latency ( $\lambda = 4.0$ )       | 30.2 s  | 26.4 s      |
| Eviction-induced restart rate ( $\lambda = 4.0$ ) | 2.88%   | 0%          |

**Table 1** Eviction diagnostic for p512d1000 at  $\lambda = 4.0$ . Sarathi incurs eviction-induced restarts while WAIT avoids eviction.



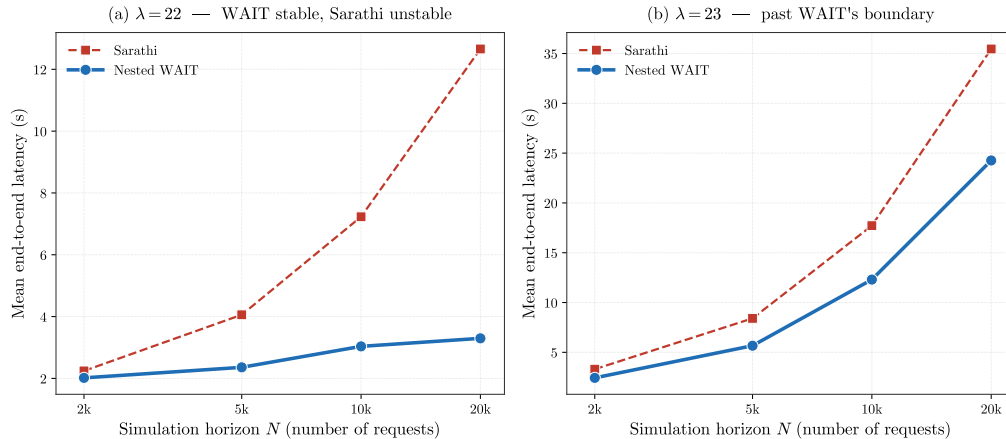
**Figure 9** Long-decode workload (p512d1000, prefill 512, decode 1000): mean end-to-end latency (left) and effective completion rate (right) versus arrival rate  $\lambda$ . At low rates the three policies are largely comparable, with only small mixed latency differences; the main separation emerges from  $\lambda \approx 3.0$  onward. Near the boundary, WAIT attains the lowest latency and sustains the highest completion rate.

*Long-horizon check near the transition.* To check the estimated transition, we vary the simulation horizon at the two arrival rates nearest Nested WAIT’s transition. Figure 10 plots mean latency on  $(0.7p_{512d20}, 0.3p_{512d50})$  for  $\lambda \in \{22, 23\}$  as  $N$  increases from 2,000 to 20,000. At  $\lambda = 22$ , Nested WAIT approaches a finite latency plateau, whereas Sarathi’s mean latency grows rapidly over the tested horizons. At  $\lambda = 23$ , both policies exhibit continued latency growth, but Nested WAIT remains 30%–50% below Sarathi at every horizon we consider. These comparisons are consistent with the observed transition from near-overloaded to overloaded operation lying between the two rates.

## 6.2. Application to the `lmsys-chat-1m` Dataset

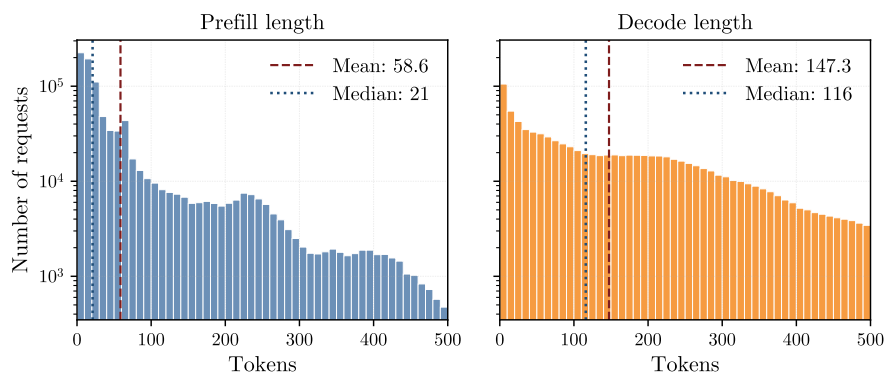
We next evaluate the policies on `lmsys-chat-1m`, a real dataset with heterogeneous prompt and response lengths, where decode lengths are unknown at admission. This empirical setting lets us assess the robustness of Nested WAIT’s state-dependent admission rule under a heterogeneous workload distribution, and characterize how the induced capacity allocation varies across prefill and decode stages.

*Dataset.* The dataset contains approximately 1 million human–LLM conversations; we treat each user message as the prompt (prefill) and the corresponding assistant response as the generated output (decode). Figure 11 shows the prefill-length and decode-length distributions after filtering to requests with both lengths below 500 tokens, which is the workload population used for sampling.



**Figure 10** Long-horizon check near the transition from near-overloaded to overloaded on  $(0.7p512d20, 0.3p512d50)$ . At  $\lambda = 22$ , Nested WAIT approaches a finite latency plateau while Sarathi grows rapidly with  $N$ ; at  $\lambda = 23$ , both policies show continued latency growth, but Nested WAIT remains lower-latency.

Prefill lengths have mean 58.6 tokens and median 21, while decode lengths have mean 147.3 tokens and median 116. Short responses remain common: about 31% of prompts decode for at most 50 tokens, whereas about 14% exceed 300. For simulation, we sample 5,000 requests from this filtered workload population. Arrivals follow a Poisson process with rate  $\lambda$ , measured in requests per second. All runs use the same simulator, hardware, and baseline settings as in Section 6.1.



**Figure 11** Filtered `lmsys-chat-1m` prefill and decode length distributions, restricted to requests with both lengths below 500 tokens. Dashed lines mark means and medians.

*Segment construction and threshold allocation for Nested WAIT.* Nested WAIT (Algorithm 2) regulates admission by partitioning the decode horizon into ordered segments, each with its own threshold. A request is counted against the threshold of its current segment: it first consumes capacity in the head segment and enters downstream segments only if it continues decoding. The



policy therefore allocates admission capacity across realized continuation stages, rather than making admission decisions from predicted final response lengths.

Nested WAIT is distribution-aware: its thresholds are constructed from the empirical workload distribution and the offered arrival rate. For a chosen number of segments  $L$ , the decode horizon 0–500 is divided into  $L$  equal-width segments, and a request belongs to the segment containing its current decode stage. Let  $\Delta l'_k$  be the number of decode stages in segment  $k$ , and let  $\lambda'_k$  be the arrival rate of requests whose final decode length lies in that segment. A request reaches segment  $k$  only if its decode length exceeds the previous segment boundary, so the continuation probability from segment  $k$  to segment  $k+1$  is  $p_k = (\sum_{r=k+1}^L \lambda'_r) / (\sum_{r=k}^L \lambda'_r)$ . Given a margin parameter  $\eta$  (the implementation parameter `seg_margin`), we set  $q_k = \min\{p_k + \eta, 0.99\}$ , making the downstream threshold ratios slightly larger than the empirical continuation probabilities. The real-valued per-stage thresholds are  $n_k = n_1 \prod_{r < k} q_r$ , where  $n_1$  is chosen to satisfy  $\sum_{k=1}^L \Delta l'_k n_k = \text{tl}$ . Thus, the system-wide batch-size cap `tl` fixes the overall threshold scale, while  $L$  and the continuation probabilities determine how that scale is allocated across decode segments. All real-data arrival-rate runs use the baseline KV-cache cap of approximately  $1.37 \times 10^5$  tokens described above. The cap `tl` should not be read as allowing `tl` longest-context requests to reside on the GPU simultaneously: resident KV-cache use depends on the realized prompt lengths and decode stages, and exceeding the memory cap triggers eviction and restart. Before scheduling, each segment-level cap  $B_k = \Delta l'_k n_k$  is rounded to an integer and spread as evenly as possible across the  $\Delta l'_k$  stages in that segment. In the real-data runs, we set  $\eta = 0.05$ , choose `tl` from  $20, 40, \dots, 200$ , choose  $L$  from  $\{1, 2, 3, 4, 5, 10, 20\}$ , and use the lowest-latency configuration on this parameter grid for each offered arrival rate. The performance is not monotone in either parameter: a cap that is too small leaves batching and downstream control underused, whereas a cap that is too large can push resident KV-cache usage toward overflow. Similarly, too few segments blur meaningful decode-stage differences, while too many segments fragment the segment-level caps and make prompts wait at segment thresholds too often. The selected setting is therefore the best tradeoff on this parameter grid for the workload and arrival rate.

To relate the real-data arrival rates to the fluid memory condition, Table 2 reports  $M^*(\lambda)$  for the same real-data sample. We compute these values by estimating the service-time parameters from our experiments and substituting the empirical prefill/decode length distribution into the fluid memory formula (8). The calculation places the fluid memory crossing  $M^*(\lambda) = C$  between  $\lambda = 70$  and  $\lambda = 80$ , at approximately 74.1 queries/s. This is consistent with the range where the real-data

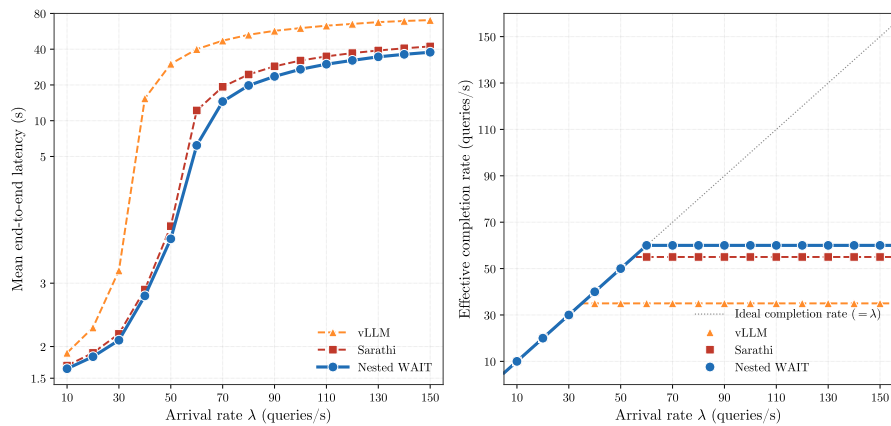


latency curves begin moving from near-overloaded to overloaded operation; the remaining gap between the continuous fluid calculation and the simulated transition is expected from integer threshold rounding, finite-horizon stochastic variation, and the discrete segment calibration used by Nested WAIT.

| Arrival rate $\lambda$ | $M^*(\lambda)$ (KV tokens) | $M^*(\lambda)/C$ | Fluid stable region |
|------------------------|----------------------------|------------------|---------------------|
| 50                     | 34,377                     | 0.25             | inside              |
| 60                     | 55,758                     | 0.41             | inside              |
| 70                     | 100,327                    | 0.73             | inside              |
| 80                     | 250,509                    | 1.83             | outside             |

**Table 2** Fluid memory requirement for the real dataset using service-time parameters estimated from the experiments. The capacity is  $C \approx 1.37 \times 10^5$  KV-cache tokens.

*Results.* Figure 12 varies the arrival rate  $\lambda$  from 10 to 150 in steps of 10. The plotted comparison follows the same qualitative ordering as the synthetic experiments. The baseline schedulers are competitive when the workload is light, but their latencies increase more quickly as the system approaches and then exceeds the range in which completions can keep pace with arrivals. Nested WAIT is most clearly separated from the baselines in these near-overloaded and overloaded regimes. This pattern is consistent with the heterogeneous decode distribution: many requests finish in the early decode stages, while a smaller group remains active for hundreds of generated tokens. Nested WAIT’s nested thresholds control how many requests may continue across successive decode-stage ranges, allowing short responses to clear early while limiting the population in later decode stages that consumes KV-cache capacity.



**Figure 12** Real-data experiment on `lmsys-chat-1m` (5,000 requests): mean end-to-end latency (left) and effective completion rate (right) versus  $\lambda$ . Nested WAIT has lower latency than the baselines, with a larger gap near overload.

## 7. Conclusion and Future Directions

This paper studies online scheduling for LLM inference, a setting in which each request's memory consumption grows during decoding and exceeding capacity triggers evictions that can destabilize a system even when capacity is theoretically sufficient for stability under the fluid equilibrium. We develop a multi-stage online scheduling model with endogenous memory growth and analyze a fluid approximation whose fluid equilibrium identifies both the fluid stability region and the memory requirement  $M^*$  needed to support it. Guided by this equilibrium, we design the (Nested) WAIT algorithm, a threshold-based admission rule that keeps the stochastic system close to the fluid composition across prefill and decode stages. The threshold mechanism performs a dimensionality reduction: it induces lower-dimensional threshold and boundary queues that capture the memory-coupled state created by endogenous KV-cache growth. Under the linear multi-stage model with endogenous memory growth and the stated memory conditions, the algorithms attain the fluid effective-throughput benchmark, measured as completed decode-token service net of eviction, with the corresponding latency and TTFT guarantees even when output lengths are unknown at arrival. Because the algorithms keep the stochastic system near the fluid-guided load-balancing allocation and avoid the eviction cascades that destabilize memory-reactive baselines, they expand the empirically observed stable operating range and reduce mean end-to-end latency most clearly near capacity and under overload. Vidur simulations configured for Llama-2-7B on an NVIDIA A100 GPU show these effects compared with the tested vLLM and Sarathi configurations, and supplemental real-GPU validation reported in the appendix provides implementation evidence as well as support for the simulator's accuracy.

Several questions remain open. First, our threshold design relies on knowledge of the prompt-type arrival distribution; developing adaptive variants that learn this distribution online, or versions that remain robust under bounded misspecification, would extend the algorithm's applicability. Second, practical deployments face arrival rates that drift over time and occasionally surge. Theorem 3 covers a finite-horizon time-varying setting with bounded window loads and bounded first-segment waiting times. Sharper volatility, such as demand spikes that temporarily exceed nominal capacity or extended low-arrival intervals that call for timeout rules, requires further analysis. Two concrete questions are how fast thresholds can re-tune while preserving asymptotic optimality and how transient surges interact with the safety buffer in Nested WAIT.

Finally, extending the analysis to multi-GPU deployments introduces inter-device communication cost and a choice of parallelization strategy (data, tensor, or pipeline). This regime is especially relevant for mixture-of-experts (MoE) architectures (Shazeer et al. 2017, Fedus et al. 2022, Jiang et al. 2024), in which token-level routing distributes experts across GPUs and creates skewed memory and compute load that our single-GPU analysis does not capture. A unified framework that models inter-device communication inside the iteration-time model, and coordinates admission thresholds across GPUs in response to the routing distribution, is a natural open direction.

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## Appendix A: Extensions

The WAIT and Nested WAIT algorithms assume constant arrival rates and a fixed number of segments matching the number of prompt types. In practice, however, LLM inference systems face two additional challenges: (i) **time-varying workloads**, where arrival rates fluctuate over time; (ii) **numerous prompt types**, where maintaining separate thresholds for hundreds of output lengths becomes infeasible. We address these challenges through two extensions that preserve the core algorithmic principles—eviction prevention and approaching load balance—while adapting the threshold construction to more realistic deployment settings.

First, we extend Nested WAIT to handle time-varying arrival rates  $\lambda_j^t$  by replacing constant-rate threshold conditions with accumulated arrival constraints over each iteration window. This gives a conservative finite-horizon guarantee under bounded window loads. The delay bound additionally requires that the first threshold can be collected on the service-normalized time scale, which rules out arbitrarily long low-arrival intervals (Section A.1). Second, we introduce a **segment-wise design** that groups many prompt types into a smaller number of decode-stage segments, reducing algorithmic complexity while maintaining the same threshold structure (Section A.2).

### A.1. Time-Varying Arrival Rates

Time-varying arrival rates capture moderate changes in workload intensity, such as predictable daily variation or slowly changing demand. The threshold mechanism remains valid when the accumulated arrivals during each service window stay below the corresponding completion capacity. Formally, we replace constant rates  $\lambda_j$  with time-dependent rates  $\lambda_j^t$ , assumed continuous and uniformly bounded by  $\lambda_{\max} < \infty$  to prevent unbounded bursts.

We analyze the asymptotic optimality of the Nested WAIT algorithm under time-varying arrival rates, denoted  $\lambda_j^t$  for prompt type  $j \in [m]$  at physical time  $t \in [0, T]$ .

For a service-normalized window  $h$ , define the expected number of type- $j$  arrivals in the corresponding physical interval by

$$\mathcal{A}_j^{(\zeta)}(t, h) := \zeta \int_t^{\min\{T, t+h/\zeta\}} \lambda_j^u du.$$

The truncation only affects the final partial window and can only reduce the number of arrivals. Recall from Equation (11) that  $\Delta T_{[1,2,\dots,m]}(n_1, \dots, n_m)$  represents the service-normalized iteration



cost when processing exactly  $n_k$  prompts at each stage in segment  $k$ , for all  $k \in [m]$ , as in (12), where  $n_k$  is the threshold defined in Algorithm 2.

For the Nested WAIT policy defined by  $\pi = [n_1, \dots, n_m]$ , we impose the following uniform slack conditions for all  $t \in [0, T]$ ,  $0 < h \leq \Delta T_{[1,2,\dots,m]}(n_1, \dots, n_m)$  and  $k = 1, 2, \dots, m-1$ , analogous to (12):

$$\begin{aligned} \sum_{j=1}^m \mathcal{A}_j^{(\zeta)}(t, h) &\leq \sum_{j=1}^m \mathcal{A}_j^{(\zeta)}(t, \Delta T_{[1,2,\dots,m]}(n_1, \dots, n_m)) < n_1, \\ p_k^{(\zeta)}(t, h) &< n_k/n_{k-1} < 1, \quad k = 2, \dots, m, \end{aligned} \quad (15)$$

where  $p_k^{(\zeta)}(t, h) = (\sum_{j=k}^m \mathcal{A}_j^{(\zeta)}(t, h)) / (\sum_{j=k-1}^m \mathcal{A}_j^{(\zeta)}(t, h))$  when the denominator is positive, and  $p_k^{(\zeta)}(t, h) = 0$  otherwise. The time-varying fluid benchmark is

$$\text{Throughput}_T^* := \frac{1}{T} \int_0^T \sum_{j=1}^m \lambda_j^t l'_j dt.$$

For the downstream safety buffers, set

$$p_k^* := \sup_{\substack{t \in [0, T], 0 < h \leq \Delta T_{[1,\dots,m]} \\ \sum_{j=k-1}^m \mathcal{A}_j^{(\zeta)}(t, h) > 0}} \frac{\sum_{j=k}^m \mathcal{A}_j^{(\zeta)}(t, h)}{\sum_{j=k-1}^m \mathcal{A}_j^{(\zeta)}(t, h)}, \quad k \geq 2,$$

and set  $p_k^* = 0$  if the denominator is zero for all such windows. Segments with  $p_k^* = 0$  have no stochastic downstream input and can be omitted or merged; the theorem below is stated for downstream segments with  $p_k^* > 0$ .

*First-segment waiting.* The window-load conditions above control overload and downstream boundary queues. A delay guarantee also needs a mild lower-side condition: if arrivals nearly stop, a subthreshold first-segment queue can wait for a long physical time before the next batch forms. Let  $\tau_1^{(\zeta)}(t)$  be the physical time needed, starting at time  $t$ , for the scaled aggregate external arrival process to generate  $n_1$  arrivals. For the  $O(1)$  service-normalized delay bound below, assume there is a constant  $H < \infty$ , independent of  $(\zeta, T)$ , such that

$$\sup_{t \in [0, T]} \zeta \mathbb{E}[\tau_1^{(\zeta)}(t)] \leq H.$$

This condition is implied, for example, by a uniform lower bound  $\sum_{j=1}^m \lambda_j^t \geq \lambda_{\min} > 0$ . Without such a condition, the throughput and memory statements remain meaningful, but the service-normalized delay can be dominated by waiting through a long low-arrival interval. The following theorem extends Theorem 2 to time-varying arrival rates.

**THEOREM 3.** *Suppose the thresholds  $\pi = [n_1, \dots, n_m]$  satisfy (15) and the physical memory capacity  $C$  fulfills:*

$$\begin{aligned} C &\geq M_{\text{req, tv}}^{(\zeta, \pi)} := M^\pi + \sum_{k=2}^m (l + l'_{k-1}) \left( n_k + \theta_k^{-1} \ln \left( \frac{m(1+\zeta T/d_0)}{\delta} \right) \right) \\ &= O \left( M^\pi + \sum_{k=2}^m \theta_k^{-1} (l + l'_{k-1}) \ln \left( \frac{m(1+\zeta T/d_0)}{\delta} \right) \right), \end{aligned}$$

where the memory usage  $M^\pi$  is defined as (13) and  $\theta_k > 0$  is the unique solution of

$$e^{-\theta_k n_k} (1 - p_k^* + p_k^* e^{\theta_k})^{n_{k-1}} = 1,$$

with lower bound  $\theta_k \geq 8(n_k - n_{k-1} p_k^*)/n_{k-1}$ . All quantities  $p_k^*$ ,  $\theta_k$ , and  $M_{\text{req, tv}}^{(\zeta, \pi)}$  are evaluated for the given scaled finite-horizon system  $(\zeta, T)$ . For the corresponding no-overflow Nested WAIT dynamics,

$$\text{Throughput}_T^* - \mathbb{E} [\text{Throughput}^{(\zeta, \pi)}] = O((\zeta T)^{-1}).$$

Additionally, the no-overflow dynamics are feasible on the physical GPU, i.e., memory usage remains below  $C$ , with probability at least  $1 - \delta$  over the time horizon  $[0, T]$ . If the first-segment waiting condition above also holds, then the service-normalized delay metrics satisfy

$$\mathbb{E} [\text{Latency}^{(\zeta, \pi)}], \mathbb{E} [\text{TTFT}^{(\zeta, \pi)}] = O(1).$$

The extension to time-varying rates preserves the dimensionality reduction structure from Theorem 2. The key adaptation is that threshold conditions (15) must hold uniformly over physical time  $t \in [0, T]$ , replacing constant-rate arrival counts with the scaled-window arrivals  $\mathcal{A}_j^{(\zeta)}(t, h)$  over each service-normalized iteration window. These upper-side conditions control overload and memory growth. The separate first-segment waiting condition controls the time needed for the Segment 1 entry queue to reach its threshold. For downstream segments, prompts transitioning from segment  $k - 1$  to segment  $k$  are controlled by the corresponding time-varying thinning probability. The Lindley recursion and Doob's maximal inequality apply with worst-case arrival and thinning parameters, bounding queue lengths with the same logarithmic overhead. Appendix F provides the detailed proof.

## A.2. Segment Design

When LLM systems handle many distinct output lengths, maintaining individual thresholds for each type becomes impractical: the algorithm must track queue lengths separately for every output length, increasing memory overhead and implementation complexity. The **segment-wise design** addresses this by grouping similar output lengths into  $L \leq m$  segments, where each segment  $k$  aggregates types with comparable decode lengths. This reduces algorithmic complexity from  $O(m)$  to  $O(L)$  while preserving the threshold logic: prompts still classify themselves on the fly, but now at segment boundaries rather than individual type boundaries.

We assume constant arrival rates for clarity, though the approach extends to time-varying rates as discussed in Section A.1. Consider  $m$  prompt types with decode lengths  $l'_1 < \dots < l'_m$  and arrival rates  $\lambda_1, \lambda_2, \dots, \lambda_m$ . Our objective is to merge these into  $L$  ordered segments, where  $L \leq m$ . Choose cut points  $0 = h_0 < h_1 < \dots < h_L = m$ , so segment  $k$  contains output-length classes  $h_{k-1} + 1, \dots, h_k$ .

For each segment  $k \in \{1, \dots, L\}$ , the total arrival rate is defined as  $\lambda'_k = \sum_{h_{k-1} < j \leq h_k} \lambda_j$ . Additionally, we define  $\lambda'(k_1 \rightarrow k_2) := \sum_{r=k_1}^{k_2} \lambda'_r$  for  $1 \leq k_1 \leq k_2 \leq L$ .

Given a policy  $\pi = [n_1, \dots, n_L]$  with threshold  $n_k$  for the  $k$ -th segment in Algorithm 2, we formulate a linear system analogous to Equation (12):

$$\begin{aligned} \Delta T_{[1:L]}(n_{1:L}) &< \frac{n_1}{\sum_{r=1}^L \lambda'_r}, \\ p_k &< \frac{n_k}{n_{k-1}} < 1, \quad \forall k = 2, \dots, L, \end{aligned} \tag{16}$$

where  $p_k = \frac{\lambda'(k \rightarrow L)}{\lambda'(k-1 \rightarrow L)}$  for  $k \geq 2$ . The strict first inequality gives negative drift for the first segment's entry queue, which is needed for the  $O((\zeta T)^{-1})$  throughput gap and  $O(1)$  service-normalized delay in Theorem 4. Let  $r_k = l'_{h_k}$  be the decode boundary of segment  $k$ , with  $r_0 = 0$ . Define  $\delta_1 = r_1 + 1$ ,  $\bar{s}_1 = r_1/2$ , and for  $k \geq 2$ ,  $\delta_k = r_k - r_{k-1}$  and  $\bar{s}_k = (r_{k-1} + 1 + r_k)/2$ . Then  $\Delta T_{[1,\dots,L]}(n_1, \dots, n_L) = d_0 + d_1 M^\pi(n_1, \dots, n_L)$ , where the base memory used by threshold-sized segment batches is

$$M^\pi(n_1, \dots, n_L) = \sum_{k=1}^L n_k (l + \bar{s}_k) \delta_k.$$

For heterogeneous prefill lengths, the segment logic is unchanged because boundaries are determined by decode progress. A sufficient memory condition replaces the common  $l$  in each segment by a segment-wise upper bound  $\bar{l}_k = \max\{l_j : \text{output-length class } j \text{ can enter segment } k\}$ , or by the corresponding class-weighted term when the output-length mix within the segment is known.

The segment-wise design achieves two practical goals: (i) **robustness** to varying or uncertain type distributions—if actual output lengths differ slightly from predictions, grouping into segments absorbs the variation; and (ii) **simplicity** in deployment—the scheduler maintains thresholds for a small number of decode-stage segments rather than for every possible output length. The theoretical guarantee shows that this simplification preserves the same asymptotic throughput rate, with the same logarithmic safety-buffer structure.

**THEOREM 4.** *For any fixed number of segments  $L$  and thresholds  $\pi = [n_1, \dots, n_L]$  satisfying Equation (16), if the physical memory capacity  $C$  satisfies:*

$$\begin{aligned} C &\geq M_{\text{req},L}^{(\zeta,\pi)} := M^\pi + \sum_{k=2}^L (l + r_{k-1})n_k \\ &\quad + \sum_{k=2}^L (l + r_{k-1})\theta_k^{-1} \ln \left( \frac{m(1 + \zeta T/d_0)}{\delta} \right) \\ &= O \left( M^\pi + \sum_{k=2}^L (l + r_{k-1})\theta_k^{-1} \ln \left( \frac{m(1 + \zeta T/d_0)}{\delta} \right) \right), \end{aligned} \tag{17}$$

where  $\theta_k > 0$  is the unique solution of  $e^{-\theta_k n_k} (1 - p_k + p_k e^{\theta_k})^{n_{k-1}} = 1$ , with lower bound  $8(n_k - n_{k-1}p_k)/n_{k-1}$  for  $k \geq 2$ . The logarithm uses  $m$  as a conservative upper bound on the number of segment boundaries, since  $L \leq m$ . For the corresponding no-overflow Nested WAIT dynamics, the performance satisfies:

$$\text{Throughput}^* - \mathbb{E} [\text{Throughput}^{(\zeta,\pi)}] = O \left( (\zeta T)^{-1} \right),$$

$$\mathbb{E} [\text{Latency}^{(\zeta,\pi)}] = O(1),$$

$$\mathbb{E} [\text{TTFT}^{(\zeta,\pi)}] = O(1).$$

Moreover, the no-overflow dynamics are feasible on the physical GPU, i.e., memory is not exceeded, with probability at least  $1 - \delta$  over the time horizon  $t \in [0, T]$ .

*Proof of Theorem 4.* The proof applies the Nested WAIT argument in Appendix E to aggregated segment classes. The only changes are the segment arrival rates and the segment boundaries; the boundary-thinning domination, Lindley bound for downstream boundary queues, and high-probability martingale buffer are the same as in Theorem 2. Segment  $k$  contains original output-length classes with aggregate rate  $\lambda'_k = \sum_{h_{k-1} < j \leq h_k} \lambda_j$ , and the rate of prompts that can reach segment  $k$  is  $\lambda'(k \rightarrow L) = \sum_{r=k}^L \lambda'_r$ . Hence, conditional on a threshold-sized batch crossing boundary  $r_{k-1}$ ,

the number of prompts that continue to segment  $k$  is dominated by a binomial random variable with parameters  $n_{k-1}$  and  $p_k = \lambda'(k \rightarrow L)/\lambda'(k-1 \rightarrow L)$ , exactly as in the boundary-thinning step of Theorem 2. The threshold conditions in (16) give negative drift for the Segment 1 entry queue and for every downstream boundary queue. Therefore the same Lindley-recursion bounds give uniformly bounded expected entry and boundary queues. The terminal-work and time-average queue conversions used in Appendix E then yield the  $O((\zeta T)^{-1})$  normalized throughput gap and  $O(1)$  service-normalized latency and TTFT bounds.

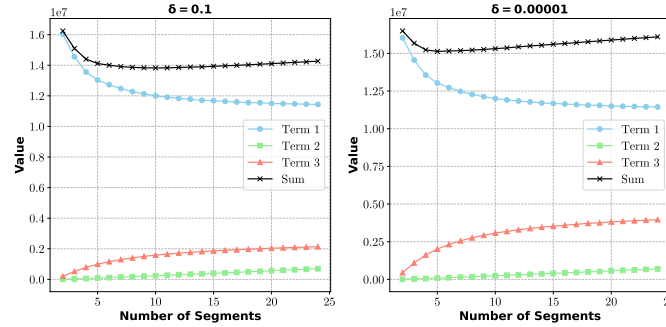
It remains to check memory. The base term  $M^\pi = \sum_{k=1}^L n_k(l + \bar{s}_k)\delta_k$  charges the threshold-controlled interior stages of each segment. The resident boundary queue at  $r_{k-1}$ ,  $k \geq 2$ , is treated separately, with deterministic component  $n_k(l + r_{k-1})$  and stochastic carryover controlled by the exponential-martingale bound used in Appendix G.5. Applying that bound over at most  $1 + \zeta T/d_0$  processing opportunities and union bounding over downstream segment boundaries gives the logarithmic buffer in (17). Since  $L \leq m$ , using  $m$  in the logarithm is conservative. On this high-probability event, the no-overflow Nested WAIT dynamics are feasible whenever  $C \geq M_{\text{req},L}^{(\zeta,\pi)}$ , completing the proof.

We validate the segment-wise design numerically using a real-data workload derived from the `lmsys-chat-1m` dataset Zheng et al. (2023) (Section 6). Decode lengths range up to 500 tokens in this workload, and the segment arrival rates are proportional to the empirical frequencies of requests whose output lengths fall in the corresponding decode-stage ranges.

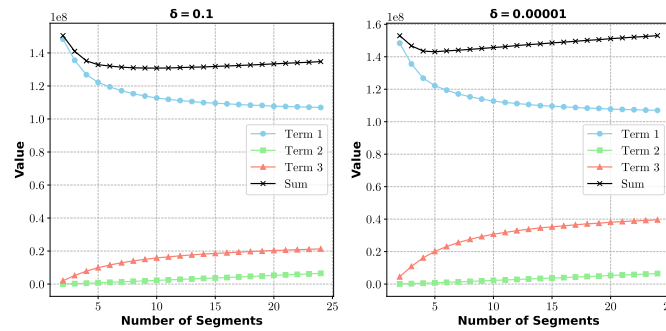
We analyze the three components of the memory bound in Equation (17): (1) the base batch memory  $M^\pi = \sum_{k=1}^L n_k(l + \bar{s}_k)\delta_k$ , (2) the deterministic boundary-queue term  $\sum_{k=2}^L n_k(l + r_{k-1})$ , and (3) the high-probability boundary-queue term  $\sum_{k=2}^L \theta_k^{-1}(l + r_{k-1}) \ln\left(\frac{m(1+\zeta T/d_0)}{\delta}\right)$ .

Figures 13, 14, and 15 plot these three memory components (y-axis) as functions of the number of segments  $L$  (x-axis), under varying arrival rates, time horizons  $T$ , and confidence levels  $\delta$ . Each panel shows: Term 1 (blue circles) representing the base threshold-batch memory  $M^\pi$ , Term 2 (green squares) representing the deterministic queue bound, Term 3 (red triangles) representing the stochastic safety buffer with logarithmic dependence on  $T$  and  $\delta$ , and their sum (black crosses) representing the total required memory  $M_{\text{req},L}^{(\zeta,\pi)}$ . Three key observations emerge: (i) Term 1 dominates Terms 2 and 3 across all parameter settings, confirming that the safety buffer remains modest relative to the base batching requirement. (ii) The total required memory (black line) follows a **U-shaped**

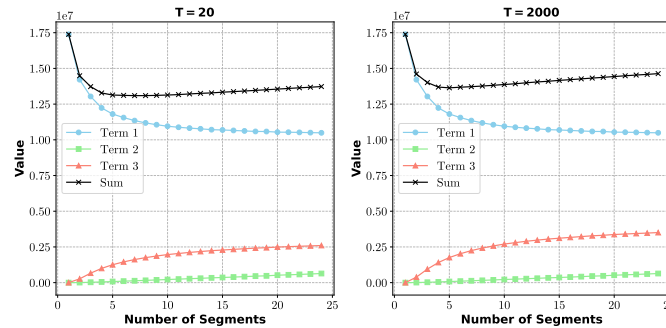
**curve** with respect to  $L$ : too few segments waste memory by over-provisioning for heterogeneous types (high Term 1 at small  $L$ ), while too many segments increase overhead from maintaining separate queues. (iii) Term 3 exhibits smooth growth as  $\delta$  decreases (comparing left vs. right panels in Figures 13 and 14) and as  $T$  increases (comparing Figure 15), consistent with the logarithmic bound in the theoretical analysis.



**Figure 13** Memory usage proportions under low arrival rate (total rate = 50,  $T = 200$ ). Left:  $\delta = 0.1$ ; Right:  $\delta = 10^{-5}$ .



**Figure 14** Memory usage proportions under high arrival rate (total rate = 500,  $T = 200$ ). Left:  $\delta = 0.1$ ; Right:  $\delta = 10^{-5}$ .



**Figure 15** Memory usage proportions with varying time horizons (total rate = 50,  $\delta = 0.001$ ). Left:  $T = 20$ ; Right:  $T = 2000$ .

These extensions demonstrate that the core threshold-based design, eviction prevention through controlled admission, remains robust to realistic deployment conditions. Time-varying arrival rates and segment-wise grouping introduce no fundamental barrier; the coupling-based analysis extends naturally with logarithmic overhead. Section 6 illustrates this implementation on a real workload, where Nested WAIT uses coarser decode-stage segments rather than maintaining a separate threshold for every possible output length.

## Appendix B: Notation Summary

**Table 3** Summary of notation

| Symbol                          | Description                                                                         |
|---------------------------------|-------------------------------------------------------------------------------------|
| $m$                             | Number of prompt types                                                              |
| $j$                             | Index of prompt type, $j \in \{1, \dots, m\}$                                       |
| $\lambda_j$                     | Arrival rate of type- $j$ prompts                                                   |
| $l_j$                           | Input length of type- $j$ prompts after prefill                                     |
| $l'_j$                          | Output length of type- $j$ prompts in the decode phase                              |
| $s$                             | Stage index: $s = 0$ for prefill and $s \in \{1, \dots, l'_j\}$ for decode          |
| $k$                             | Segment index in Nested WAIT                                                        |
| $b$                             | Batch index                                                                         |
| $n^t_{js}$                      | Number of type- $j$ prompts at stage $s$ at time $t$                                |
| $n_j$                           | Per-stage threshold for type- $j$ prompts in WAIT                                   |
| $n_k$                           | Threshold for segment $k$ in Nested WAIT                                            |
| $C$                             | GPU memory capacity                                                                 |
| $\mathcal{G}^t$                 | Set of prompts with KV cache on GPU at time $t$                                     |
| $B^t$                           | Batch of prompts processed at iteration $t$ , with $B^t \subseteq \mathcal{G}^t$    |
| $\tau$                          | Iteration time to process a batch                                                   |
| $d_0$                           | Fixed overhead time per batch iteration                                             |
| $d_1$                           | Time cost per unit of KV cache memory                                               |
| $M^*$                           | Memory requirement needed to support the fluid equilibrium                          |
| $n^*_j$                         | Equilibrium per-stage inventory of type- $j$ prompts                                |
| <b>Throughput</b> $^{(T, \pi)}$ | Effective throughput of policy $\pi$ over horizon $[0, T]$                          |
| <b>Latency</b> $^{(T, \pi)}$    | Average latency of policy $\pi$ over horizon $[0, T]$                               |
| <b>TTFT</b> $^{(T, \pi)}$       | Average time to first token of policy $\pi$ over horizon $[0, T]$                   |
| <b>Throughput</b> $^*$          | Equilibrium effective throughput, measured in completed decode tokens per unit time |
| $\Pi$                           | Class of admissible scheduling policies                                             |



## Appendix C: Proofs of Propositions

### C.1. Proof of Proposition 1

We use a service-capacity cut. The memory-dependent work required by arriving prompts already saturates the server at the stated boundary; the fixed positive overhead  $d_0$ , together with finite memory capacity, makes the service capacity strictly smaller at equality.

Consider a system with only type- $j$  prompts and finite memory capacity  $C$ . Every completed prompt must be selected once for prefill and once for each decode stage  $s = 1, \dots, l'_j$ . Across these  $l'_j + 1$  service steps, its KV-cache contribution is

$$S_j = \sum_{s=0}^{l'_j} (l_j + s) = (l'_j + 1) \left( l_j + \frac{l'_j}{2} \right).$$

Because an iteration with selected memory  $M$  has duration  $d_0 + d_1 M \geq d_1 M$ , completing  $C_j(T)$  type- $j$  prompts by time  $T$  requires at least  $d_1 S_j C_j(T)$  units of calendar time, up to the bounded contribution of prompts initially present in the system. Hence

$$C_j(T) \leq \frac{T}{d_1 S_j} + O(1).$$

This proves a linear backlog gap when  $\lambda_j d_1 S_j > 1$ .

It remains to handle the boundary case  $\lambda_j d_1 S_j = 1$ . Because the GPU memory capacity is finite, there is a finite upper bound  $B_C$  on the number of type- $j$  prompts that can be served in one iteration; for instance  $B_C \leq C/l_j$ . Therefore completing  $C_j(T)$  prompts requires at least  $C_j(T)/B_C$  nonempty iterations. Each such iteration incurs the fixed overhead  $d_0$ . Combining memory-dependent service and fixed overhead gives

$$T \geq d_1 S_j C_j(T) + d_0 \frac{C_j(T)}{B_C} - O(1),$$

and hence

$$C_j(T) \leq \frac{T + O(1)}{d_1 S_j + d_0/B_C}.$$

At the boundary,  $\lambda_j = 1/(d_1 S_j)$ , so this upper bound is strictly smaller than the expected arrivals  $\lambda_j T$  by a linear amount. Thus in both the strict and boundary cases, expected arrivals exceed possible completions by  $\Omega(T)$ , which implies  $\mathbb{E}[\mathbf{Latency}^{(T,\pi)}] = \Omega(T)$ .

## C.2. Proof of Proposition 2

The proof compares completed decode-token service with the cumulative decode-token work brought by exogenous arrivals. The fluid calculation identifies the memory requirement and batch composition under which this upper bound is attained in equilibrium.

In the fluid model, the equilibrium parameters are:

$$\Delta T = \frac{d_0}{1 - d_1 \sum_{j=1}^m \lambda_j (l'_j + 1)(l_j + l'_j/2)}, \quad n_j^* = \Delta T \cdot \lambda_j,$$

$$M^* = \Delta T \cdot \sum_{j=1}^m \lambda_j (l'_j + 1)(l_j + l'_j/2), \quad \text{Throughput}^* = \sum_{j=1}^m \lambda_j l'_j.$$

In any stochastic system, throughput is counted only after prompts complete. Over time horizon  $[0, T]$ , let  $\text{Arrivals}_j^T$  denote the total arrivals of type- $j$  prompts. The total completed output tokens cannot exceed the total output length of all arrivals:

$$\text{Throughput}^{(\pi, T)} \leq \sum_{j=1}^m \text{Arrivals}_j^T \cdot l'_j.$$

Taking expectations and dividing by  $T$ :

$$\mathbb{E}[\text{Throughput}^{(T, \pi)}] \leq \sum_{j=1}^m \mathbb{E} \left[ \frac{\text{Arrivals}_j^T}{T} \right] \cdot l'_j = \sum_{j=1}^m \lambda_j l'_j = \text{Throughput}^*.$$

## C.3. Proof of Proposition 4

We use a single-type instance at the fluid boundary. The point of the construction is to remove all type-mix complications: the only remaining source of delay is the stochastic fluctuation of a critical entry queue. This is exactly the regime in which the square-root reflected-random-walk scaling is unavoidable.

Consider one prompt type with prefill length  $l$ , decode length  $l' \geq 1$ , and arrival rate  $\lambda$ . Let

$$S = (l' + 1) \left( l + \frac{l'}{2} \right)$$

be the total KV-cache mass accumulated by one prompt over its prefill and decode stages. Choose  $d_0, d_1, \lambda$  so that the fluid equilibrium has iteration length

$$\Delta^* = \frac{d_0}{1 - d_1 \lambda S}$$

and  $n = \lambda \Delta^*$  is an integer. The fluid memory requirement is then  $M^* = \lambda \Delta^* S = nS$ . Set the physical capacity to  $C = M^*$ . At this capacity, the fluid equilibrium has room for exactly the threshold-sized batch composition with  $n$  prompts at each stage and no additional buffer.

Now consider the  $\zeta$ -scaled system over a calendar horizon  $T$ , and divide time into comparison intervals of length  $\Delta^*/\zeta$ . The number of such intervals is

$$B_\zeta = \left\lfloor \frac{\zeta T}{\Delta^*} \right\rfloor.$$

The number of external arrivals in each interval is i.i.d.  $\text{Poisson}(n)$ . Any policy whose normalized effective-throughput gap is  $o(1)$  must, up to  $o(B_\zeta)$  intervals, admit and advance work at the fluid rate. Otherwise, the missed service opportunities alone would create a nonvanishing effective-throughput gap. Since the memory capacity is exactly  $M^* = nS$ , such a policy cannot create a persistent positive service slack above this rate without exceeding the boundary memory profile. Thus the entry process contains, as a lower-bound workload, a critical bulk-service queue with arrivals  $X^b \sim \text{Poisson}(n)$  and service capacity  $n$  per comparison interval:

$$Q^{b+1} = (Q^b + X^b - n)^+, \quad Q^0 = 0.$$

This queue is the most favorable nonanticipative way to clear the entry backlog under the boundary service rate: serving less only increases delay or sacrifices throughput, while serving more persistently is not feasible at  $C = M^*$ .

The reflected random walk above has mean-zero increments  $X^b - n$  with positive finite variance. Standard results for critical reflected random walks, equivalently for the waiting time process of a critical  $GI/D/1$ -type queue, give

$$\mathbb{E}[Q^b] = \Theta(\sqrt{b}), \quad \frac{1}{B} \sum_{b=0}^{B-1} \mathbb{E}[Q^b] = \Theta(\sqrt{B}).$$

For completeness, this follows from the Lindley representation for reflected random walks ([Assmussen 2003](#), Proposition 6.3) and the random-walk maximum identity of [Strait \(1974\)](#): if  $S_k = \sum_{r=1}^k (X^r - n)$ , then the reflected workload is governed by maxima of  $S_k$ , and the central limit theorem yields  $\mathbb{E}[S_k^+] \asymp \sqrt{k}$  because the increments are nondegenerate and centered.

The time-average of  $Q^b$  lower-bounds the accumulated waiting before first service. Therefore TTFT is already at least a constant multiple of the average critical backlog in comparison-slot units; end-to-end latency is no smaller. The theorem reports service-normalized delay, and one comparison interval has scaled length  $\Delta^*$ . Since the number of arrivals and completions under any  $o(1)$ -throughput-gap policy is  $\Theta(B_\zeta)$ , the preceding time-average backlog bound implies

$$\mathbb{E}[\text{TTFT}^{(\zeta, \pi)}], \mathbb{E}[\text{Latency}^{(\zeta, \pi)}] \geq c \sqrt{B_\zeta} = \Omega((\zeta T)^{1/2})$$

for some constant  $c > 0$  depending only on the constructed instance. This proves the claimed lower bound.

### C.4. Proof of Proposition 3

We construct a two-type instance in which the fluid equilibrium fits exactly in memory but FCFS does not preserve the corresponding prefill/decode-stage composition. We partition the sample path into disjoint two-iteration blocks and compare the useful completions in each block with the fluid completion count. The key event is that FCFS admits arrivals that are feasible at the moment of admission but become infeasible, with constant probability, after they advance downstream.

Let type 1 have  $(l_1, l'_1, \lambda_1) = (1, 1, 1)$  and let type 2 have  $(l_2, l'_2, \lambda_2) = (1, 2, 1)$ . Choose  $d_1 < 1/9$  and  $d_0 = 1 - 9d_1$ . Then the fluid balance equations

$$d_0 + d_1(3n_1 + 6n_2) = n_1 = n_2$$

give  $n_1^* = n_2^* = 1$ . The associated memory requirement is

$$M^* = (1 + 2) + (1 + 2 + 3) = 9.$$

Set  $C = M^* = 9$ . The fluid batch composition therefore contains one type 1 prompt at prefill, one type 1 prompt at its only decode stage, one type 2 prompt at prefill, and one type 2 prompt at each of its two decode stages. This composition completes  $l'_1 + l'_2 = 3$  useful decode tokens per iteration.

Consider now the FCFS sample path immediately following an iteration epoch, and suppose that the next two-iteration block begins with this composition. Let  $(X_1, X_2)$  be the external arrivals during the first iteration;  $X_1$  and  $X_2$  are independent Poisson(1) random variables. Enumerating the next two FCFS state transitions, the only arrival combinations that do not force memory overflow within the block are

$$(0, 0), \quad (1, 0), \quad (0, 1), \quad (1, 1), \quad (2, 0).$$

All other combinations create too many resident prompts after the newly admitted work advances to later decode stages. Hence the overflow probability in such a block is

$$p = 1 - \frac{1}{e^2} - \frac{1}{e^2} - \frac{1}{e^2} - \frac{1}{e^2} - \frac{1}{2e^2} = 1 - \frac{9}{2e^2} > 0.$$

On this overflow event, FCFS must evict at least one resident prompt before the block can realize the fluid two-iteration completion count of  $2(l'_1 + l'_2) = 6$  useful decode tokens. The evicted prompt either has already generated decode tokens, which are discarded under the effective-throughput convention, or must restart from prefill before it can contribute completed decode tokens. Since

the constructed block has fixed length and the overflow event has probability  $p > 0$ , the conditional expected useful-completion deficit in such a block is bounded below by a positive constant.

The same block lower bound applies uniformly in a neighborhood of the fluid-boundary composition. If a block begins below the displayed composition in some prefill/decode stage, then FCFS cannot complete more useful decode work in that block than the fluid composition does. If a block begins with extra resident work, the system is no farther from overflow than in the displayed composition, so the same arrival event yields the same restart loss after a bounded adjustment for the initial excess. We can therefore discard a bounded initial transient and consider the disjoint blocks that start in this neighborhood. On each such block, conditioning on the state at the block boundary, the arrival event above has the same positive probability and creates a useful-completion deficit bounded away from zero. Consequently, the expected useful-completion deficit is bounded below by  $\epsilon$  per relevant block, for some  $\epsilon > 0$  depending only on the constructed instance. Dividing the accumulated deficit over  $\Theta(T)$  blocks by the horizon gives

$$\text{Throughput}^* - \mathbb{E}[\text{Throughput}^{(T, \text{FCFS})}] = \Omega(1).$$

### C.5. Proof of Proposition 5

We construct a two-type instance in which prompts are indistinguishable until the first decode boundary. At the known-type fluid memory level, there is no room for positive survivor-count fluctuations at that boundary. The lower bound is therefore an information constraint: before the boundary revelation, any non-predictive policy must either reserve capacity and lose a constant amount of throughput, or admit at the fluid scale and face a constant probability of overflow.

Consider two prompt types with the same prefill length and the same first decode token. Type 1 completes after the first decode token, whereas type 2 continues to one additional decode stage. Let a newly admitted prompt be type 2 with probability  $p \in (0, 1)$ , independently across prompts, and suppose the type is not revealed until the prompt either completes at the first boundary or survives into the second segment. This is the information structure faced by any non-predictive online policy.

Choose the arrival rate and the memory capacity so that, under known output lengths, the fluid equilibrium uses exactly the available memory  $C = M^*$ . In that known-type equilibrium, the downstream segment contains the mean number of survivors and there is no additional memory reserved for positive survivor-count fluctuations. Let  $\mathcal{F}$  be the sigma-field generated by the policy's observations just before a decision epoch, including past arrivals, past boundary revelations, current queues, and any internal randomization already realized by the policy. Conditional on  $\mathcal{F}$ , suppose

the policy admits  $b$  prompts whose types are still hidden. The admission decision is  $\mathcal{F}$ -measurable, while the hidden continuation labels are independent of  $\mathcal{F}$ . Therefore, conditional on  $\mathcal{F}$ , the number of admitted prompts that survive the first boundary is

$$Y_b \sim \text{Binomial}(b, p)$$

conditional on  $\mathcal{F}$ .

There are two cases. First, suppose the policy admits an unrevealed admission batch large enough to match a fixed fraction of the known-type fluid balance in the block. Then  $b$  is bounded below by a positive constant in this fixed instance, and the boundary-survivor count has a positive probability of exceeding its fluid mean allocation by at least one prompt:

$$\Pr\{Y_b > \lfloor pb \rfloor\} \geq \rho$$

for some constant  $\rho > 0$  depending only on  $p$  and the chosen instance. Since  $C = M^*$  contains no downstream safety buffer, this event forces memory overflow. Under the paper's effective-throughput convention, an overflow discards useful partial work through eviction and restart, so the expected completed decode-token service falls below the fluid benchmark by a constant amount.

Second, suppose the policy avoids this overflow event by admitting strictly fewer unrevealed prompts than required by the known-type fluid balance. Then it leaves a constant amount of admissible service capacity unused in this instance. Since the fluid benchmark **Throughput\*** equals the offered decode-token load that can be completed at the known-type equilibrium, this conservative admission again creates a constant throughput gap.

Because the argument holds after conditioning on the policy's filtration  $\mathcal{F}$ , randomized policies are covered by averaging over their internal randomization. Repeating the block construction over the horizon gives a constant expected useful-completion deficit per relevant block, up to bounded end effects. Combining the two cases gives an instance for which every non-predictive online policy incurs

$$\text{Throughput}^* - \mathbb{E}[\text{Throughput}^{(\zeta, \pi)}] = \Omega(1).$$

## Appendix D: Proof of Theorem 1

This proof establishes asymptotic optimality of the WAIT algorithm when output lengths are known. The argument proceeds in two parts:

1. **Single-type case:** We construct a coupled dominating process using the Lindley recursion and apply Kingman's bound for queue length expectations.

2. **Multiple-type case:** We construct an auxiliary embedded full-threshold process and couple it to event-driven WAIT type by type. This embedded process separates stochastic entry accumulation from the event-driven batch composition while preserving the service opportunities created by the WAIT thresholds.

Key techniques include: Lindley recursion for queue length bounds (Asmussen 2003, Proposition 6.3), coupling construction to dominate the original process, Kingman's bound for negative drift processes (Kingman 1962), and standard random walk results (Strait 1974).

Throughout this appendix, we use  $b$  to denote the index of the embedded full-threshold review epochs, which is distinct from the stage index  $s \in \{0, 1, \dots, l'_j\}$  used in the main text for decode stages. We use  $t$  for continuous time and  $B$  for the number of embedded review intervals.

We first consider the single-type case, then extend to multiple types.

**D.0.1. Single-Type Case** Consider a single prompt type with threshold  $n$ . For general scheduling algorithms, analyzing this system in continuous time is challenging: KV cache grows dynamically during decode, arrivals are stochastic, and eviction risks complicate the dynamics.

The WAIT policy's *threshold mechanism achieves a dimensionality reduction*. By waiting until exactly  $n$  prompts accumulate before initiating a batch, the algorithm enforces a fixed batch size, which ensures constant processing time  $\Delta T$  per batch. This decouples the continuous-time memory-constrained dynamics into a tractable discrete-time process: a memory-constrained queueing system becomes a simple random walk. The reduction is enabled by the algorithm design and is what makes the subsequent drift analysis apply.

Specifically, let  $\tilde{\lambda}$  denote the continuous-time arrival rate and let  $\mu = \tilde{\lambda}\Delta T$  be the expected number of arrivals during a threshold-sized batch. The boundary case is  $\mu = n$ . In this case the embedded process observed at batch-completion epochs, indexed by  $b \in \mathbb{N}$ , is a zero-drift threshold queue. The throughput loss arises from “stuck” iterations where insufficient prompts are available.

**LEMMA 1 (Queue Length and Stuck Time).** *This lemma connects queue accumulation to throughput loss through stuck iterations in the critical case  $\mu = n$ . Let  $X^b$  denote the number of*

arrivals during batch  $b$ , where  $X^b \sim \text{Poisson}(n)$ . Let  $W^b$  denote the queue length after batch  $b$ , with  $W^0 = 0$ . The state transition is:

$$W^{b+1} = W^b + X^b - n \cdot \mathbf{1}\{W^b + X^b \geq n\}.$$

Define  $B_{\text{stuck}}$  as the number of stuck iterations, where  $W^b + X^b < n$ . Then:

$$\mathbb{E}[W^B] = n \cdot \mathbb{E}[B_{\text{stuck}}].$$

The proof is in Appendix G.1. This lemma connects queue accumulation to throughput loss: stuck iterations directly translate to waiting prompts. To bound  $\mathbb{E}[W^B]$ , we construct a coupled dominating process that admits tractable analysis via classical queueing techniques.

**LEMMA 2 (Coupled Dominating Process).** *The coupled process starts with a safety buffer ( $2n$  versus  $0$ ) and maintains dominance throughout. Define  $\{\tilde{W}^b\}$  by:*

$$\tilde{W}^0 = 2n, \quad \tilde{W}^{b+1} = \max\{2n, \tilde{W}^b + X^b - n\}.$$

Then  $\tilde{W}^b \geq W^b + n$  for all  $b \in \mathbb{N}$ .

The proof is in Appendix G.2.

*Lindley recursion.* To analyze the coupled process, we apply the Lindley recursion representation (Asmussen 2003, Proposition 6.3):

**LEMMA 3 (Lindley Representation).** *Let  $S^k = \sum_{r=1}^k (X^r - n)$  be the partial sum process. Then:*

$$\tilde{W}^B = 2n + \max(S^B, S^B - S^1, \dots, S^B - S^{B-1}, 0).$$

This classical queueing representation is made applicable by the WAIT policy's threshold design. Define  $\xi^k = X^k - n$ . We use time-reversal symmetry to simplify the maximum expression:

$$\max(S^B, S^B - S^1, \dots, S^B - S^{B-1}, 0) = \max_{1 \leq k \leq B} \left( \sum_{r=k}^B \xi^r \right)^+ \stackrel{d}{=} \max_{1 \leq k \leq B} \left( \sum_{r=1}^k \eta^r \right)^+, \quad \eta^r \stackrel{d}{=} X^1 - n.$$

Thus  $\tilde{W}^B \stackrel{d}{=} 2n + \max_{1 \leq k \leq B} (\sum_{r=1}^k \eta^r)^+$ .

Since  $\tilde{W}^B \geq W^B + n$ , we have  $\mathbb{E}[W^B] \leq \mathbb{E}[\tilde{W}^B] - n$ . Let  $Y^B = \max_{1 \leq k \leq B} (\sum_{r=1}^k \eta^r)^+$ . By (Strait 1974, Lemma 2):



**LEMMA 4 (Random Walk Maximum).**

$$\mathbb{E}[Y^B] = \mathbb{E}\left[\max_{1 \leq k \leq B} (S^k)^+\right] = \sum_{k=1}^B \frac{1}{k} \mathbb{E}[(S^k)^+].$$

This standard result (Strait 1974) decomposes the maximum into a sum over batch indices. For  $\mathbb{E}[(S^k)^+]$ , the Cauchy-Schwarz inequality and  $\text{Var}(S^k) = nk$  imply  $\mathbb{E}[(S^k)^+] \leq \sqrt{nk}$ . Hence:

$$\mathbb{E}[Y^B] \leq \sqrt{n} \sum_{k=1}^B k^{-1/2} = O(\sqrt{nB}).$$

**LEMMA 5 (Throughput Gap Bound).**

$$\mathbb{E}[W^B] - n \leq \mathbb{E}[Y^B] \leq \sqrt{n} \sum_{k=1}^B \frac{1}{\sqrt{k}} = O(\sqrt{nB}).$$

By Lemma 1, the expected number of stuck iterations is  $\mathbb{E}[B_{\text{stuck}}] = \mathbb{E}[W^B]/n = O(\sqrt{B})$ , so the normalized throughput loss from stuck iterations is  $O(B^{-1/2})$ . The same bound applies to every prefix  $b \leq B$ , and therefore

$$\frac{1}{B} \sum_{b=1}^B \mathbb{E}[W^b] \leq \frac{1}{B} \sum_{b=1}^B O(\sqrt{b}) = O(\sqrt{B}).$$

This time-averaged backlog bound is the quantity used for the latency and TTFT estimates.

*Scaling with the horizon.* Let  $\Delta^\pi$  denote the fixed processing time of a threshold-sized batch under policy  $\pi$ . In the scaled system, the corresponding processing time is  $\Delta^\pi/\zeta$ . Over a calendar horizon  $T$ , set

$$B_\zeta = \left\lceil \frac{\zeta T}{\Delta^\pi} \right\rceil.$$

Then  $B_\zeta = \zeta T/\Delta^\pi + O(1)$ . The terminal backlog bound gives normalized throughput loss  $O(B_\zeta^{-1/2}) = O((\zeta T)^{-1/2})$ . For delay, the theorem reports service-normalized time, so one scaled threshold-batch interval has length  $\Delta^\pi$ ; the time-averaged backlog bound therefore gives latency and TTFT contributions  $O(B_\zeta^{1/2}) = O((\zeta T)^{1/2})$ . The bounded number of service stages is lower order. This yields the single-type throughput, latency, and TTFT orders stated in the theorem.

**D.0.2. Multiple-Type Case** We now extend to  $m$  known output-length classes. The additional issue is that event-driven WAIT need not process the full class composition in every realized batch: if only a subset of class queues has reached its threshold, only that subset is batched. The proof handles this by introducing an auxiliary embedded full-threshold process observed at deterministic review epochs. The actual policy is the event-driven WAIT policy from the main text, which batches all currently eligible types whenever the server becomes idle. The embedded process is not used to compare the total number of realized batches or the total fixed overhead paid by the two systems. Its role is to put each type- $j$  threshold queue on a common full-threshold service scale, obtain a Lindley-type recursion, and then transfer the resulting bounds back to the actual policy through a sample path coupling of service opportunities.

For a threshold vector  $\pi = [n_1, \dots, n_m]$ , let

$$\Delta^\pi \equiv \Delta T_{[1, \dots, m]} = d_0 + d_1 M^\pi, \quad M^\pi = \sum_{j=1}^m n_j (l'_j + 1) \left( l_j + \frac{l'_j}{2} \right). \quad (18)$$

This is the processing time of the full threshold composition, i.e., the batch that contains  $n_j$  prompts from every decode stage of every type  $j$ . The load-balance and memory conditions are

$$\begin{aligned} \lambda_j \Delta^\pi &\leq n_j \quad \text{for all } j \in [m], \\ M^* &\leq M^\pi \leq C. \end{aligned} \quad (19)$$

The first line says that, over one full-threshold iteration, the expected type- $j$  arrivals do not exceed the  $n_j$  type- $j$  prompts that can be advanced from each active stage. The second line is the corresponding feasibility condition: the threshold composition is at least on the scale of the fluid memory requirement but does not exceed the physical memory  $C$ .

**LEMMA 6 (GPU-resident memory invariant).** *Starting from the empty system, WAIT maintains*

$$N_{js}(t) \leq n_j, \quad j \in [m], s = 1, \dots, l'_j,$$

where  $N_{js}(t)$  is the number of resident type- $j$  prompts at decode stage  $s$ . Stage-0 prompts that have not been selected for prefill have no resident KV cache.

The proof is by induction over batch completions. If type  $j$  is not served in a realized batch, its GPU-resident prompts at decode stages are unchanged. If type  $j$  is served, the invariant before the batch implies that each decode stage contains at most  $n_j$  prompts, so WAIT selects all prompts

currently present in that stage. After the batch completes, stage  $s + 1$  contains only the prompts advanced from stage  $s$ , again at most  $n_j$ . Thus no decode stage can exceed  $n_j$  resident prompts.

The invariant gives the memory-feasibility statement needed by the policy. If  $a_{j0}(t) \leq n_j$  denotes the number of type- $j$  prompts selected for prefill in the current batch, then at any decision epoch,

$$\sum_{j=1}^m \sum_{s=1}^{l'_j} N_{js}(t)(l_j + s) + \sum_{j=1}^m a_{j0}(t)l_j \leq \sum_{j=1}^m n_j \sum_{s=0}^{l'_j} (l_j + s) = M^\pi \leq C.$$

Thus unselected resident prompts are included in the memory accounting, while unselected stage-0 prompts remain outside the GPU KV-cache state.

For any realized event-driven WAIT batch  $r$ , let  $a_{js}^{(r)} \leq n_j$  be the number of selected type- $j$  prompts at decode stage  $s$ , and let  $J_r$  be the set of selected types. The selected batch composition has size

$$M_r = \sum_{j \in J_r} \sum_{s=0}^{l'_j} a_{js}^{(r)}(l_j + s) \leq \sum_{j=1}^m n_j \sum_{s=0}^{l'_j} (l_j + s) = M^\pi.$$

Therefore the physical processing time of the realized batch in the  $\zeta$ -scaled system is

$$S_r^{(\zeta)} = \frac{d_0 + d_1 M_r}{\zeta} \leq \frac{\Delta^\pi}{\zeta}.$$

This bound concerns processing time, while Lemma 6 gives memory feasibility. Actual batches may contain fewer types and have shorter duration, but no realized WAIT batch is larger than the full-threshold composition. The fixed overhead  $d_0$  is therefore handled through a calendar-time coupling of service opportunities, not by matching realized batches one for one. If a type is absent from a realized batch, then either its entry queue has not yet reached threshold, or the threshold is reached while the server is processing another batch. In the latter case the waiting time until the next review is bounded by the remaining processing time of that in-service batch, which is at most one full-threshold interval.

In the  $\zeta$ -scaled system, one full-threshold review interval has physical length  $\Delta^\pi/\zeta$ . Define deterministic review epochs

$$u_b = b\Delta^\pi/\zeta, \quad b = 0, 1, 2, \dots$$

The embedded process reviews the entry queues only at these epochs. We use an end-of-interval accounting convention: arrivals accumulate over  $(u_b, u_{b+1}]$ , and at  $u_{b+1}$  the embedded process

removes one type- $j$  entry batch of size  $n_j$  whenever the type- $j$  entry queue has reached  $n_j$ ; it also advances up to  $n_j$  resident type- $j$  prompts from each active decode stage. Equivalently, one can view the selected work as occupying the following full review interval. A batch removed at review  $u_{b+1}$  can therefore shift completion accounting by at most  $l'_j + 1$  embedded intervals for type  $j$ , which contributes only  $O((\zeta T)^{-1})$  to normalized throughput and  $O(1)$  to service-normalized average delay. The full-threshold interval length is feasible by the resident memory invariant above, and it is deliberately chosen as the service scale at which the type-wise threshold queues have Lindley recursions.

Throughout this argument,  $b$  indexes embedded full-threshold review intervals, not realized event-driven WAIT batches.

**LEMMA 7 (Type-wise sample path coupling of service opportunities).** *Couple actual event-driven WAIT and the auxiliary embedded full-threshold process on the same arrival sample path, both starting empty. Let  $L_\zeta = \Delta^\pi / \zeta$  and  $u_b = bL_\zeta$ . For type  $j$ , let  $R_j^\pi(t)$  be the cumulative number of type- $j$  threshold batches of size  $n_j$  started by actual WAIT by time  $t$ , and let  $R_j^{\bar{\pi}}(b)$  be the cumulative number of type- $j$  batches removed from the embedded entry queue at reviews  $u_1, \dots, u_b$ . Then*

$$R_j^\pi(u_{b+1}) \geq R_j^{\bar{\pi}}(b), \quad b \geq 0, j \in [m].$$

*The same one-interval delayed dominance holds for type- $j$  stage advancements.*

To see this, fix a type  $j$  and index its threshold batches in order of eligibility. Let  $\tau_{jq}$  be the first time at which the  $q$ -th type- $j$  entry batch of size  $n_j$  is present in the actual entry queue, and let  $\sigma_{jq}$  be the time at which actual WAIT starts that batch. Since actual WAIT batches all eligible types whenever the server is idle, the batch can wait after  $\tau_{jq}$  only if another batch is already in service. Every such in-service batch has duration at most  $L_\zeta$  by the bound above. Hence, on every sample path,

$$\sigma_{jq} \leq \tau_{jq} + L_\zeta.$$

Now suppose the embedded process has removed  $q$  type- $j$  batches by review epoch  $u_b$ . Because the two systems see the same type- $j$  arrival process, the  $q$ -th actual type- $j$  threshold batch has become eligible by time  $u_b$ , so  $\tau_{jq} \leq u_b$ . The preceding inequality gives  $\sigma_{jq} \leq u_b + L_\zeta = u_{b+1}$ . Therefore actual WAIT has started at least as many type- $j$  threshold batches by  $u_{b+1}$  as the embedded process

has removed by  $u_b$ , which proves the stated dominance. This argument is in calendar time and includes the fixed overhead paid by any subset batch.

The stage-advancement statement follows from the same sample path coupling of service opportunities. Use the number of type- $j$  service opportunities as the induction variable. After  $r$  such opportunities, any type- $j$  batch that has entered service has advanced at least  $\min\{r, l'_j\}$  decode stages, up to the same one-interval lag. Since a type- $j$  service opportunity advances a threshold batch from every active type- $j$  stage, the entry dominance propagates stage by stage and gives the stated dominance for completions.

Lemma 7 is the structural step that transfers the embedded-process queue bounds back to event-driven WAIT. It does not require the actual system to use fewer batches than the embedded process. Actual WAIT may realize different subset compositions across time, but for each type the threshold batch removed at an embedded review epoch has a corresponding actual service opportunity within one full-threshold review interval. Thus it is enough to bound the embedded comparison queues: the one-interval lag and the bounded in-service pipeline contribute only  $O((\zeta T)^{-1})$  to normalized throughput and  $O(1)$  to the service-normalized delay metrics.

For the embedded process, let  $\bar{W}_{(j)}^b$  be the residual type- $j$  entry queue after the  $b$ -th review, and let

$$\bar{X}_{(j)}^b = A_j(u_b, u_{b+1}] \sim \text{Poisson}\left(\zeta \lambda_j \frac{\Delta^\pi}{\zeta}\right) = \text{Poisson}(\lambda_j \Delta^\pi).$$

The arrivals  $\{\bar{X}_{(j)}^b\}$  are independent across embedded intervals and do not depend on other type queues. With the end-of-interval accounting convention above, the residual entry queue evolves as

$$\bar{W}_{(j)}^{b+1} = \bar{W}_{(j)}^b + \bar{X}_{(j)}^b - n_j \mathbf{1}\{\bar{W}_{(j)}^b + \bar{X}_{(j)}^b \geq n_j\}.$$

This is the point at which the batch-composition issue is reduced to a stochastic comparison: the only coupling across types is the deterministic feasibility condition  $M^\pi \leq C$ , while the embedded process gives each type its own threshold queue.

**LEMMA 8 (Embedded comparison queue bound).** *For each type  $j$ , define*

$$\tilde{W}_{(j)}^0 = 2n_j, \quad \tilde{W}_{(j)}^{b+1} = \max\{2n_j, \tilde{W}_{(j)}^b + \bar{X}_{(j)}^b - n_j\}.$$

*Then  $\tilde{W}_{(j)}^b \geq \bar{W}_{(j)}^b$  for all  $b \geq 0$  and  $j \in [m]$ .*

The proof is in Appendix G.3. Let  $\mu_j = \lambda_j \Delta^\pi$ . The increment in the dominating process is  $\bar{X}_{(j)}^b - n_j$ , whose mean is  $\mu_j - n_j \leq 0$  by (19). If  $\mu_j = n_j$ , the same random-walk maximum argument used in the single-type proof gives

$$\mathbb{E}[\bar{W}_{(j)}^B] \leq \mathbb{E}[\bar{W}_{(j)}^B] = O(B^{1/2}).$$

The same bound holds for every prefix  $b \leq B$ , so

$$\frac{1}{B} \sum_{b=0}^{B-1} \mathbb{E}[\bar{W}_{(j)}^b] = O(B^{1/2}).$$

If  $\mu_j < n_j$ , the dominating process has strictly negative drift. Since

$$\left| \mathbb{E}[\bar{X}_{(j)}^b - n_j] \right| = n_j - \mu_j,$$

by Kingman's bound (Kingman 1962):

$$\sup_{B \geq 0} \mathbb{E}[\bar{W}_{(j)}^B] \leq 2n_j + \frac{\mathbb{E}[(\bar{X}_{(j)}^b - n_j)^2]}{2(n_j - \mu_j)} < \infty.$$

Combining the zero-drift and negative-drift cases, under the nonpositive drift condition in (19),

$$\sum_{j=1}^m \mathbb{E}[\bar{W}_{(j)}^B] = O(B^{1/2}), \quad \frac{1}{B} \sum_{b=0}^{B-1} \sum_{j=1}^m \mathbb{E}[\bar{W}_{(j)}^b] = O(B^{1/2}).$$

Under strict slack  $\mu_j < n_j$  for all  $j$ , both quantities are  $O(1)$ . If equality holds for at least one type and strict inequality holds for the others, the zero-drift types dominate the aggregate bounds, so the first part of the theorem applies. Types with zero arrival rates can be omitted from  $[m]$ .

It remains to translate these  $B$ -interval bounds into the theorem's finite-horizon metrics. The number of completed embedded full-threshold intervals in the physical horizon  $[0, T]$  is exactly

$$B_\zeta(T) = \max\{b : u_b \leq T\} = \left\lfloor \frac{\zeta T}{\Delta^\pi} \right\rfloor,$$

and the residual time is less than one embedded full-threshold interval. This is the number of deterministic review intervals in the embedded process, not the number of actual event-driven WAIT batches. Since  $B_\zeta = \zeta T / \Delta^\pi + O(1)$ ,  $B_\zeta^{-1/2} = O((\zeta T)^{-1/2})$  and  $B_\zeta^{-1} = O((\zeta T)^{-1})$ .

For throughput, terminal entry backlog controls the completion shortfall. More precisely, for the embedded process,

$$\text{Throughput}^* - \mathbb{E}[\text{Throughput}^{(\zeta, \bar{\pi})}] \leq \frac{1}{\zeta T} \sum_{j=1}^m l'_j \mathbb{E} \left[ \bar{W}_{(j)}^{B_\zeta} + A_j(u_{B_\zeta}, T) + (l'_j + 2)n_j \right].$$

The second term accounts for final partial-interval arrivals, and the last term absorbs the bounded in-service pipeline, startup effects, and the one-interval lag from Lemma 7. Since  $T - u_{B_\zeta} < \Delta^\pi / \zeta$ ,  $\mathbb{E}[A_j(u_{B_\zeta}, T)] \leq \lambda_j \Delta^\pi = O(1)$ . Thus the terminal-backlog term gives an  $O((\zeta T)^{-1/2})$  gap under nonpositive drift and an  $O((\zeta T)^{-1})$  gap under strict slack.

For delay, terminal backlog is not the right object; the relevant quantity is the time average of the entry queues. Let  $\bar{Q}^b = \sum_{j=1}^m \bar{W}_{(j)}^b$ . In physical time, each embedded full-threshold interval has length  $\Delta^\pi / \zeta$ . The theorem, however, reports service-normalized delay, multiplying elapsed physical time by  $\zeta$ . Thus each embedded interval contributes a constant scaled time  $\Delta^\pi$ . The entry-wait integral in service-normalized units is bounded by

$$\Delta^\pi \sum_{b=0}^{B_\zeta-1} \bar{Q}^b + O(B_\zeta),$$

where the  $O(B_\zeta)$  term accounts for the embedded-interval convention and bounded decode-stage waiting. The expected number of completed prompts is  $\Theta(B_\zeta)$ : cumulative threshold-batch removals equal cumulative arrivals minus terminal backlog, and the missed removals are  $O(B_\zeta^{1/2})$  at the boundary and  $O(1)$  under strict slack. Therefore

$$\mathbb{E}[\mathbf{Latency}^{(\zeta, \bar{\pi})}], \mathbb{E}[\mathbf{TTFT}^{(\zeta, \bar{\pi})}] \leq O\left(\frac{1}{B_\zeta} \sum_{b=0}^{B_\zeta-1} \mathbb{E}[\bar{Q}^b]\right) + O(1).$$

The prefix bound gives

$$\mathbb{E}[\mathbf{Latency}^{(\zeta, \bar{\pi})}], \mathbb{E}[\mathbf{TTFT}^{(\zeta, \bar{\pi})}] = O(B_\zeta^{1/2}) = O((\zeta T)^{1/2})$$

under nonpositive drift. The corresponding calendar-time delay is smaller by a factor  $1/\zeta$ . Under strict slack, the same service-normalized expression is  $O(1)$ . Lemma 7 transfers these throughput, latency, and TTFT bounds from  $\bar{\pi}$  to the actual event-driven WAIT policy, completing the proof of Theorem 1.

## Appendix E: Proof of Theorem 2

The proof analyzes the no-overflow Nested WAIT dynamics and then shows that the memory condition makes those dynamics feasible with high probability. The main difference from WAIT is that downstream arrivals are generated endogenously: a prompt reaches segment  $k$  only if it survives the boundary  $l'_{k-1}$ . The proof therefore follows the queues at these segment boundaries. Segment

1 is a threshold-renewal queue with external Poisson arrivals. For  $k \geq 2$ , each upstream boundary-crossing opportunity produces a binomially thinned boundary batch. The threshold conditions give negative drift for these embedded queues. Standard reflected-random-walk and exponential-martingale bounds then control their expected size and high-probability excursions; the nonstandard part is the memory accounting that separates fresh boundary arrivals from carryover residual queues.

Let

$$\Lambda_k = \sum_{j=k}^m \lambda_j, \quad p_k = \frac{\Lambda_k}{\Lambda_{k-1}}, \quad k \geq 2,$$

and write

$$\Delta^\pi = \Delta T_{[1, \dots, m]}(n_1, \dots, n_m) = d_0 + d_1 M^\pi.$$

The theorem assumes  $\Lambda_1 \Delta^\pi < n_1$  and  $p_k < n_k / n_{k-1} < 1$  for  $k \geq 2$ . In the  $\zeta$ -scaled system, every realized iteration has physical duration at least  $d_0 / \zeta$ , so the number of realized processing opportunities over  $[0, T]$  is at most

$$B_\zeta^{\max} = \left\lceil \frac{\zeta T}{d_0} \right\rceil \leq 1 + \frac{\zeta T}{d_0}.$$

This is an upper bound on actual event opportunities, not a full-threshold comparison-slot count.

### E.1. Segment-1 Entry Queue

Prompts waiting to enter segment 1 have not been prefilled, so they are outside the GPU-resident KV-cache state. The only subtlety is that the server may idle while waiting for the segment-1 threshold. We therefore index the segment-1 queue by threshold renewals rather than by realized processing iterations. Let  $R_q$  be the residual number of not-yet-prefilled prompts immediately after the  $q$ -th segment-1 service start, after  $n_1$  prompts have been removed for prefill. During the following service period, the number of external arrivals is stochastically dominated by

$$\bar{Y}_{(1)}^q \sim \text{Poisson}(\Lambda_1 \Delta^\pi),$$

because the physical service period is at most  $\Delta^\pi / \zeta$  and arrivals occur at rate  $\zeta \Lambda_1$ . If the residual plus these arrivals is below  $n_1$ , the system idles until the next arrivals complete the threshold and then immediately starts the next segment-1 service. Thus

$$R_{q+1} \leq (R_q + \bar{Y}_{(1)}^q - n_1)^+.$$



This recursion is dominated by the reflected random walk

$$\tilde{R}_{(1)}^0 = 2n_1, \quad \tilde{R}_{(1)}^{q+1} = \max\{2n_1, \tilde{R}_{(1)}^q + \tilde{Y}_{(1)}^q - n_1\}.$$

Since  $\mathbb{E}[\tilde{Y}_{(1)}^q] - n_1 = \Lambda_1 \Delta^\pi - n_1 < 0$ , the standard Lindley-recursion bound for a random walk with negative drift (see, e.g., [Kingman 1962](#), [Asmussen 2003](#)) gives

$$\sup_{q \geq 0} \mathbb{E}[R_q] \leq 2n_1 + \frac{\mathbb{E}[(\tilde{Y}_{(1)}^q - n_1)^2]}{2(n_1 - \Lambda_1 \Delta^\pi)} < \infty.$$

The idle time before the next segment-1 service is the time needed to collect fewer than  $n_1$  arrivals, whose expected physical length is  $O(1/\zeta)$ ; in service-normalized time this contributes  $O(1)$ .

## E.2. Boundary Queues for Downstream Segments

For  $k \geq 2$ , consider an opportunity at which segment  $k - 1$  advances prompts across boundary  $l'_{k-1}$ . Let  $A_{k-1}^r \leq n_{k-1}$  be the number of prompts that cross this boundary at the  $r$ -th such opportunity. Conditional on the filtration just before output-length revelation at the boundary, the indicators that these prompts continue to segment  $k$  are independent Bernoulli random variables with probability  $p_k = \Lambda_k / \Lambda_{k-1}$ . This uses the i.i.d. output-length mix and the fact that service before the boundary is nonanticipating with respect to future output length; before a boundary is reached, prompts with possible remaining output-length classes are exchangeable within the surviving mixture. Therefore

$$Y_{(k)}^r \mid A_{k-1}^r \sim \text{Binomial}(A_{k-1}^r, p_k), \quad Y_{(k)}^r \preceq \tilde{Y}_{(k)}^r,$$

where the dominating variables  $\tilde{Y}_{(k)}^r \sim \text{Binomial}(n_{k-1}, p_k)$  can be chosen independent across boundary opportunities.

Let  $V_{(k)}^r$  be the carryover boundary queue at stage  $l'_{k-1}$  after the threshold review at the start of the  $r$ -th opportunity in which segment  $k - 1$  is processed. Thus  $V_{(k)}^r$  is a post-review residual, not the pre-review boundary population. This convention preserves the nested, prefix structure of the algorithm. Whenever segment  $k - 1$  is active, all earlier segment thresholds have been met. If the boundary queue for segment  $k$  also contains at least  $n_k$  prompts at that review, Nested WAIT includes segment  $k$  in the same prefix batch and removes a threshold batch from the boundary. During the batch, at most  $A_{k-1}^r \leq n_{k-1}$  additional prompts may survive segment  $k - 1$  and join the boundary. These fresh survivors are part of the just-processed upstream threshold batch, so their KV-cache memory is charged to the threshold-sized batch term  $M^\pi$ . Only the carryover residual

remains as additional boundary memory. With the binomial domination above, the pre-review queue for the next opportunity is  $V_{(k)}^r + \bar{Y}_{(k)}^r$ , and the post-review residual satisfies

$$V_{(k)}^{r+1} \leq V_{(k)}^r + \bar{Y}_{(k)}^r - n_k \mathbf{1}\{V_{(k)}^r + \bar{Y}_{(k)}^r \geq n_k\}.$$

The residual queue is dominated by

$$\tilde{W}_{(k)}^0 = n_k, \quad \tilde{W}_{(k)}^{r+1} = \max\{n_k, \tilde{W}_{(k)}^r + \bar{Y}_{(k)}^r - n_k\}.$$

This domination follows by induction over boundary-crossing opportunities. If  $V_{(k)}^r + \bar{Y}_{(k)}^r \geq n_k$ , the post-review residual is at most  $V_{(k)}^r + \bar{Y}_{(k)}^r - n_k$ , which is bounded by  $\tilde{W}_{(k)}^r + \bar{Y}_{(k)}^r - n_k$ . If  $V_{(k)}^r + \bar{Y}_{(k)}^r < n_k$ , the residual is below  $n_k$ , while the dominating recursion is always at least  $n_k$ . Hence  $V_{(k)}^r \leq \tilde{W}_{(k)}^r$  for all  $r$ .

The increments  $\bar{Y}_{(k)}^r - n_k$  have strictly negative drift because

$$\mathbb{E}[\bar{Y}_{(k)}^r - n_k] = n_{k-1}p_k - n_k < 0.$$

Applying the same negative-drift reflected-random-walk bound yields, uniformly in  $r$ ,

$$\mathbb{E}[V_{(k)}^r] \leq n_k + \frac{\mathbb{E}[(\bar{Y}_{(k)}^r - n_k)^2]}{2(n_k - n_{k-1}p_k)} < \infty, \quad k = 2, \dots, m.$$

Thus all external and downstream boundary queues have bounded expected size, uniformly over  $\zeta$  and  $T$ .

We also use the stage-cap invariant induced by the threshold rule. Starting from the empty system, every threshold-controlled interior stage of segment  $k$  contains at most  $n_k$  resident prompts. A batch that includes segment  $k$  advances at most  $n_k$  prompts from each preceding stage into the next stage; a batch that excludes segment  $k$  leaves its interior stages unchanged. The only resident queues not covered by this interior-stage cap are the downstream boundary queues  $V_{(k)}$ ,  $k \geq 2$ , which are controlled by the residual-queue bounds above.

### E.3. Throughput and Delay

We first translate the queue bounds into the no-overflow performance estimates. Let  $W_{(1)}(T)$  denote the segment-1 entry residual at time  $T$ , let  $V_{(k)}(T)$  be the downstream resident boundary queue for  $k \geq 2$ , and let  $I_k(T)$  be the number of prompts inside the threshold-controlled interior of segment  $k$ . The threshold rule gives  $I_k(T) \leq n_k \delta_k$ , where  $\delta_k$  is the number of interior stages in segment  $k$ .

There are constants  $\alpha_k, \beta_k$ , depending only on the fixed output lengths, such that terminal unfinished decode work is bounded by

$$\sum_{k=1}^m \alpha_k W_{(k)}(T) + \sum_{k=1}^m \beta_k I_k(T) + A_{\text{tail}}(T),$$

where  $W_{(k)} = V_{(k)}$  for  $k \geq 2$ , and  $A_{\text{tail}}(T)$  accounts for arrivals in the final partial opportunity. The expected final-partial-opportunity arrivals are  $O(1)$  because that opportunity has physical length at most  $\Delta^\pi/\zeta$ . Consequently,

$$\begin{aligned} \text{Throughput}^* - \mathbb{E}[\text{Throughput}^{(\zeta, \pi)}] &\leq \frac{1}{\zeta T} \mathbb{E} \left[ \sum_{k=1}^m \alpha_k W_{(k)}(T) + \sum_{k=1}^m \beta_k I_k(T) + A_{\text{tail}}(T) \right] \\ &= O((\zeta T)^{-1}). \end{aligned}$$

For delay, let  $Q(t)$  be the total number of prompts in the segment-1 entry queue, downstream boundary queues, and threshold-controlled segment interiors. The bounds above, together with the  $O(1/\zeta)$  expected idle period needed to collect fewer than a threshold batch, imply

$$\mathbb{E} \left[ \int_0^T Q(t) dt \right] = O(T).$$

The expected number of completed prompts over  $[0, T]$  is  $\Theta(\zeta T)$ , because terminal unfinished work is  $O(1)$  in expectation. Hence the calendar-time average latency and TTFT are  $O(1/\zeta)$ . The theorem reports service-normalized delays,  $\mathbf{Latency}^{(\zeta, \pi)} = \zeta \mathbf{Latency}_{\text{cal}}^{(\zeta, \pi)}$  and  $\mathbf{TTFT}^{(\zeta, \pi)} = \zeta \mathbf{TTFT}_{\text{cal}}^{(\zeta, \pi)}$ , so

$$\mathbb{E}[\mathbf{Latency}^{(\zeta, \pi)}], \mathbb{E}[\mathbf{TTFT}^{(\zeta, \pi)}] = O(1).$$

#### E.4. High-Probability Memory Bound

The base term

$$M^\pi = \sum_{k=1}^m n_k (l + \bar{s}_k) \delta_k$$

accounts for threshold-sized segment batches. With the convention used in Theorem 2, segment 1 covers stages  $s = 0, \dots, l'_1$ , while for  $k \geq 2$ , the resident boundary queue at  $s = l'_{k-1}$  is tracked separately and segment  $k$ 's interior covers  $s = l'_{k-1} + 1, \dots, l'_k$ . The memory accounting separates two boundary terms. Fresh survivors that have just crossed from segment  $k - 1$  are a subset of the upstream threshold batch. They have the same KV-cache size as the prompts charged to the  $s = l'_{k-1}$  stage of segment  $k - 1$  in  $M^\pi$ , so charging them to the base term does not add memory. The only

additional memory is the carryover residual left at downstream boundaries across opportunities. Therefore, if  $V_{(k)}^r \leq c_k$  for all downstream boundaries, then total resident KV-cache memory is at most

$$M^\pi + \sum_{k=2}^m (l + l'_{k-1})c_k.$$

The deterministic  $n_k$  component of  $c_k$  is on the same scale as the base term: since  $n_k < n_{k-1}$ ,  $(l + l'_{k-1})n_k$  is bounded by the contribution of the terminal stage  $s = l'_{k-1}$  of segment  $k - 1$  already included in  $M^\pi$ .

Fix  $k \geq 2$ . Since  $p_k < n_k/n_{k-1} < 1$ , Appendix G.5 records the standard exponential-martingale construction for binomial increments and gives a unique  $\theta_k > 0$  satisfying

$$e^{-\theta_k n_k} (1 - p_k + p_k e^{\theta_k})^{n_{k-1}} = 1.$$

For  $S_{(k)}^i = \sum_{r=1}^i (\bar{Y}_{(k)}^r - n_k)$ , the process  $e^{\theta_k S_{(k)}^i}$  is a martingale. Doob's maximal inequality (see, e.g., Durrett 2019) gives

$$\mathbb{P}\left(\max_{0 \leq i \leq r} S_{(k)}^i \geq x\right) \leq e^{-\theta_k x}.$$

Equivalently, the finite-horizon reflected-random-walk bound in Appendix G.5 gives, for the dominating boundary queue,

$$\mathbb{P}\left(\max_{0 \leq r \leq B} (\tilde{W}_{(k)}^r - n_k) \geq x\right) \leq B e^{-\theta_k x}, \quad B \geq 1.$$

Since  $V_{(k)}^r \leq \tilde{W}_{(k)}^r$ , set

$$c_k = n_k + \theta_k^{-1} \ln\left(\frac{m(1 + \zeta T/d_0)}{\delta}\right).$$

Applying this finite-horizon bound with  $B = B_\zeta^{\max}$ , and then unioning over  $k = 2, \dots, m$ , gives

$$\mathbb{P}\left(V_{(k)}^r \leq c_k \text{ for all } k = 2, \dots, m, r \leq B_\zeta^{\max}\right) \geq 1 - \delta.$$

On this event, the no-overflow Nested WAIT dynamics are feasible on the physical GPU whenever the memory condition in Theorem 2 holds. If a physical implementation instead evicts on the complement of this event, the unconditional throughput gap acquires an additional  $O(\delta)$  failure term; the theorem reports the no-overflow performance bounds and the separate high-probability feasibility guarantee.

Finally, Appendix G.5 also gives

$$\theta_k \geq \frac{8(n_k - n_{k-1}p_k)}{n_{k-1}},$$

which is the stated drift-to-buffer scaling. This completes the proof.

## Appendix F: Proof of Theorem 3

The proof is a conservative version of the Nested WAIT argument. Instead of matching each instantaneous arrival pattern exactly, we dominate each service-window arrival count and each downstream thinning probability by worst-case quantities over the finite horizon. This gives the same bounded-queue and logarithmic-buffer structure as the constant-rate proof. The first-segment waiting condition is used only when converting bounded queues into service-normalized delay bounds; the throughput and memory conclusions use only the upper-side window-load conditions and the finite-horizon safety buffer. The structure proceeds as follows:

1. **First segment analysis:** Poisson arrivals with time-varying rates are dominated by a time-invariant process using the worst-case scaled-window arrival count  $\Lambda^\pi = \sup_b \{ \sum_{j=1}^m \mathcal{A}_j^{(\zeta)}(t(b), \Delta T_{[1, \dots, m]}) \}$ . We couple this queue to a Lindley process and apply Kingman's inequality.
2. **Subsequent segments analysis:** Binomial arrivals with time-varying thinning probabilities  $p_k(b)$  are dominated by the worst-case continuation probability  $p_k^*$  defined in Section A.1. The same Lindley coupling then controls each downstream boundary queue.
3. **Throughput, delay, and memory:** Bounded expected terminal work yields the normalized throughput gap. The service-normalized delay bound additionally uses the first-segment waiting condition to control idle waiting before the first segment. High-probability memory bounds follow from the same martingale construction with worst-case parameters.

We use  $b$  to denote the *batch index* and  $k \in \{1, \dots, m\}$  to denote the *segment index*. The total number of batches is denoted by  $B$ .

### F.1. First Segment Analysis

For the first segment ( $k = 1$ ), we use the same threshold-renewal convention as in Appendix E. Idle periods before a segment-1 service start contain fewer than  $n_1$  prompts, so they contribute only a bounded amount of terminal unfinished work and do not affect the memory bound. The first-segment waiting condition in Section A.1 is needed later only to bound these idle periods in service-normalized time. At renewal epochs, the state transition rule is

$$W_{(1)}^{b+1} = W_{(1)}^b + Y_{(1)}^b - n_1 \cdot \mathbf{1}\{W_{(1)}^b + Y_{(1)}^b \geq n_1\}, \quad Y_{(1)}^b \sim \text{Poisson}\left(\sum_{j=1}^m \mathcal{A}_j^{(\zeta)}(t(b), \Delta T_b)\right),$$

where  $t(b)$  denotes the physical start time of batch  $b$  and  $\Delta T_b$  denotes the service-normalized processing time of that batch, so its physical duration is  $\Delta T_b / \zeta$ .

*Worst-case domination.* Time-varying arrival rates prevent exact load balance because the arrival composition changes over the horizon. We therefore use a conservative domination argument. Since  $\Delta T_b \leq \Delta T_{[1, \dots, m]}$  (the service-normalized time for a full batch containing all  $m$  types), the arrivals during any realized service period are dominated by arrivals over a full service-normalized window. This gives the process  $\{\bar{W}_{(1)}^b\}$ :

$$\bar{W}_{(1)}^{b+1} = \bar{W}_{(1)}^b + \bar{Y}_{(1)}^b - n_1 \cdot \mathbf{1}\{\bar{W}_{(1)}^b + \bar{Y}_{(1)}^b \geq n_1\}, \quad \bar{Y}_{(1)}^b \sim \text{Poisson}\left(\sum_{j=1}^m \mathcal{A}_j^{(\zeta)}(t(b), \Delta T_{[1, \dots, m]})\right).$$

By (15), the aggregate arrival rate during any batch is bounded by the worst-case aggregate rate:

$$\sum_{j=1}^m \mathcal{A}_j^{(\zeta)}(t(b), \Delta T_{[1, \dots, m]}) \leq \sup_{\forall b} \left\{ \sum_{j=1}^m \mathcal{A}_j^{(\zeta)}(t(b), \Delta T_{[1, \dots, m]}) \right\} < n_1.$$

Define  $\Lambda^\pi = \sup_{\forall b} \left\{ \sum_{j=1}^m \mathcal{A}_j^{(\zeta)}(t(b), \Delta T_{[1, \dots, m]}) \right\} < n_1$  as the worst-case aggregate arrival count over a full service-normalized iteration window. This enables a time-invariant dominating process that provides tractable bounds:

$$\tilde{W}_{(1)}^{b+1} = \tilde{W}_{(1)}^b + \tilde{Y}_{(1)}^b - n_1 \cdot \mathbf{1}\{\tilde{W}_{(1)}^b + \tilde{Y}_{(1)}^b \geq n_1\}, \quad \tilde{Y}_{(1)}^b \sim \text{Poisson}(\Lambda^\pi).$$

This time-invariant process enables the same queueing bound used in the constant-rate proof. Define  $X_{(1)}^b = \tilde{Y}_{(1)}^b - n_1$ . The process  $\{\tilde{W}_{(1)}^b\}$  is dominated by a Lindley process:

$$\bar{W}_{(1)}^0 = 2n_1, \quad \bar{W}_{(1)}^{b+1} = \max\{2n_1, \bar{W}_{(1)}^b + X_{(1)}^b\}, \quad \forall b \in \mathbb{N}.$$

Thus we have the dominance chain:

$$W_{(1)}^b \leq \bar{W}_{(1)}^b \leq \tilde{W}_{(1)}^b \leq \bar{W}_{(1)}^b, \quad \forall b \in \mathbb{N}.$$

Since  $n_1 > \Lambda^\pi$ , the process  $\{\bar{W}_{(1)}^b\}$  has negative drift. By Kingman's inequality (Kingman 1962), with

$$\text{Var}(X_{(1)}^b) = \Lambda^\pi, \quad \left| \mathbb{E}[X_{(1)}^b] \right| = n_1 - \Lambda^\pi,$$

we obtain

$$\mathbb{E}[W_{(1)}^b] \leq 2n_1 + \frac{\mathbb{E}[(\tilde{Y}_{(1)}^b - n_1)^2]}{2(n_1 - \Lambda^\pi)}.$$

## F.2. Subsequent Segments Analysis ( $k \geq 2$ )

For segment  $k$  ( $2 \leq k \leq m$ ), arrivals come from prompts that completed segment  $k - 1$ . The batch index  $b$  for segment  $k$  counts only batches containing segment  $k - 1$ . These indices differ across segments but are bounded by the total batch count  $B$ . As in Appendix E,  $W_{(k)}^b$  is the post-review carryover residual at the downstream boundary; fresh survivors from segment  $k - 1$  are charged to the upstream threshold batch, and only residual carryover receives the extra safety buffer.

*Time-varying thinning probabilities.* Consider the  $b$ -th arrival to segment  $k$ , which comes from the  $b$ -th batch containing segment  $k - 1$ . The thinning probability  $p_k(b)$  depends on the prompt composition in this batch and varies over time. Although exact arrival times are unknown, the relevant prompts accumulate over one or more scaled windows  $(t, h)$  before the batch containing segment  $k - 1$  begins. To handle this time variation, we use the worst-case thinning probability. Conditional on the filtration before boundary revelation, the survivor indicators in a boundary batch are independent Bernoulli random variables with possibly different probabilities  $q_i \leq p_k^*$ . Their sum is therefore a Poisson-binomial random variable stochastically dominated by  $\text{Binomial}(n_{k-1}, p_k^*)$ . By (15) and the definition of  $p_k^*$  in Section A.1,

$$p_k(b) \leq p_k^* < \frac{n_k}{n_{k-1}}, \quad \forall b \in \mathbb{N}.$$

The arrival process is

$$Y_{(k)}^b \sim \text{Binomial}(n_{k-1}, p_k(b)), \quad \forall b \geq 0,$$

and the state transition rule is

$$W_{(k)}^{b+1} = W_{(k)}^b + Y_{(k)}^b - n_k \cdot \mathbf{1}\{W_{(k)}^b + Y_{(k)}^b \geq n_k\},$$

where  $b$  indexes batches containing segment  $k - 1$ , since new arrivals to segment  $k$  occur only when segment  $k - 1$  is processed.

Define  $X_{(k)}^b = Y_{(k)}^b - n_k$ . Although the distribution  $Y_{(k)}^b \sim \text{Binomial}(n_{k-1}, p_k(b))$  is time-varying, the coupling construction follows the same induction as Appendix G.4:

**LEMMA 9 (Coupling for Time-Varying Case).** *Define a coupled process  $\{\tilde{W}_{(k)}^b\}$  by:*

$$\begin{aligned} \tilde{W}_{(k)}^0 &= n_k, \\ \tilde{W}_{(k)}^{b+1} &= \max\{n_k, \tilde{W}_{(k)}^b + X_{(k)}^b\}, \\ X_{(k)}^b &= Y_{(k)}^b - n_k. \end{aligned}$$

*Then  $\tilde{W}_{(k)}^b \geq W_{(k)}^b$  for all  $b \in \mathbb{N}$ .*

To apply Kingman's inequality (Kingman 1962), we construct a time-invariant dominating process:

$$\begin{aligned}\bar{W}_{(k)}^0 &= n_k, \\ \bar{W}_{(k)}^{b+1} &= \max\{n_k, \bar{W}_{(k)}^b + \bar{X}_{(k)}^b\}, \\ \bar{X}_{(k)}^b &= \bar{Y}_{(k)}^b - n_k, \\ \bar{Y}_{(k)}^b &\sim \text{Binomial}(n_{k-1}, p_k^*), \quad \forall b \in \mathbb{N}.\end{aligned}$$

Then we have the dominance chain:

$$\bar{W}_{(k)}^b \geq \tilde{W}_{(k)}^b \geq W_{(k)}^b, \quad \forall b \in \mathbb{N}.$$

The process  $\bar{W}_{(k)}^b$  has the Lindley representation:

$$\begin{aligned}\bar{W}_{(k)}^0 &= n_k, \\ \bar{W}_{(k)}^{b+1} &= n_k + \max_{0 \leq i \leq b} \{\bar{S}_{(k)}^i\}, \\ \text{where } \bar{S}_{(k)}^0 &= 0, \bar{S}_{(k)}^i = \sum_{r=1}^i \bar{X}_{(k)}^r, \quad 1 \leq i \leq b.\end{aligned}$$

Since

$$\mathbb{E} \left[ \bar{X}_{(k)}^b \right] = n_{k-1} \cdot p_k^* - n_k < 0, \quad \forall b \in \mathbb{N},$$

the coupled process  $\{\bar{W}_{(k)}^b\}$  has negative drift. With

$$\text{Var}(\bar{X}_{(k)}^b) = n_{k-1} p_k^* (1 - p_k^*), \quad \left| \mathbb{E} \left[ \bar{X}_{(k)}^b \right] \right| = n_k - p_k^* n_{k-1},$$

Kingman's inequality (Kingman 1962) yields:

$$\mathbb{E} \left[ \max_{0 \leq i \leq b} \{\bar{S}_{(k)}^i\} \right] \leq \frac{n_{k-1} p_k^* (1 - p_k^*)}{2(n_k - n_{k-1} p_k^*)}.$$

Thus, the expected queue length is bounded:

$$\mathbb{E} \left[ W_{(k)}^b \right] \leq \mathbb{E} \left[ \tilde{W}_{(k)}^b \right] \leq \mathbb{E} \left[ \bar{W}_{(k)}^b \right] \leq n_k + \frac{\mathbb{E}[(\bar{Y}_{(k)}^b - n_k)^2]}{2(n_k - n_{k-1} p_k^*)}.$$

### F.3. Throughput, Delay, and High-Probability Bounds

The bounded queue estimates above imply that the expected terminal unfinished work is  $O(1)$ . This includes the first-segment entry queue, the downstream boundary queues, the bounded segment



interiors, and the final partial service opportunity. Dividing this terminal work by the scaled horizon  $\zeta T$  gives

$$\text{Throughput}_T^* - \mathbb{E}[\text{Throughput}^{(\zeta, \pi)}] = O((\zeta T)^{-1}).$$

This argument does not require a lower bound on the arrival rate. If arrivals fall below the first threshold near the end of the horizon, fewer than  $n_1$  prompts can remain in the external entry queue, which is still a bounded terminal effect.

For service-normalized delay, the queue-integral argument from Appendix E applies to the service periods and downstream boundary queues because their expected lengths are uniformly bounded. The only additional issue in the time-varying setting is waiting for the Segment 1 entry queue to reach its threshold during low-arrival intervals. The first-segment waiting condition gives  $\sup_t \zeta \mathbb{E}[\tau_1^{(\zeta)}(t)] < \infty$ , so these idle waiting periods add only  $O(1)$  in service-normalized time. Hence

$$\mathbb{E}[\text{Latency}^{(\zeta, \pi)}], \mathbb{E}[\text{TTFT}^{(\zeta, \pi)}] = O(1)$$

whenever the first-segment waiting condition holds.

For high-probability memory bounds, the analysis follows the same martingale construction applied to the time-invariant coupled process  $\{\bar{W}_{(k)}^b\}$  defined above. Each realized iteration has duration at least  $d_0/\zeta$ , so the number of realized opportunities over  $[0, T]$  is at most  $B_\zeta^{\max} = \lceil \zeta T/d_0 \rceil$ . By the same argument as in Appendix E, with  $\bar{X}_{(k)}^r = \bar{Y}_{(k)}^r - n_k$ ,  $\bar{Y}_{(k)}^r \sim \text{Binomial}(n_{k-1}, p_k^*)$ , and  $p_k^* < n_k/n_{k-1} < 1$ , the memory bound is:

$$\mathbf{M}^{(\zeta, T)} = M^\pi + \sum_{k=2}^m n_k \cdot (l + l'_{k-1}) + \sum_{k=2}^m \theta_k^{-1} \ln \left( \frac{m(1 + \zeta T/d_0)}{\delta} \right) \cdot (l + l'_{k-1}),$$

where  $\theta_k$  ( $2 \leq k \leq m$ ) is the unique positive solution to:

$$e^{-\theta_k n_k} (1 - p_k^* + p_k^* e^{\theta_k})^{n_{k-1}} = 1. \quad (20)$$

Since  $p_k^*$  is used to dominate every realized thinning probability, the same exponential-martingale argument as Appendix G.5 applies with this worst-case binomial increment. The lower bound for  $\theta_k$  follows from the same calculation.

## Appendix G: Proof of Lemmas

Throughout this section, we use  $b$  for the batch index and  $B$  for the total number of batches.

### G.1. Proof of Lemma 1

Write  $n$  for the critical single-type threshold, so  $X^b \sim \text{Poisson}(n)$ . Taking expectations on both sides of the state transition equation gives

$$\mathbb{E}[W^{b+1}] = \mathbb{E}[W^b] + n - n\mathbb{P}(W^b + X^b \geq n).$$

Summing from  $b = 0$  to  $B - 1$ , and noting that  $W^0 = 0$ , we obtain

$$\mathbb{E}[W^B] = nB - n\mathbb{E}[B - B_{\text{stuck}}] = n \cdot \mathbb{E}[B_{\text{stuck}}].$$

### G.2. Proof of Lemma 2

We prove by induction on the batch index  $b$ . The coupled process  $\{\tilde{W}^b\}$  starts with a safety buffer of  $2n$ , compared to the real process starting at 0, ensuring it remains above  $W^b + n$  throughout. For the inductive step, we consider two cases based on whether the system can process a batch or experiences a stuck iteration.

*Base case.* For  $b = 0$ ,  $\tilde{W}^0 = 2n \geq n = W^0 + n$ .

*Inductive step.* Assume  $\tilde{W}^b \geq W^b + n$  for some  $b \geq 0$ . We show  $\tilde{W}^{b+1} \geq W^{b+1} + n$  by considering two cases:

**Case 1 (Batch processed):** If  $W^b + X^b \geq n$ , the queue has accumulated enough prompts to process a batch, so  $W^{b+1} = W^b + X^b - n$ . Then

$$\tilde{W}^{b+1} = \max\{2n, \tilde{W}^b + X^b - n\} \geq \tilde{W}^b + X^b - n \geq (W^b + n) + X^b - n = W^{b+1} + n.$$

**Case 2 (Stuck iteration):** If  $W^b + X^b < n$ , insufficient prompts are available and the system cannot process a batch. The queue simply accumulates arrivals:  $W^{b+1} = W^b + X^b$ . The safety buffer ensures dominance:

$$\tilde{W}^{b+1} = \max\{2n, \tilde{W}^b + X^b - n\} \geq 2n > W^{b+1} + n.$$

By induction,  $\tilde{W}^b \geq W^b + n$  for all  $b \in \mathbb{N}$ .

### G.3. Proof of Lemma 8

The lemma compares the type- $j$  entry queue in the periodic comparison system with a reflected random walk that always subtracts  $n_j$  after adding one slot of arrivals, but never falls below the safety level  $2n_j$ . Fix type  $j$  and write  $\bar{W}^b = \bar{W}_{(j)}^b$ ,  $\tilde{W}^b = \tilde{W}_{(j)}^b$ ,  $n = n_j$ , and  $\bar{X}^b = \bar{X}_{(j)}^b$ . We prove by induction that  $\tilde{W}^b \geq \bar{W}^b$ .

The claim holds at  $b = 0$ , since  $\tilde{W}^0 = 2n$  and  $\bar{W}^0 = 0$ . Suppose  $\tilde{W}^b \geq \bar{W}^b$ . If  $\bar{W}^b + \bar{X}^b \geq n$ , then the comparison queue serves  $n$  prompts and

$$\bar{W}^{b+1} = \bar{W}^b + \bar{X}^b - n.$$

Using the induction hypothesis,

$$\tilde{W}^{b+1} = \max\{2n, \tilde{W}^b + \bar{X}^b - n\} \geq \tilde{W}^b + \bar{X}^b - n \geq \bar{W}^{b+1}.$$

If  $\bar{W}^b + \bar{X}^b < n$ , then no type- $j$  service occurs in the comparison queue and  $\bar{W}^{b+1} = \bar{W}^b + \bar{X}^b < n$ . In this case  $\tilde{W}^{b+1} \geq 2n > \bar{W}^{b+1}$ . Thus the domination holds for all  $b$ , and the argument applies independently to every type  $j$ .

#### G.4. Nested WAIT Coupling Bound

This lemma is used when a downstream boundary queue receives binomially thinned arrivals from the previous segment. We index the queue after the threshold review at the start of an opportunity in which the previous segment is processed. If the downstream boundary already has a threshold batch, Nested WAIT removes that batch in the same prefix batch; newly surviving prompts generated by the previous segment are treated as fresh boundary input for the next review.

Let  $W_{(k)}^b$  be the residual resident boundary queue for segment  $k \geq 2$  after this review, and let  $Y_{(k)}^b$  be the number of newly surviving prompts at the next boundary-crossing opportunity. The queue satisfies

$$W_{(k)}^{b+1} \leq W_{(k)}^b + Y_{(k)}^b - n_k \mathbf{1}\{W_{(k)}^b + Y_{(k)}^b \geq n_k\}.$$

Define  $X_{(k)}^b = Y_{(k)}^b - n_k$  and

$$\tilde{W}_{(k)}^0 = n_k, \quad \tilde{W}_{(k)}^{b+1} = \max\{n_k, \tilde{W}_{(k)}^b + X_{(k)}^b\}.$$

We prove by induction that  $\tilde{W}_{(k)}^b \geq W_{(k)}^b$ . At  $b = 0$ , the domination holds because  $\tilde{W}_{(k)}^0 = n_k$  and  $W_{(k)}^0 = 0$ . Suppose it holds at  $b$ . If  $W_{(k)}^b \geq n_k$ , then

$$W_{(k)}^{b+1} \leq W_{(k)}^b + Y_{(k)}^b - n_k \leq \tilde{W}_{(k)}^b + X_{(k)}^b \leq \tilde{W}_{(k)}^{b+1}.$$

If  $W_{(k)}^b < n_k$  but  $W_{(k)}^b + Y_{(k)}^b \geq n_k$ , a threshold batch is removed at the next boundary review, and

$$W_{(k)}^{b+1} \leq W_{(k)}^b + Y_{(k)}^b - n_k \leq \tilde{W}_{(k)}^b + X_{(k)}^b \leq \tilde{W}_{(k)}^{b+1}.$$

If  $W_{(k)}^b + Y_{(k)}^b < n_k$ , then

$$W_{(k)}^{b+1} = W_{(k)}^b + Y_{(k)}^b < n_k \leq \tilde{W}_{(k)}^{b+1}.$$

Thus the domination holds for all  $b \in \mathbb{N}$ .

### G.5. Martingale Root Bound for Nested WAIT

This auxiliary calculation records the exponential tilting parameter used in the high-probability memory bound. Let  $0 < p_k < n_k/n_{k-1} < 1$ , let  $\bar{Y} \sim \text{Binomial}(n_{k-1}, p_k)$ , and set  $X = \bar{Y} - n_k$ . The moment-generating equation

$$\mathbb{E}[e^{\theta X}] = e^{-\theta n_k} (1 - p_k + p_k e^{\theta})^{n_{k-1}} = 1$$

has a unique positive solution  $\theta_k$ . This is the standard Cramer–Lundberg root for a negative-drift random walk with bounded positive jumps (see, e.g., [Asmussen 2003](#)). With  $S_{(k)}^i = \sum_{r=1}^i X_{(k)}^r$ , the process  $e^{\theta_k S_{(k)}^i}$  is therefore a mean-one martingale, and Doob’s maximal inequality (see, e.g., [Durrett 2019](#)) yields

$$\mathbb{P}\left(\max_{0 \leq i \leq r} S_{(k)}^i \geq x\right) \leq e^{-\theta_k x}.$$

We will use the corresponding finite-horizon bound for the reflected process

$$Z_{(k)}^0 = 0, \quad Z_{(k)}^{b+1} = \max\{0, Z_{(k)}^b + X_{(k)}^b\}.$$

If  $\max_{0 \leq b \leq B} Z_{(k)}^b \geq x$ , then some excursion of the random walk has gained at least  $x$ : there are  $0 \leq a < b \leq B$  such that  $\sum_{r=a}^{b-1} X_{(k)}^r \geq x$ . Applying the preceding maximal inequality from each possible excursion start and unioning over at most  $B$  starts gives

$$\mathbb{P}\left(\max_{0 \leq b \leq B} Z_{(k)}^b \geq x\right) \leq B e^{-\theta_k x}, \quad B \geq 1.$$

For completeness, we also record the elementary drift-to-root bound used in the theorem. Writing  $D = n_k - n_{k-1}p_k > 0$  and  $g(\theta) = \log \mathbb{E}[e^{\theta X}]$ , we have  $g(0) = 0$ ,  $g'(0) = -D$ , and  $g''(\theta) \leq n_{k-1}/4$ . Applying Taylor’s theorem at the positive root  $g(\theta_k) = 0$  gives

$$\theta_k \geq \frac{8(n_k - n_{k-1}p_k)}{n_{k-1}}. \quad (21)$$

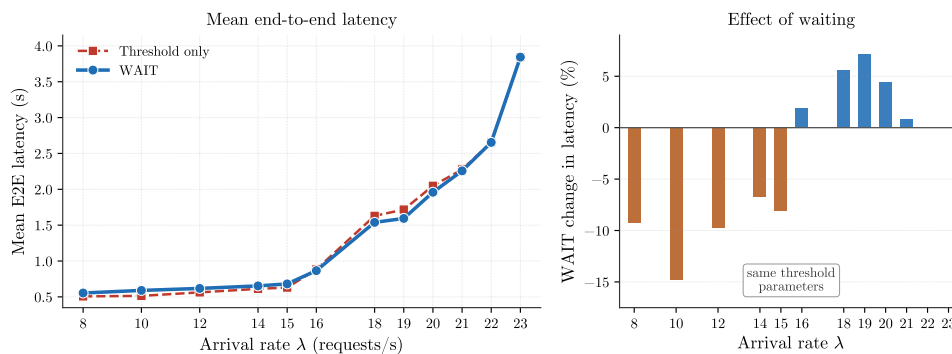
## Appendix H: Additional Experiments

This appendix reports supplemental robustness and implementation-fidelity analyses for Section 6. Section [H.1](#) isolates the role of the accumulation requirement by comparing WAIT with a threshold-only variant. Section [H.2](#) validates the simulator against direct GPU measurements and reports supplemental physical-GPU comparisons on single-type and real-data workloads. Section [H.3](#) examines repeated-run variability near the observed transition from near-overloaded to overloaded operation, and Section [H.4](#) evaluates the policy in a prefill-decode-disaggregated deployment.

## H.1. Threshold without waiting

We isolate the role of waiting on the single-type p512d20 workload by comparing full WAIT with a threshold-only variant, denoted *WAIT-no-wait*. For each arrival rate, the two policies use the same parameter setting and hence the same threshold upper bound. The threshold-only variant enforces this upper bound but does not wait for enough requests to accumulate before service. The comparison therefore holds the admission-capacity scale fixed and changes only whether service is delayed to form a full threshold batch.

Figure 16 reports the comparison across arrival rates. At low arrival rates, waiting introduces a modest latency cost: requests may wait to accumulate even when the server has enough capacity to process them immediately. This cost shrinks as the arrival rate increases. Near the overload boundary, waiting becomes useful because it regulates the timing at which work enters service, reducing latency relative to the threshold-only variant. Once the system is overloaded, the upper bound becomes the dominant control, and the two variants again behave similarly.



**Figure 16** Effect of waiting on the single-type p512d20 Vidur workload. At each arrival rate, full WAIT and the threshold-only variant use the same parameter setting; the comparison only changes whether service waits for enough requests to accumulate before scheduling.

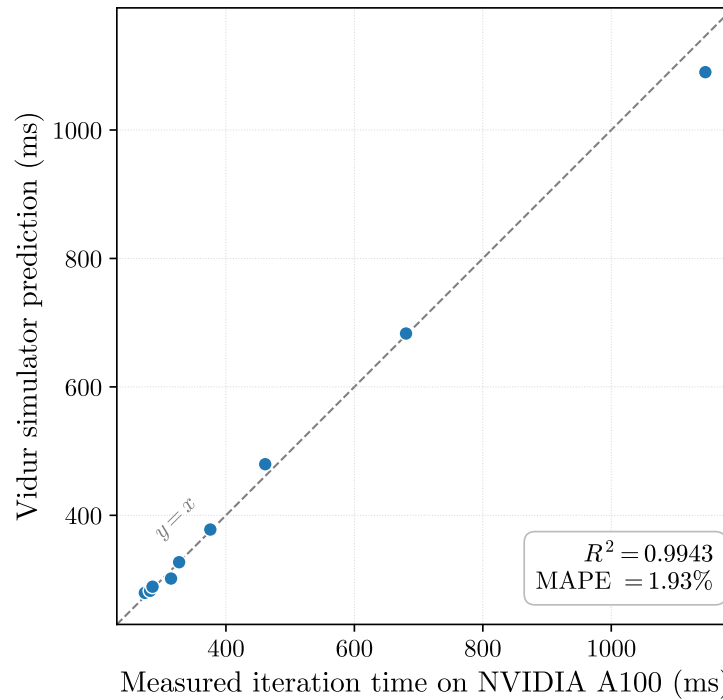
## H.2. Real-GPU validation

We complement the Vidur experiments in Section 6 with measurements on a physical NVIDIA A100 80 GB serving Llama-2-7B. These runs serve two purposes. First, direct GPU measurements calibrate the iteration-time model used by Vidur. Second, we implement WAIT’s admission logic using SGLang 0.5.7, an open-source LLM serving framework (SGLang Team 2024), and test the resulting scheduler in end-to-end GPU runs. We report three comparisons: iteration-time measurements against Vidur’s prediction, a single-type GPU workload corresponding to Section 6.1, and a real-data GPU workload corresponding to Section 6.2.

*Simulator vs GPU.* To calibrate the simulator in the operating region used in Section 6, we compare Vidur’s iteration-time predictions with direct measurements on the same A100 80 GB GPU running Llama-2-7B under the SGLang baseline scheduler. Vidur’s raw Llama-2-7B attention profiling covers batch sizes up to 128. Within this profiled range, the validation uses batch sizes  $B \in \{1, 2, 4, \dots, 128\}$  with prefill length 256 and decode length 20; we also test extrapolation to  $B = 256$ . In Figure 17, each point is one batch-size configuration: the horizontal coordinate is the measured GPU batch inference time, and the vertical coordinate is Vidur’s prediction. Points near the  $y = x$  line therefore indicate accurate simulator predictions. On the plotted grid, the fitted linear iteration-time model has mean absolute percentage error 1.93% and  $R^2 = 0.9943$ ; extrapolation to  $B = 256$  has absolute percentage error 4.93%. The fitted intercept and slope are  $d_0 \approx 276$  ms and  $d_1 \approx 0.0115$  ms per KV-cache token, respectively, consistent with the time model in Equation (1). This agreement supports the Vidur-based latency and operating-range comparisons in Section 6 for the workloads studied here.

*Single-type workload on GPU.* Using the same single-type workload as Section 6.1, we run end-to-end SGLang experiments with Poisson arrivals at  $\lambda \in \{1, 2, \dots, 15\}$  requests/s and 500 prompts per rate. The SGLang baseline scheduler uses `chunked_prefill_size=256` and `prefill_max_requests=8`; its default configuration triggers memory-overflow failures at several mid-to-high rates and is therefore not a feasible baseline on this workload. WAIT uses the same parameter values at all rates. Its system-wide batch-size cap is  $tl = 21$ , and its per-stage WAIT threshold is  $n_1 = 1$ . The chunk size is 384, so a prefill of length 512 is processed as  $K = \lceil 512/384 \rceil = 2$  prefill blocks before the request enters the 20 decode-token stages. Thus, the single-type pipeline has  $K + 20 = 22$  stages: WAIT schedules at most one request from each active stage, while  $tl = 21$  caps how many requests can be served concurrently across these stages. The SGLang patch implements this policy through the batch-size cap and the per-request prefill chunk cap; a transition block of size 10 switches the scheduler back to SGLang’s native rule in lightly loaded states. Figure 18 reports mean end-to-end latency as  $\lambda$  varies. Under this GPU configuration, WAIT is consistently lower-latency than the feasible SGLang baseline scheduler, with an average reduction of 29.7% across the arrival-rate grid.

*Real dataset on GPU.* Using the same lmsys-chat-1m workload distribution as Section 6.2, we run paired SGLang experiments with Poisson arrivals at  $\lambda \in \{0.1, 0.2, \dots, 0.7\}$  requests/s and 200 prompts per rate. The server is restarted before each rate so that the SGLang baseline and Nested WAIT are compared from the same clean GPU state. Nested WAIT uses chunk size 384

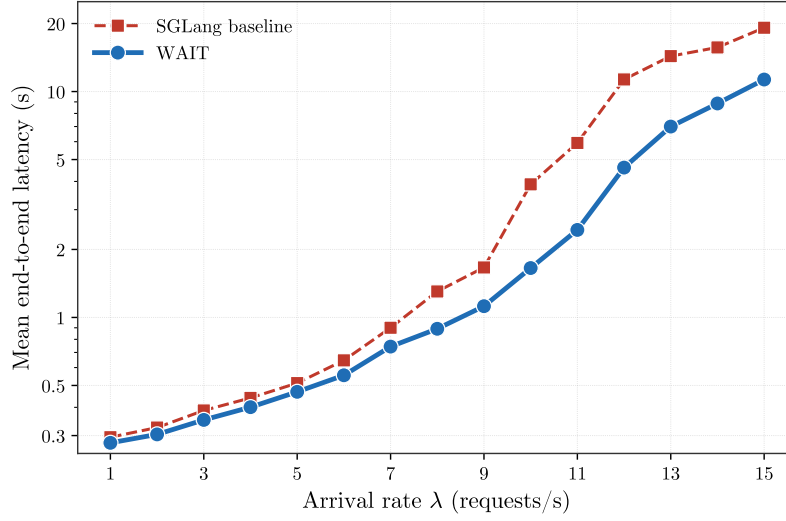


**Figure 17** Simulator calibration against NVIDIA A100 80 GB GPU measurements. Each marker is one batch-size configuration: its horizontal coordinate is the measured GPU batch inference time, and its vertical coordinate is the Vidur prediction. The comparison uses prefill length 256 and decode length 20, with powers-of-two batch sizes through the profiled range  $B \leq 128$  and one extrapolation check to  $B = 256$ . The dashed line is the  $y = x$  reference. The fitted linear iteration-time model in Equation (1) achieves  $R^2 = 0.9943$  and mean absolute percentage error 1.93% on the plotted grid.

and a rate-specific system-wide batch-size cap  $tl$ . For the real-data runs,  $tl$  is distributed over the decode horizon through the listed segment partitions, using the same continuation-based allocation described in Section 6.2. Table 4 reports the batch-size caps and decode-stage partitions used in these GPU runs. Figure 19 reports mean end-to-end latency in absolute units. Across the tested rates, Nested WAIT has lower latency than the SGLang baseline, with the largest reduction at  $\lambda = 0.6$  and an average reduction of 3.6%.

### H.3. Repeated-run variability

The main arrival-rate figures in Section 6 report averages over 10 independent simulation replications at each plotted configuration. To examine sampling variability near the observed transition from near-overloaded to overloaded operation, we additionally repeat the near-transition configurations 20 times and report the corresponding deviations.



**Figure 18** Single-type GPU experiment on SGLang 0.5.7 (NVIDIA A100 80 GB, Llama-2-7B, random prompts with input length 512 and output length 20): mean end-to-end latency versus arrival rate. WAIT is compared with the feasible SGLang baseline scheduler configuration used in this experiment and yields a mean latency reduction of 29.7% across the arrival-rate grid.

| $\lambda$ (req/s) | tl | Decode-stage segments            |
|-------------------|----|----------------------------------|
| 0.1               | 40 | [0, 175), [175, 350), [350, 500] |
| 0.2               | 50 | [0, 210), [210, 420), [420, 500] |
| 0.3               | 60 | [0, 200), [200, 400), [400, 500] |
| 0.4               | 70 | [0, 175), [175, 350), [350, 500] |
| 0.5               | 70 | [0, 200), [200, 400), [400, 500] |
| 0.6               | 70 | [0, 175), [175, 350), [350, 500] |
| 0.7               | 95 | [0, 250), [250, 500]             |

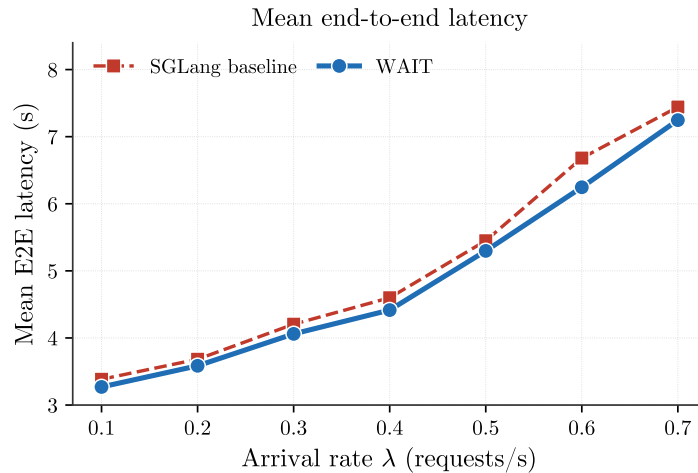
**Table 4** Nested WAIT parameter settings for the real-data GPU experiment on SGLang 0.5.7 with the lmsys-chat-1m workload distribution, NVIDIA A100 80 GB, Llama-2-7B, and 200 prompts per rate. The column tl gives the system-wide batch-size cap; the decode-stage ranges specify the segment partition over the 500-token decode horizon. The per-stage thresholds are induced from these two quantities by the continuation-based allocation rule in Section 6.2.

The repeated-run diagnostic preserves the same latency ordering as the main figures across independent arrival draws.

| Workload    | $\lambda$ | vLLM (s)         | Sarathi (s)     | WAIT (s)                          |
|-------------|-----------|------------------|-----------------|-----------------------------------|
| Single-type | 22        | 18.23 $\pm$ 2.74 | 1.85 $\pm$ 0.33 | <b>1.03 <math>\pm</math> 0.04</b> |
| Single-type | 23        | 26.78 $\pm$ 3.35 | 5.08 $\pm$ 1.72 | <b>1.29 <math>\pm</math> 0.10</b> |
| Two-type    | 22        | 56.70 $\pm$ 2.22 | 4.06 $\pm$ 1.66 | <b>2.36 <math>\pm</math> 0.57</b> |
| Two-type    | 23        | 80.25 $\pm$ 2.37 | 8.41 $\pm$ 1.74 | <b>5.66 <math>\pm</math> 1.88</b> |

**Table 5** Repeated-run reproducibility check near the observed transition from near-overloaded to overloaded operation. Each entry reports mean end-to-end latency together with the sample standard deviation across repeated simulation runs.



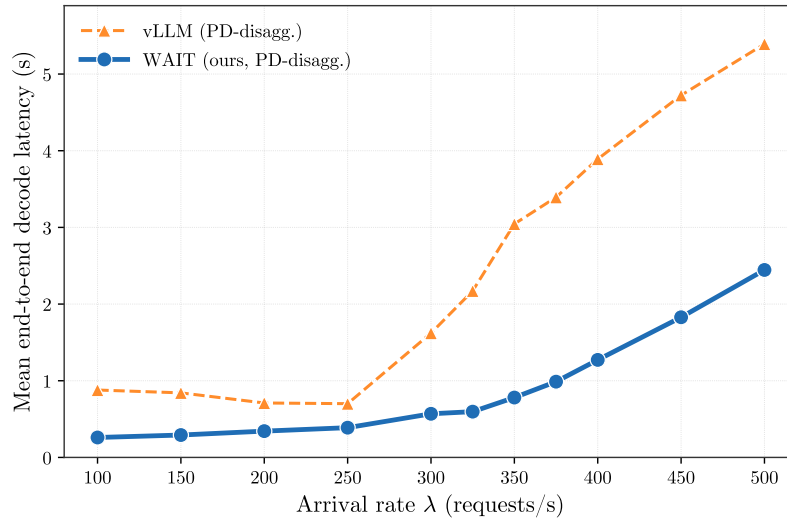


**Figure 19** Real-data GPU experiment on SGLang 0.5.7 with the lmsys-chat-1m workload distribution: mean end-to-end latency versus arrival rate for the SGLang baseline scheduler and Nested WAIT. The Nested WAIT runs use the rate-specific parameter settings in Table 4.

#### H.4. Prefill-decode-disaggregated deployment

Production LLM serving increasingly uses a *prefill-decode disaggregation* (PD) architecture, in which prefill and decode are executed by separate services and the KV cache is transferred between them (Patel et al. 2024, Zhong et al. 2024). This architecture aligns more closely with the memory-dependent timing model in Section 2. Because the decode server no longer mixes prefill and decode work, each iteration processes one token for each active post-prefill job, and the iteration time is driven primarily by active KV-cache usage. The linear iteration-time model in (1) is therefore especially appropriate in this setting. The corresponding control decision is how many post-prefill jobs to keep in concurrent decoding; the WAIT threshold implements this decision as an upper bound on the decode-side service population.

We implement this setting in Vidur by using a decode-only request generator: each request enters the simulated decode server after prefill has completed, with post-prefill context length 630 tokens and decode length 20 (p630d20). Arrivals follow a Poisson process with  $\lambda \in \{100, 150, \dots, 500\}$  requests/s. In this decode-only setting, Sarathi’s chunked-prefill rule no longer changes the scheduling decision, so vLLM’s PD scheduler (vLLM (PD)) represents the baseline. Figure 20 reports mean end-to-end decode latency: WAIT improves over vLLM at every rate, with the relative gap ranging from 45% at  $\lambda = 100$  to 55% at  $\lambda = 500$ . Thus, in this tested decode-only setting, the threshold rule improves delay by controlling the decode-side batch composition.



**Figure 20** Prefill-decode-disaggregated deployment (p630d20, decode-only): mean end-to-end decode latency versus arrival rate. WAIT attains between 45% and 55% latency reduction over the vLLM PD baseline across the tested arrival-rate range.